INTERACTION EFFECTS IN THE WEAKLY REPULSIVE BOSE GAS: TWO STUDIES

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ABSTRACT OF THE DISSERTATION

Interaction Effects in the Weakly Repulsive Bose Gas: Two Studies

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Two investigations on correlation effects in a uniform three-dimensional Weakly Interacting Bose Gas (WIBG) are presented in this thesis.

The first topic is a study of squeezing effects, in the sense of quantum optics, in the zero-temperature WIBG. After a review of relevant quantum-optics concepts (coherence and squeezing), we describe a variational investigation of squeezing in the Bose-condensed system.

The variational study reveals that squeezing in the zero-momentum (condensate) mode is not present at mean field level but comes from diagrams beyond one-loop order. On the other hand, for nonzero momenta, mixed-mode squeezing between opposite-momenta pair modes appears at mean field already. An alternate variational calculation is also formulated, to provide a more physical picture of the mixed-mode squeezing.

The second study involves the dependence of the Bose-Einstein condensation temperature (T_c) on the interaction. The shift of T_c from the non-interacting gas value, $T_{\rm c}^{(0)} = \left(2\pi/m[\zeta(\frac{3}{2})]^{2/3}\right) n^{2/3}$, is a long-disputed problem. We present a critique of various studies on the topic, and describe our own efforts.

Our approach to $T_{\rm c}$ involves a quasiparticle description of the Bose gas in the symmetry-unbroken phase. We regulate ultraviolet divergences and rule out any effect of quasiparticle lifetime, up to second order. Our method reduces the calculation of $T_{\rm c}$ to the determination of the quasiparticle spectrum at criticality. Two approaches to the quasiparticle spectrum are detailed out.

The first technique is to use the chemical potential as an explicit cutoff for regularizing critical-point infrared divergences. This procedure yields a $T_{\rm c}$ shift approximately proportional to $a\sqrt{|\ln a|}$, where a is the scattering length.

The second procedure is a self-consistent treatment of the spectrum, which enables a numeric calculation of both the quasiparticle spectrum at $T_{\rm c}$, and of $T_{\rm c}$ itself.

Preface

During the period spent at Rutgers, I studied several aspects of the weakly interacting Bose gas (WIBG) under the direction of my PhD advisor, Andrei E. Ruckenstein. I learned about quantized vortices precessing in trapped Bose condensates. For about two years, I probed at the "T_c problem". Later, I read about light scattering and coherence phenomena in the condensates, which eventually led to the work on "squeezing".

Other than the WIBG work, I attempted a number of things in collaboration with my colleagues S. Pankov and I. Paul. Most didn't work out, but the three of us published a paper on lattice structures of a Wigner crystal on liquid substrates.

Over the last year at Rutgers, I started projects on exciton physics in semiconductors, particularly on exciton condensation in biased coupled quantum wells. Some of the projects died ingloriously, but some I hope to continue later.

For the PhD thesis, it seemed appropriate to write on my work on the Bose gas. Over the decades, the Bose fluid has attracted some of the greatest minds of condensed matter physics. It has been and continues to be a fruitful source and testing-ground of many-body concepts and techniques. I am happy to be able to write something on this enduring topic.

Chapters 1 and 2 are review & survey material, background for the following chapters. They were written explicitly for this thesis.

The work described in chapter 3 has not been published, and appears here for the first time.

Chapter 4 is adapted from an article placed on the Los Alamos archives in February 2003 [1], where I reviewed the state of the T_c problem (the aspects I understood). The article was meant as a service for the community of theorists working on the T_c problem, and involves analysis of existing work, rather than new reults. A modified version serves in this thesis as an overview of the topic.

Chapter 5 is based on an article written with A. E. Ruckenstein. The paper is available on the Los Alamos archives [2], but not yet published.

M. Haque

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I have to thank my many friends at Rutgers, who made life away from my home country not only bearable but often delightful.

Above all, I am fortunate to have had Indranil Paul near me, who shared my joys and sorrows over the years, and with whom I explored culture and physics. I like to think that we represented the relationship between East and West Bengal, as it should be.

I spent many hours discussing physics and non-physics with Sergey Pankov. A rewarding experience!

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To those who took care of me while I was injured, far better than my mother could ever have done, I don't quite have the words to express my gratitude: Indranil and Sahana, Sergey, Max, Subarana & Saadi, David Brookes, William, Mick, Nayana Shah, Marcello Civelli. Because of these wonderful people, having an accident far away from home and family did not seem such a bad thing!

Physics was much more fun in discussion with peers. Indranil and Sergey, in addition to being close friends, also taught me the most. I am also grateful to Craig, Ivan Skachko, Carlos Bolech, Andrei Lopatin, Harald Jeschke, Edouard and Pankaj for physics lessons, and to Rossen for teaching computing tricks. My presentations were critiqued by Craig, Indranil, Edouard, Carlos, Nayana, Rossen. I hope I am a better speaker now, thanks to these people. Craig, David Brookes and Sahana proofread my physics writings: hopefully I write better because of them!

Throughout my Ph.D. years, I received continuous emotional support from my parents and siblings in Bangladesh, from teachers at my undergraduate institution (Susanta K. Das, Yasmeen Haque, Arun K. Basak, M. Zafar Iqbal), and from my Bangladeshi friends at Rutgers, Subarna Khan and Saadi.

Finally, Banu Ilal deserves all credit for feeding, housing, and sustaining me through the difficult final months of my PhD.

Dedication

Dedicated to

All who strive for justice, equality, and a secular world

And to

Peoples of the Third World.

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List of Abbreviations

WIBG	Weakly Interacting Bose Gas
HFB	Hartree-Fock-Bogoliubov
BEC	Bose-Einstein Condensate, or Bose-Einstein Condensation
RG	Renormalization Group
$\int_{\mathbf{k}}$	$\displaystyle\int \frac{d^3k}{(2\pi)^3}$
UV	Ultraviolet (high momentum)
IR	Infrared (low momentum)
3D	Three-dimensional, or three dimensions
1D (2D)	One-(two-)dimensional, or one (two) dimension(s)

Chapter 1

Weakly Interacting Bose Gas

The work described in this thesis involves the dilute uniform (homogeneous) Bose gas with a weak repulsive two-body interaction beween the constituent bosons. We present two studies of interaction effects in the weakly interacting Bose gas (WIBG).

The first topic (chapters 2 and 3) is a study of a particular kind of coherence phenomenon, known as squeezing. Squeezed states were first studied in the context of quantum optics, in the 1970s and 1980s. It is only recently that squeezing has been discussed in the context of Bose-Einstein condensation (BEC), and our work is a contribution towards this effort. We describe a variational study of squeezing effects in the zero-temperature Bose-condensed WIBG.

The second topic is a study of the effect of the repulsive interaction on the critical temperature (T_c) for Bose-Einstein condensation. This is a longstanding problem that has remained unsolved for several decades. Interest in this problem has re-surfaced with the resurgence of BEC research in recent years. In chapters 4 and 5, we describe attempts to understand the T_c shift in the WIBG.

Einstein's deduction in 1924, that an ideal Bose gas at low temperatures would condense into the lowest single-particle state , was used in 1938 by Fritz London, who suggested [3] that the newly-discovered superfluid transition of ^{4}He was related to the BEC phenomenon. The WIBG is a modification of the ideal Bose gas, introduced by theorists struggling to understand the superfluid physics of liquid helium. Liquid helium is a strongly interacting dense liquid rather than a dilute weakly interacting gas; nevertheless it was expected that some of the insights from studying the WIBG would carry over to liquid heium. In the 2-3 decades after Bogoliubov's seminal 1947 paper [4], the WIBG model became an object of intense interest in its own right. During this period, much of the early many-body theory was developed in the context of the weakly repulsive Bose gas.

Prolonged experimental efforts, directed at producing a Bose-Einstein condensate by laser-cooling a trapped atomic gas, were finally successful in 1995. The study of the WIBG received an enormous boost as a result. With new experiments being reported at an extraordinary rate, the amount of theoretical work in the field has continued to increase exponentially. Other than the physics relevant to the new experiments, older unsettled questions have received fresh attention during this period of heightened interest. The $T_{\rm c}$ problem is one such question.

The idea of squeezed states developed around 1975-1985, in studying novel quantum states of light and coherence phenomena using laser fields. The associated language and concepts open up a new set of questions concerning the Bose gas. We report in chapter 3 our efforts at identifying and addressing several such questions.

This chapter is an overview of the weakly repulsive Bose gas model. In Sec. 1.1 we introduce the WIBG Hamiltonian, and give a quick historical review.

Some known results on the WIBG are summarized briefly in Secs. 1.2 and 1.3. This description serves as the background for topics discussed in later chapters.

Much of the material reviewed in these two sections can be found in the 1964 review by Hohenberg and Martin [5], or in texts from the mid-1960s onwards [6–13]. The terminology used here is more modern. Also, the material on the many-body $\mathcal{T}$-matrix and its temperature-dependence is new, about five years old at the time of writing [14–16].

In Sec. 1.4 we give an introduction to the $T_{\rm c}$ problem, as a prelude to chapters 4 and 5. Since trapped atomic gases have become so important in the study of the WIBG, in Sec. 1.5 we survey selected experiments related to BEC in the trapped-cloud systems. Finally, we give an outline of the rest of the thesis in 1.6.

1.1 The model

We first present the WIBG Hamiltonian and make some preliminary comments, before giving a historical survey of theoretical (1.1.1) and experimental (1.1.2) research on the model. In second-quantized form, the Hamiltonian is

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} \hat{c}^{\dagger}_{\mathbf{k}} \hat{c}_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{p},\mathbf{q},\mathbf{k}} U(\mathbf{k}) \hat{c}^{\dagger}_{\mathbf{p}+\mathbf{k}} \hat{c}^{\dagger}_{\mathbf{q}-\mathbf{k}} \hat{c}_{\mathbf{p}} \hat{c}_{\mathbf{q}} \,. \tag{1.1}$$

$\epsilon_{\mathbf{k}} = k^2/2m$ is the free-gas spectrum, and $\hat{c}(\hat{c}^{\dagger})$ are bosonic annihilation (creation) operators. The simplest interaction to use is a momentum-independent one, $U(\mathbf{k}) = U$, i.e., a point interaction $U(\mathbf{r}) = U\delta(\mathbf{r})$ in real space.

The use of a point interaction often gives rise to unphysical ultraviolet divergences, some of which may be cured by using the two-body scattering amplitude, or the two-body t-matrix, instead of the interaction potential U. At first order, U is equal to the t-matrix. Therefore, using the low-energy limit for t [17], one finds $U = t = 4\pi\hbar^2 a/m$, where a is the s-wave scattering length. Corrections to this may be required at higher order. This is discussed in more detail in chapter 5, where we use the correction to remove ultraviolet divergences.

An alternative way to deal with ultraviolet divergences is to use the "pseudopotential" method introduced by Huang and Yang [18], which moderates the $U(\mathbf{r}) = U\delta(\mathbf{r})$ approximation with a differential operator: $U(\mathbf{r}) = U\delta(\mathbf{r})\frac{\partial}{\partial r}$. We will not be using this method in the present thesis.

The scattering length a, compared to other lengths relevant to the system, gives dimensionless measures of the interaction. One such measure is the ratio of a to the average inter-particle distance, $an^{1/3}$, where n is the density. (In two dimensions, the corresponding quantity is $an^{1/2}$.) Thus a bose gas maybe considered to be "weakly interacting," even if a is large, when the density is low enough. This is why the WIBG is often known as the *dilute* interacting Bose gas.

Using the thermal wavelength $\lambda = \sqrt{2\pi\hbar^2/mk_{\rm B}T}$, the ratio a/λ may also be used as the measure of the interaction, but only at nonzero temperatures. In the discussion of the $T_{\rm c}$ problem (chapters 4 and 5), both $an^{1/3}$ and a/λ are used in the literature.

The work in this thesis concerns only the three-dimensional (3D) Bose gas. The physics of two- and one-dimensional WIBGs are interesting topics in their own right, but will not be discussed here except in passing. The uniform WIBG does not undergo Bose-Einstein condensation at lower dimension. The investigations here involve BEC (in one case the $T = 0$ condensed gas, and in the other case the temperature of condensation). Therefore our calculations pertain to 3D.

1.1.1 Brief history

Theoretical studies of the WIBG start with Bogoliubov [4], who used the idea of macroscopic occupancy of the $\mathbf{k} = 0$ mode to do a mean-field approximation, replacing the $\mathbf{k} = 0$ operators ($\hat{c}_0$, $\hat{c}_0^\dagger$) by their c-number expectation value $\sqrt{N_0}$. This method yields the gapless Bogoliubov spectrum for the zero-temperature WIBG, $E_{\mathbf{k}} = [\epsilon_{\mathbf{k}}^2 + 2Un_0\epsilon_{\mathbf{k}}]^{1/2}$.

Beliaev [19, 20] introduced the formalism of Green's functions and diagrammatics in the study of the Bose-condensed gas. Other than reformulating Bogoliubov's work in field-theoretic language, he calculated terms in perturbation theory beyond mean field, and introduced the use of the $\mathcal{T}$-matrix (Sec. 1.2.3). His work was later refined and extended by Popov [21], who devised methods for higher temperatures as well. A review of this line of work, in modern language, can be found in [15].

In the West, a great number of studies were reported during the early period, up to 1965. We mention here some of the most influential.

- Penrose and Onsager developed [22,23] the idea of Off-Diagonal Long Range Order (ODLRO), in terms of the factorization of density matrices, to describe Bose condensation. The ODLRO idea was later expanded by Yang [24].

- Brueckener and Sawada performed an early field-theoretic description of the Bose-condensed system, in 1957 [25]. Like Beliaev, they use the $\mathcal{T}$-matrix. This work produced a spurious gap in the spectrum, an anomaly that torments many approximations for the WIBG.

- Hugenholtz and Pines [26] produced another early field-theoretic calculation. The celebrated Hugenholtz-Pines (HP) theorem, which we discuss in Sec. 1.2.2, was introduced and proved to all orders (for zero temperature) in this work.

- Girardeau and Arnowitt performed a variational calculation [27] emphasizing opposite-momenta pair correlations (as we do, in more detail and in a new language, in this thesis).

- Huang, Yang and collaborators did a series of calculations [18, 28, 29] in which they introduced their pseudopotential method of using a derivative with the s-wave approximation.

- The understanding of the many-boson system obtained during these early years is reviewed and formalized in the 1964 classic Hohenberg & Martin article [5].

Of the variety of theoretical work on the WIBG after the early years until around 1990, we point out as highlights the investigations on spatially nonuniform systems (development and use of the Gross-Pitaevskii equations [30–34]), vortices and vortex lattices (representative references: [35–40]), two-dimensional [12, 41–47] and one-dimensional [48–50] systems, numerical work via variational

[51, 52] and Monte Carlo [53, 54] methods, theory of BEC in particular systems, e.g., in spin-polarized hydrogen [33, 55–58] and in excitonic systems [59–61], etc.

With the prospect of the creation of a quantum-degenerate WIBG in the laboratory, work on this model has intensified since the early 1990s, and has exploded since the actual reports [62–64] in 1995 of successful cooling to form condensates.

The theoretical literature produced during this time is vast, but the majority of these calculations concern trapped condensates, or issues related to current experiments. Trapped-gas theory has been reviewed, e.g., in [65–67]. We will omit detailed discussion of this type of theoretical work, since this thesis is concerned with interaction effects in the *uniform* WIBG. A survey of current-era experiments is given in Sec. 1.5.

Of studies relevant to the homogeneous WIBG performed during the current period, I point out the following: renormalization group (RG) and effective-action (functional) formulations [14, 16, 68–70], descriptions of the Bose-condensed state avoiding symmetry-breaking (number-conserving descriptions) [71–75], and the use of quantum-optics concepts to study the quantum state of condensates [76–80].

1.1.2 Physical realizations

Liquid ^{4}He was the first (and for a long time the only) bosonic fluid that could be cooled down to quantum degeneracy. Below the "lambda point" (2.2 K), liquid helium goes superfluid and displays a number of remarkable properties. Helium is a dense liquid with strong interactions among the particles, very different from the ideal non-interacting gas for which Einstein had first predicted the BEC phenomenon. London's idea [3], that ^{4}He superfluid properties are related to BEC, was therefore not quite obvious. In fact, Landau's paper on helium superfluidity [81], which explained many aspects of the phenomenon, never mentions the role of bosonic statistics.

The development and acceptance of London's idea (of superfluid helium being a Bose condensate) took some time. The WIBG model grew as a compromise between, on the one hand, the textbook ideal Bose gas with no interactions, and on the other, liquid helium the strongly interacting liquid. The WIBG does indeed display (explain) many of the properties of liquid helium, e.g., the existence of a second-order transition, the linear spectrum at long wavelengths, quantized vortices, etc. It does not reflect certain other properties, such as the roton minimum in the superfluid helium spectrum, which occur in ^{4}He because of its large density.

As the theoretical WIBG model itself became a subject of intense interest, it became natural to use liquid helium to study the model, rather than thinking of WIBG studies as a path to understanding helium superfluidity. In search of a physical realization of the WIBG, Reppy and coworkers [82–85] at Cornell used helium in Vycor (a porous glass) to produce extremely dilute liquid helium, in both two or three dimensions. (Helium will act as a WIBG if it is dilute enough, when na^3 is reduced far below unity.) In addition to other measurements starting from around 1980, Reppy's group recently used this arrangement to measure the transition temperature (T_c) for the Bose condensation of the WIBG [85].

Superconductors and superfluid ^{3}He may be regarded as a condensate of pseudo-bosonic Cooper pairs, although there are many differences with the WIBG picture of condensation. (For example, the Cooper pairs themselves break up at the temperature where the condensate vanishes). With the advent of high-T_c superconductors, the picture of superconductivity as a BEC phenomenon has become important again, the idea being that in some parameter regimes there is a crossover from a momentum-space pairing (BCS picture) to real-space pairing (BEC picture). The BCS-BEC crossover idea was introduced in 1969 [86] and has received recent attention in the context of the cuprate superconductors.

Another classic context in which BEC physics has been sought for many years is that of excitons in semiconductors. The idea is that electron-hole pairs excited by a laser beam form bound states (excitons or biexcitons) which are approximately bosonic, and might be made to condense if the density is high enough (intense laser) and the temperature is low enough. The electon-hole pair may be closely or loosely bound compared to the density, so that a BCS and a WIBG description are possible in different geometries or different parameter regimes.

Since the first suggestions in the early 1960s [87, 88], experimental attempts toward a degenerate excitonic WIBG have continued for four decades, in a number of systems and geometries. (Relatively recent reviews can be found in [89] and in the exciton-related articles in [90]). Currently, use of parallel quantum wells, in which the electrons and holes are forced by a biasing field to be spatially separated, have generated intense interest [91–94].

Spin-polarized atomic hydrogen is the system whose study led to attempts to cool alkali gases, which eventually led to the first realizations of WIBGs [62–64,95]. Considering the current importance of these systems, we will discuss these systems separately in more detail, in 1.5.1.

1.2 Field-Theoretic Language

The Green's function formalism has to be specialized for the case of a Bose-condensed gas, which we summarize in this section. This will allow us to discuss $U(1)$ symmetry breaking (1.2.2), and various levels of approximations (Sec. 1.3).

1.2.1 Green's functions and diagrammatics

The peculiarity of the condensed system is the macroscopic occupancy of a particular ($\mathbf{k} = 0$) single-particle state. The usual way to deal with this is to treat the second-quantization operators for this particular mode as commuting numbers. Propagators involving condensate ($\mathbf{k} = 0$) bosons are therefore treated specially.

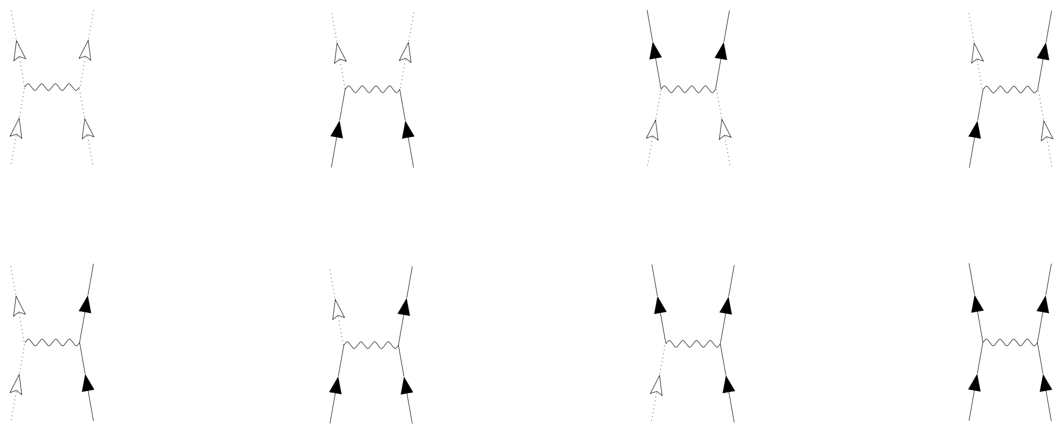


Figure 1.1: Eight different vertices are required when the mean-field prescription $\hat{c}_0$, $\hat{c}_0^\dagger \to \sqrt{N_0}$ is used. (Solid line represents non-condensate propagator, dotted lines are $\mathbf{k} = 0$ propagators or factors of $\sqrt{N_0}$, and wavy lines represent the interaction.) The first vertex goes in the calculation of the ground state. The next four, involving two condensate lines, are the ones retained in mean field treatments.

Therefore, for the interaction term in Eqn. (1.1), we have eight different kinds of vertices in the field theory, depending on the number of condensate lines (Fig. 1.1).

We want to describe the Green's functions for both zero and finite temperatures. We will use a generic "time" argument $\tilde{t}$, which will mean, for $T = 0$, the usual time, and for $T \neq 0$, the periodic variable τ with inverse-temperature dimensions. Similarly, the variable z will represent ordinary frequecies for $T = 0$ and discrete Matsubara frequencies for $T \neq 0$. Also, retarded and advanced Green's functions can be obtained with the usual prescription $z \to \omega \pm 0^+$.

The usual Green's function is defined as the correlator of a time ordered product, $G(\mathbf{r}, \tilde{t}) = -i\langle T\hat{\psi}(\mathbf{r}, \tilde{t})\hat{\psi}^\dagger(0, 0)\rangle$. To treat the Bose gas with a macroscopic occupation of one single-particle state, it becomes convenient to also consider correlators of the form $\langle\hat{\psi}\hat{\psi}\rangle$ and $\langle\hat{\psi}^\dagger\hat{\psi}^\dagger\rangle$. We therefore use the tensor $\hat{\Psi} = \begin{pmatrix} \hat{\psi} \\ \hat{\psi}^\dagger \end{pmatrix}$

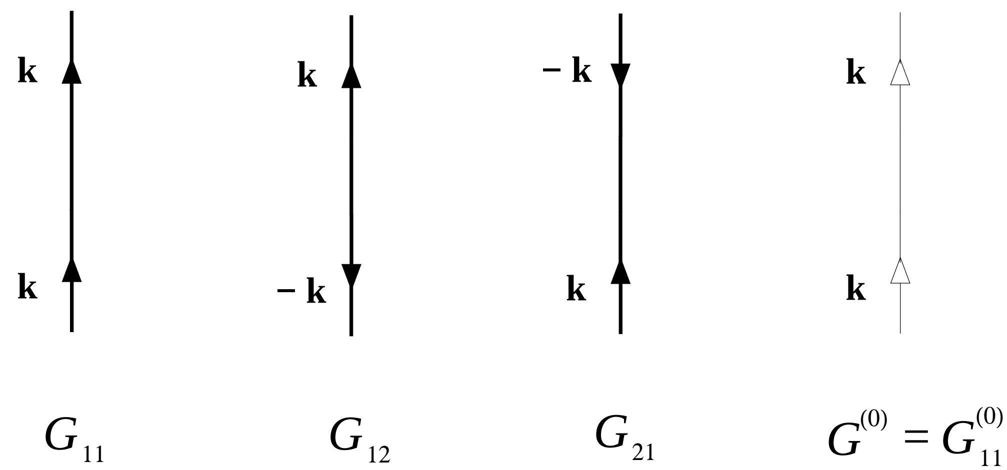


Figure 1.2: Diagrammatic representation of normal and anomalous propagators. Thick lines represent full (interacting) propagators; thin line represents free propagators. For the free theory there are no anomalous quantities.

to define a 2×2 matrix Green's function

$$G(x) = -i\langle T\hat{\Psi}(x)\hat{\Psi}^\dagger(0)\rangle \;=\; -i\begin{pmatrix}\langle T\hat{\psi}(x)\hat{\psi}^\dagger(0)\rangle & \langle T\hat{\psi}(x)\hat{\psi}(0)\rangle \\ \langle T\hat{\psi}^\dagger(x)\hat{\psi}^\dagger(0)\rangle & \langle T\hat{\psi}^\dagger(x)\hat{\psi}(0)\rangle\end{pmatrix},$$

with $x = (\mathbf{r}, \tilde{t})$. Hereafter we will work in Fourier space with $(\mathbf{k}, z)$ arguments.

The Green's function matrix actually has only two independent elements, as $G_{22}(\mathbf{k}, z) = G_{11}(-\mathbf{k}, -z)$ and $G_{21}(\mathbf{k}, z) = G_{12}(-\mathbf{k}, -z)$. G_{11} and G_{12} are called respectively the normal and anomalous Green's functions.

The self-energy is similarly a 2×2 matrix with two independent elements, the normal self-energy Σ_{11} and the anomalous self-energy Σ_{12}. Figs. 1.2 and 1.3 show the Green functions and self-energies for a Bose-condensed system.

Dyson's equation, $G = G^{(0)} + G^{(0)}\Sigma G$, is now a matrix equation, representing four equations relating the G_{ij}'s and the Σ_{ij}'s, two of which are independent. These are called the Dyson-Beliaev equations.

Using $G^{(0)}_{11}(\mathbf{k}, z) = 1/(z - \epsilon_{\mathbf{k}} + \mu)$ and $G^{(0)}_{12} = 0$, the Dyson-Beliaev equations

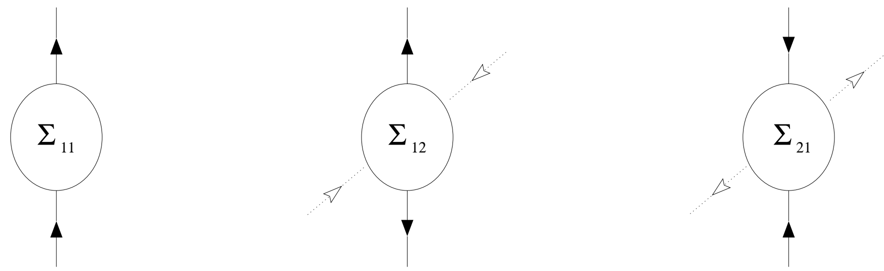


Figure 1.3: Diagrammatic representation of normal and anomalous self-energies. Dotted lines are propagators involving zero-momentum (condensate) bosons

can be solved to yield

$$G_{11} = \frac{z + \epsilon_{\mathbf{k}} - \mu + \Sigma_{22}}{[z + \epsilon_{\mathbf{k}} - \mu + \Sigma_{22}]\,[z - \epsilon_{\mathbf{k}} + \mu - \Sigma_{11}] + [\Sigma_{12}]^2}\ , \tag{1.2}$$

$$G_{12} = \frac{-\Sigma_{12}}{[z + \epsilon_{\mathbf{k}} - \mu + \Sigma_{22}]\,[z - \epsilon_{\mathbf{k}} + \mu - \Sigma_{11}] + [\Sigma_{12}]^2}\ . \tag{1.3}$$

As in the usual Green's function formalism, the pole of the Green's function gives the excitation spectrum of the system.

1.2.2 The Hugenholtz-Pines theorem, $U(1)$ symmetry breaking, and gapless spectrum

BEC is, of course, a classic example of spontaneous symmetry breaking. The $U(1)$ symmetry $\hat{\psi}(\mathbf{r}) \to \hat{\psi}(\mathbf{r})\, e^{i\theta}$, present in the Hamiltonian (1.1), is broken in the condensed phase as $\hat{\psi}(\mathbf{r})$ develops an expectation value and chooses a particular value of the phase θ.

In the language developed in 1.2.1, spontaneous symmetry breaking manifests itself as the Hugenholtz-Pines (HP) theorem:

$$\mu \;=\; \Sigma_{11}(\mathbf{k} = 0, z = 0) \;-\; \Sigma_{12}(\mathbf{k} = 0, z = 0) \tag{1.4}$$

The HP theorem was first proved by Hugenholtz and Pines [26] for $T = 0$. A more general proof, valid for all T and outlined briefly below, was given by Hohenberg and Martin [5], from the simple requirement that the field operator $\hat{\psi}(\mathbf{r})$ acquire a nonzero expectation value.

In [5], a source term $\int_{\mathbf{r}}[\eta^*(\mathbf{r})\psi(\mathbf{r})+\eta(\mathbf{r})\psi^\dagger(\mathbf{r})]$ was first used in the Hamiltonian to generate a nonzero $\langle\hat{\psi}(\mathbf{r})\rangle$. Using linear response and a small shift in the phase of η, one can derive $G(0,0)^{-1}\tau_3\langle\hat{\Psi}\rangle = \binom{\eta}{\eta^*}$. (Here $G(\mathbf{k},\omega)$ is the 2×2 Beliaev matrix and τ_3 is the Pauli matrix.) The condition to preserve the nonzero $\langle\hat{\psi}(\mathbf{r})\rangle$ when the fictitious source η vanishes is thus $G(0,0)^{-1} = 0$, which, upon use of the Dyson-Beliaev equations, leads to the HP theorem (1.4).

According to Goldstone's theorem [96–98], the spontaneous breaking of a continuous symmetry, in the absence of coupling to a gauge field, implies the existence of a gapless mode. The spectrum of the Bose-condensed state thus needs to be gapless, since the existence of a nonzero $\hat{\psi}(\mathbf{r})$ breaks a $U(1)$ symmetry.

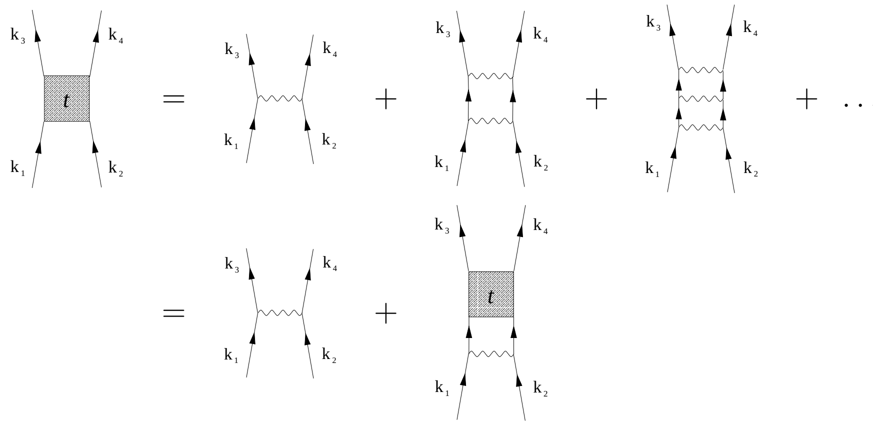
It is not surprising, therefore, that the same condition (1.4) can also be obtained by imposing the requirement that the spectrum be gapless. In Eqn. (1.2), setting the denominator to zero (pole of G_{11} gives spectrum), and imposing the condition $z(\mathbf{k}\to 0) = 0$, gives us back the Hugenholtz-Pines theorem (1.4).

1.2.3 The two-body t matrix and many-body $\mathcal{T}$ matrix

We will introduce here two quantities, each having loosely the meaning of an "effective interaction," which recur in WIBG theory. The first is the two-body t-matrix, which is the sum of all ladder diagrams. For our purposes, it is equivalent to the vacuum scattering amplitude f of two-particle scattering theory, which describes the effect of repeated interactions among two particles in vacuum. The second quantity is the many-body $\mathcal{T}$-matrix, which also incorporates the effect of the surrounding medium on the interactions.

We will denote the two quantities by t and $\mathcal{T}$ respectively. (Shi and Griffin [15] in their review use the symbols Γ and $\tilde{\Gamma}$ respectively. Stoof [16,99] uses T^{2B} and T^{MB}.)

The diagrammatic definition of the two-body t matrix:



Note that the internal propagators are all non-interacting propagators (thin lines). The corresponding equation is

$$t(\mathbf{k}_1, z_1; \mathbf{k}_2, z_2; ; \mathbf{k}_3, z_3; \mathbf{k}_4, z_4) = U(\mathbf{k}_1 - \mathbf{k}_2) + \int \frac{d^3q dz_q}{(2\pi)^4} U(\mathbf{q}) G^{(0)}(\mathbf{k}_1 - \mathbf{q}, z_1 - z_q) G^{(0)}(\mathbf{k}_2 + \mathbf{q}, z_2 + z_q) \times t(\mathbf{k}_1 - \mathbf{q}, z_1 - z_q; \mathbf{k}_2 + \mathbf{q}, z_2 + z_q; ; \mathbf{k}_3, z_3; \mathbf{k}_4, z_4) \,. \quad (1.5)$$

Actually, one only needs two momenta ($\mathbf{k}_1 - \mathbf{k}_2$ and $\mathbf{k}_3 - \mathbf{k}_4$) and a single complex energy to specify the t-matrix. The t-matrix is related to the bare potential by the Lippman-Schwinger equation

$$t(\mathbf{k}, \mathbf{k}'; z) = U(\mathbf{k} - \mathbf{k}') + \int_{\mathbf{q}} \frac{U(\mathbf{k} - \mathbf{q})\, t(\mathbf{q}, \mathbf{k}'; z)}{z - 2\epsilon_{\mathbf{q}}}$$

In chapter 5, we will use the on-shell version of this relation to remove ultraviolet divergences from our quasiparticle description of the Bose gas.

The two-body t-matrix is related to the vacuum scattering amplitude f by the Beliaev-Galitskii relation [20, 100]; at low densities they coincide. We will not need to distinguish between the two in our work.

The quantity physically measured in a scattering experiment in vacuum is t (or f) rather than the bare interaction $U(\mathbf{k})$. The zero-momentum limit of the t-matrix is proportional to the s-wave scattering length a:

$$t(\mathbf{k}_1; \mathbf{k}_2; ; \mathbf{k}_1; \mathbf{k}_2) = t = \frac{4\pi a}{m} \,.$$

The many-body $\mathcal{T}$-matrix is also a sum of ladder diagrams; however the internal propagators in the ladders are now full propagators rather than the bare propagators summed in the two-body t-matrix. Eqn. 1.5 will give $\mathcal{T}$ instead of t when the $G^{(0)}$'s are replaced by G's.

1.3 Approximations: Mean Field and Beyond

Having introduced the relevant quantities, in this section we describe some of the approximations used in describing the WIBG. In 1.3.1, the properties of the zero-temperature Bose gas are described at the level of various mean field approximations. This will be useful as a reference when, in chapter 3, we develop a description of the $T = 0$ BEC in terms of a squeezed coherent state, and compare our results with standard WIBG properties.

In 1.3.2, we review some finite-temperature physics of the WIBG; this is a conceptual prelude to our work on the BEC transition temperature (chapters 4 and 5). The temperature-dependence of the zero-energy $\mathcal{T}$ matrix is discussed separately in 1.3.3, because of the importance of this quantity in current studies of the BEC transition.

Finally, a discussion of proceeding beyond mean field is given in 1.3.4. We gloss over perturbative techniques for the Bose-condensed system, and focus instead on variational treatments, which are more relevant to this thesis.

1.3.1 Hartree-Fock-Bogoliubov approximation at $T = 0$

Mean-field schemes for the WIBG were developed first by Bogoliubov [4], and later improved by Popov [21], Girardeau and Arnowitt [27], and others.

Bogoliubov's original approximation involves the mean-field substitution $\hat{c}_0, \hat{c}_0^\dagger \rightarrow N_0^{1/2}$, along with a neglect of all terms involving more than two non-condensate operators. The Hamiltonian thus becomes quadratic and can be solved by a

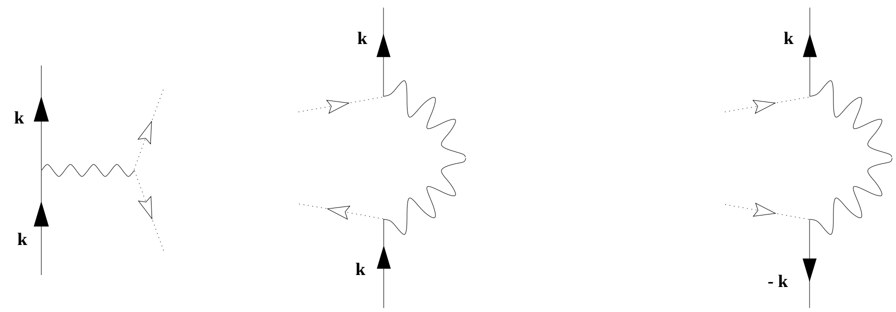


Figure 1.4: Diagrams included in the Bogoliubov approximation, for the normal self-energy Σ_{11} (first two diagrams) and for the anomalous self-energy Σ_{12} (last diagram).

canonical transformation. In the language of Green's functions, this involves using for the normal self-energy the Hartree and Fock diagrams with condensate atoms only (Fig. 1.4).

$$\Sigma_{11}(\mathbf{k}, z) = n_0 \left[U(0) + U(\mathbf{k})\right] , \qquad \Sigma_{12}(\mathbf{k}, z) = n_0 U(\mathbf{k}) \, .$$

Using these self-energies, one obtains the Bogoliubov spectrum

$$E_{\mathbf{k}} = \sqrt{\epsilon_{\mathbf{k}}^2 + 2n_0 U(\mathbf{k})\epsilon_{\mathbf{k}}} \, ,$$

which at long wavelengths has a linear phonon spectrum with sound velocity $c = [n_0 U(0)/m]^{1/2}$.

Even at the mean field level, the Bogoliubov treatment can be improved by taking into account interactions between non-condensate bosons. In other words, we include in the normal self-energy (Σ_{11}) the diagrams in Fig. 1.5 in addition to the ones of Fig. 1.4. This improved mean-field theory is known as the Hartree-Fock-Bogoliubov (HFB) approximation.

Popov [21] was one of the first to have included non-condensate contributions at mean field (and beyond); so the name Popov approximation (and sometimes Girardeau-Arnowitt approximation) is also used for this level of treatment.

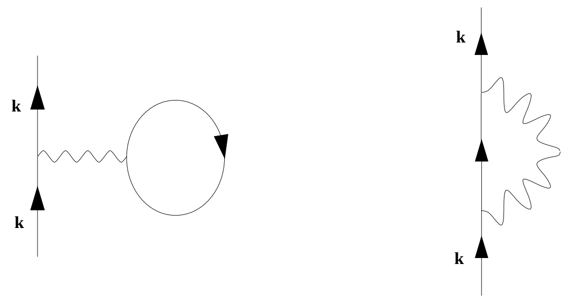


Figure 1.5: Diagrams included in the normal self-energy Σ_{11} in the HFB approximation, in addition to those already included in the Bogoliubov approximation.

The Popov improvement is particularly important at non-zero temperatures, when a significant fraction of the WIBG is outside the condensate. For now we describe this approximation at $T = 0$. Assuming $\mathbf{k}$-independent interaction $U(\mathbf{k}) = U(0) = U$,

$$\Sigma_{11}(\mathbf{k}, z) = 2n_0 U + 2U \int \frac{d^3k_1 dz_1}{(2\pi)^4} G_{11}^{(0)}(\mathbf{k}_1, z_1)\; e^{iz_1 0^+} = 2nU\,,$$

$$\Sigma_{12}(\mathbf{k}, z) = n_0 U\,.$$

In general, when we refer to the $T = 0$ HFB approximation, we mean the approximations $\Sigma_{11} = 2Un$ and $\Sigma_{12} = Un_0$.

At non-zero temperatures, Popov and HFB refer to different approximations [101], but they are identical at $T = 0$ and we are not interested much in the distinctions between finite-temperature mean-field schemes (next section).

1.3.2 Finite temperatures

For temperatures above the condensation temperature T_c, the WIBG is normal, anomalous self-energies and Green's functions vanish, and the occupancy of the lowest ($\mathbf{k} = 0$) single-particle state, N_0, becomes microscopic. Perturbation theory reduces to the more familiar form with a single kind of propagator and self-energy. This is the formalism we use in our discussion analysis of T_c, in chapter

5, where we approach the transition from above.

The fact that the anomalous self-energies vanish at and above T_c allows one to use the Hugenholtz-Pines theorem for specifying the transition point as $\mu = \Sigma(0,0)$.

At nonzero temperatures below T_c, the basic structure of the theory remains the same as that for $T = 0$, but it becomes important to account for condensate depletion and for the correlations between non-condensate particles. A number of approximations have arisen to deal with such effects. Discussion and classifications of the various approximations may be found, e.g,. in Refs. [5, 15, 101].

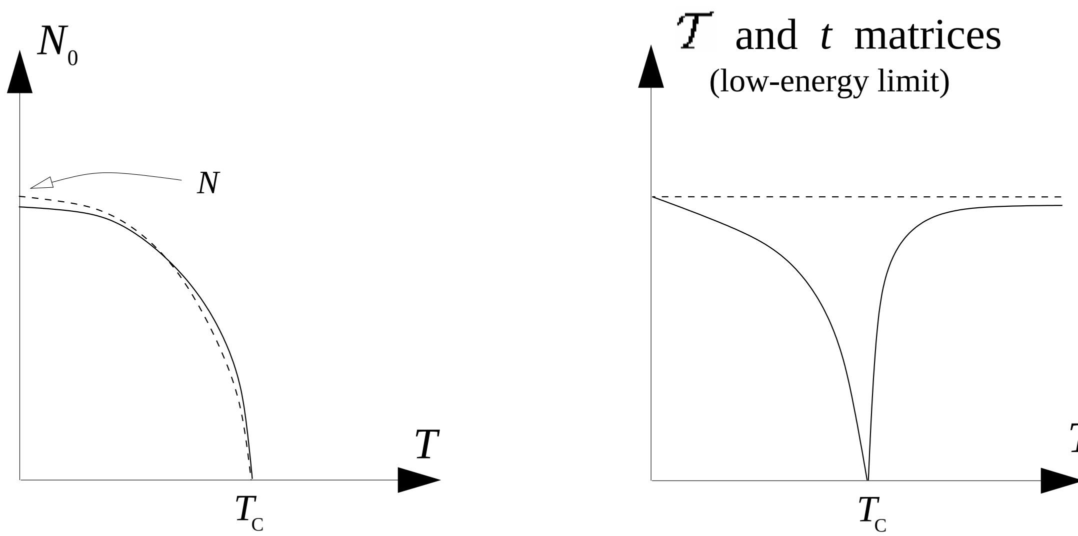


Figure 1.6: Schematic temperature-dependences of: (A) $\mathbf{k} = 0$ state occupancy for ideal Bose gas (dashed curve) and WIBG (full curve); (B) low-energy limits of the two-body t-matrix (dashed curve) and the many-body $\mathcal{T}$-matrix (full curve).

Fig. 1.6 shows schematically the variation of N_0 with temperature, for the ideal and interacting Bose gases. The interacting gas has some depletion even at $T = 0$ ($N_0(T = 0) < N$), and the low-T variation is $\sim -T^2$ rather than the $\sim -T^{3/2}$ of the ideal gas [12, 15]. We point out that the two $N_0(T)$ curves are made to vanish at the same temperature marked T_c; this reflects the facts that (1) the shift of T_c in the WIBG (compared to the ideal gas $T_c^{(0)}$) is small, and its magnitude not quite agreed upon, and (2) even the direction of the shift was under debate until a few years ago. This shift is the subject of chapters 4 and 5.

Also shown in Fig. 1.6 is the temperature-variation of the zero-energy many-body $\mathcal{T}$-matrix [14–16], discussed in more detail in 1.3.3.

1.3.3 $\mathcal{T}$-matrix near T_c

The vanishing of the low-energy $\mathcal{T}$-matrix at the transition point is a dramatic many-body effect, which has never been fully accounted for in T_c calculations. We believe this to be the main limitation of quasi-particle treatments near the critical point. A renormalization group (RG) calculation [16] shows that the effective two-body interaction follows the many-body $\mathcal{T}$-matrix curve rather than the temperature-independent two-body t-matrix.

The $\mathcal{T}$-matrix curve is a testament to the continuing excitement of WIBG theory today: this striking feature was discovered only in 1997, by Bijlsma & Stoof [14], via a variational calculation followed by numerics. An approximate derivation has been given by Shi & Griffin [15], which we summarize here.

The zero-energy limits of the two quantities will be denoted by t_0 and $\mathcal{T}_0$. Comparing definitions, one finds: $\mathcal{T}_0 = t_0/[1 + \alpha(T)t_0]$, with

$$\alpha(T) = \int_{\mathbf{q}} \left[\frac{1}{\beta} \sum_n G_{11}(\mathbf{q}, i\omega_n) G_{11}(-\mathbf{q}, -i\omega_n) - \frac{1}{2\epsilon_{\mathbf{q}}} \right] .$$

In the $\mathcal{T}$-matrix approximation, $\Sigma_{11} = 2n\mathcal{T}_0$, $\Sigma_{12} = n_0\mathcal{T}_0$. The Beliaev-Dyson equations, with the help of the H-P theorem, allow one to write $G_{11}(\mathbf{k}, i\omega_n)$ as

$$G_{11} = -\frac{i\omega_n + \epsilon_{\mathbf{k}} + n_0\mathcal{T}_0}{\omega_n^2 + \epsilon_{\mathbf{k}}^2 + 2n_0\mathcal{T}_0} ,$$

below T_c, and as $G_{11} = -1/(i\omega_n - \epsilon_{\mathbf{k}} + n_0\mathcal{T}_0)$, above T_c. Using these to evaluate α, one can solve for $\mathcal{T}_0$ both above and below T_c. Close to T_c, one obtains

$$\mathcal{T}_0 \approx \begin{cases} \dfrac{\gamma t_0}{(n^{1/3}a)} \left[1 - (T/T_c)^{3/2}\right] & \text{if } T < T_c, \\ \dfrac{\gamma t_0}{(n^{1/3}a)} \left[(T/T_c)^{3/2} - 1\right] & \text{if } T > T_c, \end{cases}$$

Here $\gamma = [\zeta(3/2)]^{4/3}/4\pi \approx 0.286$. This treatment captures the essential behavior of $\mathcal{T}_0$ near T_c: decrease to zero from both sides, with sharper decrease for smaller $n^{1/3}a$.

The $\mathcal{T}$ matrix vanishes *completely* at T_c only for zero momenta and frequency. For nonzero momenta and frequencies, $\mathcal{T}$ retains nonzero values at all temperatures. Some consequences were pointed out in Ref. [14].

1.3.4 Beyond mean field

Above T_c, perturbation theory is more straightforward (no anomalous terms). An example of second-order perturbation theory is our quasiparticle calculation of chapter 5.

In the Bose-condensed phase, perturbative calculations beyond mean field becomes quite cumbersome because of the anomalous terms. Second-order calculations were performed early in the theory of WIBG, starting with Beliaev [19,20]; details of the bookkeeping complications can be found in [15]. In recent years, a functional formulation of the WIBG has also been popular [69,102,103] ; such formulations have sometimes allowed incorporation of some higher-order diagrams which are not easy to include otherwise.

Fortunately, in this thesis, we will not deal with perturbative calculations in the ordered phase. Instead, we are interested in variational formulations of the Bose-condensed phase, which is a different route to beyond-mean-field physics.

In the early days (1959-64), Valatin and Butler [104, 105] and Girardeau and Arnowitt [27] both suggested wavefunctions created by operators similar to

$$\prod_{\mathbf{k}} \exp\left[\gamma_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}} \hat{c}^\dagger_{-\mathbf{k}} - \gamma^*_{\mathbf{k}} \hat{c}_{\mathbf{k}} \hat{c}_{-\mathbf{k}}\right] .$$

In the modern language of quantum optics, these are "squeezed" states. We will discuss these variational approaches in more detail in chapter 2, after introducing the quantum-optics terminology of quantum states. More recently, several studies have been performed ([76, 78], chapter 3 of this thesis), in the same tradition of using variational wavefunctions to account for the correlation between opposite-momenta pair excitations in the Bose-condensed WIBG.

Another variational approach to the Bose gas is a real-space one: based on the use of a wavefunction of the Jastrow (or Bijl-Dingle-Jastrow) type

$$\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N) \;\propto\; \prod_{i<j} f(r_{ij}) \;\propto\; \exp\left(\sum_{i<j} u(r_{ij})\right) .$$

Such an approach has been useful for the WIBG [51, 106–113] as well as for numerical calculations on other quantum fluids ([51,114]; also references in [107]). The method is also known as the method of correlated basis functions (CBF).

The relationship between the CBF (Jastrow) and perturbation-theoretic methods for the Bose-condensed state tends to be a difficult question. It is not always clear what classes of diagrams are accounted for in these variational procedures [107–109].

In dealing with our variational description of the Bose gas (chapter 3), we are faced with a similar issue: we find that the squeezing of the zero-momentum mode comes from some contribution beyond the Hartree-Fock-Bogoliubov level (Sec. 3.2.3). In this thesis we do not attempt to identify which diagrams are responsible for $\mathbf{k} = 0$ squeezing; this remains a future challenge.

In a recent variational calculation [14], M. Bijlsma and H. Stoof set up a functional formulation of the theory and then minimized the thermodynamic potential, treating the normal and anomalous self-energies as variational parameters. They find that this variational procedure captures Σ_{11} up to HFB (one-loop) order, while Σ_{12} goes beyond mean field, and is obtained in the $\mathcal{T}$-matrix approximation.

In our variational procedure in chapter 3, the beyond-HFB $\mathbf{k} = 0$ squeezing appears in the anomalous self-energy; thus we also get a Σ_{11} correct to HFB order, and a Σ_{12} including physics from beyond mean field.

1.4 Bose-Einstein Condensation Temperature

The Bose-Einstein Condensation Temperature $T_c^{(0)}$ of a non-interacting gas is obtained from the ideal-gas relation

$$N = N_0 + \frac{V}{\lambda^3}\, g_{3/2}(e^{\beta\mu}) \,,$$

by noting that, for a fixed density, the $\mathbf{k} = 0$ state will have to be macroscopically occupied below the temperature $T_c^{(0)} = \left(2\pi/m[\zeta(\frac{3}{2})]^{2/3}\right) n^{2/3}$. Here $g_\nu(x)$ stands for the Bose-Einstein integral functions [115].

Once a repulsive interaction is introduced between the bosons, one is faced with the question of how much the transition temperature shifts due to the repulsive interaction. Simple as this question is, it has resisted a conclusive answer for over four decades. An early statement of the "T_c problem" appears in Abrikosov, Gorkov and Dzyaloshinski's classic many-body text [13] of 1963: "*There is no general rule which allows us to determine the direction in which the transition temperature is shifted when the interaction is turned on.*" (Sec. 27, chapter 5.)

To many observers, the problem appeared to be settled in 1982 when Toyoda [116] calculated that the effect of a small repulsion is to *decrease* T_c, with a shift $\Delta T_c \propto -\sqrt{a}$, where a is the scattering length. A negative shift fit nicely with the observation that the helium λ-point temperature is lower than the BEC temperature for an ideal gas of the same density. Both the direction and the a-dependence of the shift are now known to be wrong, but it was not until the new alkali-gas era of BEC research, after 1995, that this fundamental question was re-examined seriously.

Overall, calculations on the T_c problem have produced widely dissimilar results: increases and decreases of T_c proportional to a, $\sqrt{a}$, $a \ln a$, etc. have been reported by various authors. It is only during the past several years that the community has been converging toward a consensus.

The second half of this thesis concerns the T_c problem. In chapter 4, we give a review of the diverse approaches, draw connections between them, and examine some misconceptions and fine points. In chapter 5, we analyze T_c by developing a quasi-particle description of the WIBG just above the critical point.

1.5 Experiments in Atomic Clouds

1.5.1 Realizations

The fact that interactions in liquid helium drastically reduce the occupancy of the lowest one-particle state motivated the search for a WIBG system with a higher condensate fraction. Early on [117], the possibility of using spin-polarized hydrogen for this purpose was recognized, and Stwalley and Nosanow's 1976 work [118] generated significant interest in this idea. Spin-polarized hydrogen was first stabilized by Silvera and Walraven [119], who coated the surface of the container with superfluid helium.

Surface interaction effects continued to be a deterrent for the stability of high densities, and this led a group at MIT, led by Greytak and Kleppner, to develop purely magnetic traps for atoms. Among other techniques, this group also developed evaporative cooling of magnetically trapped gases. The trapping and evaporative cooling methods developed by this group are applicable both to hydrogen and alkali gases. However, alkali gases have some added advantages, such as being amenable to highly efficient laser-cooling procedures. Efforts, therefore, were focused on alkali gases.

After two decades of development of experimental techniques, the first gases to be cooled to degeneracy temperatures and display Bose-Einstein condensation were alkali gases (rubidium [62], sodium [63] and lithium [64,120]). Condensation of spin-polarized hydrogen came later, in 1998 [95].

Other atomic species, in various hyperfine states, are also under active investigation for Bose condensation. In 2000, a different isotope of rubidium was Bose-condensed [121], and in 2001, two groups reported the condensation of metastable ^{4}He atoms in an excited state [122, 123] (the lowest spin-triplet hyperfine state). According to the Cornell & Wieman Nobel lecture [124], it is expected that almost any magnetically trappable species can be Bose-condensed. One thus expects a number of new quantum-degenerate WIBGs in the coming years.

1.5.2 Selective look at experiments

Although the research discussed in this thesis does not pertain to any particular trapped-gas experiment, the influence of these experimental developments on WIBG research is so great that it seems appropriate to present a brief survey.

Description of experiments exploring coherence and squeezing aspects of the WIBG (e.g., interference and four-wave mixing experiments) will be deferred to the next chapter (Sec. 2.4), until after we have discussed coherence and squeezing formally.

Excitations. Soon after the first creation of BECs, excitations involving shape oscillations of various kinds were excited and measured. Since then, a wide variety of excitation studies have been performed, such as the measurement of the quasiparticle spectrum, observation of Beliaev and Landau damping, creation of solitonic excitations, etc.

Rotating BEC: vortices. The first rotating condensed WIBGs, with a single quantized vortex precessing around the center of the trap, were created around the end of 1999. Since then, vortex lattices have been created, and excitations of the vortex lattices are also being studied.

Quantum Hall physics. It is expected that, at angular velocities close to the trapping frequency, there should be a regime where the rotating condensate is in a composite-boson "quantum hall" state. At the time of the writing of this thesis, experiments in JILA (Boulder, Colorado) seem to be very close to achieving such a state.

Tunable interactions and Feshbach resonances. A magnetic field can change the interaction between the atoms in a trapped gas, and thus the scattering length of a condensate can be tuned. In particular, crossing through certain values of the magnetic field even changes the *sign* of the interaction. This discovery has made possible an entirely new class of studies, such as the dynamics of a many-body condensate after the interaction strength is changed rapidly.

Lower Dimensions. By making the traps extremely flat or extremely elongated, alkali-gas experimenters have managed to produce 2D and 1D quantum-degenerate systems, with quasi-condensates.

Superfluid-"insulator" transition. By trapping cold atoms in an optical lattice, and varying the strength of the lattice, it is now possible to observe a quantum phase transition between a superfluid and a localized phase. (The localization transition is similar but not identical to the freezing of superfluid helium.) Various dynamic phenomena, e.g., involving a fast transition from one phase deep into the other, have also been demonstrated.

Light scattering, superradiance. Light scattering for the atomic-gas condensates has the status that neutron scattering did for superfluid helium. Light scattering has been used to probe the structure factor, the phonon spectrum and other excitations. In addition, a type of highly directional scattering of light ("super-radiant" scattering), involving Bose-stimulated transfer of a large fraction of the atoms to a particular mode, has been observed: this has no analog in liquid helium physics.

1.6 Outline of Thesis

In this introductory chapter, we have reviewed aspects of the physics of the weakly repulsive Bose gas. The coverage is of course highly selective: the attempt was to try to concentrate on those aspects which have some relevance to the work described in the following chapters.

Chapters 2 and 3 will concern a description of the WIBG in terms of the quantum-optics concepts of coherence and squeezing. Chapter 2 introduces these concepts and surveys their use in the study of condensed-matter systems, in particular the interacting Bose gas. In chapter 3, our variational calculations are presented.

Chapters 4 and 5 will discuss the problem of determining the transition temperature (T_c) of the WIBG.

Chapter 4 is a review and critique of the many attempts to solve the problem. We also provide a re-derivation, in modern language, of a T_c calculation first performed by Kanno in 1969 [125]. Most of the material in this chapter is adapted from the preprint [1].

Chapter 5 presents our T_c calculations, a quasiparticle formulation which reduces the problem of the shift in T_c to the problem of determining the quasiparticle spectrum at the transition point. Once the problem is formulated this way, two ways of determining the spectrum (and hence T_c) are pointed out. The results of these two approaches are given in chapter 5 while details are deferred to appendix A.

Chapter 2

Coherence and Squeezing

Part of the fascination with Bose-Einstein condensates comes from the possibility of observing coherent phenomena at a macroscopic level. In this chapter we describe some aspects of coherence. A related concept, squeezing of minimum-uncertainty states, is also introduced here. This idea is the basis of the calculations described in chapter 3, where we explore squeezing in the context of the WIBG.

In Sec. 2.1, we introduce coherent and squeezed states as they appear in quantum optics. We also explain briefly (2.1.4) how coherence and squeezing are relevant to the study of the WIBG.

In Sec. 2.2 we discuss the use of coherent states in condensed matter. In particular we focus on the idea of describing Bose condensates in terms of coherent states (2.2.1). This is an enduring idea from the beginning days of many-body theory. We also mention some attempts to *avoid* a coherent-state description of the condensed WIBG (2.2.2).

In Sec. 2.3 we review the application of squeezed states in condensed matter systems, and then focus on squeezing in a single Bose condensate (2.3.1). Some of these calculations are relatively new (last decade or so). The present thesis continues work in this direction (chapter 3).

In Sec. 2.4 we review some of the current (alkali-gas) experiments exploring coherence and squeezing aspects of Bose gas physics. These expereiments are:

interference, four-wave mixing, Josephson-type tunnelling and coherence experiments.

2.1 Quantum States of Bosonic Systems (Light)

This section is a selective review of a topic in quantum optics [126–131]. The focus in quantum optics is to produce and study novel photonic states. On the other hand, our interest lies in the possibility of describing condensed-matter systems (specifically, the WIBG) using these states.

We will introduce the concepts of coherent and squeezed states mainly for the case of a single mode, characterized by creation (annihilation) operator $\hat{c}$ ($\hat{c}^\dagger$). When we discuss mixed-mode squeezed states in 2.1.3, we will use subscripts m, n to denote the different modes.

In order to define coherent and squeezed states, it is convenient to first define number (Fock) states

$$|n\rangle = \frac{1}{\sqrt{n!}} \left(\hat{c}^\dagger\right)^n |\text{vac}\rangle \, .$$

(Fock states are themselves quite interesting, but we will talk about them only briefly, in Sec. 2.1.5.)

We will be considering fluctuations (variances) of operators. For single-mode discussions, convenient operators are the hermitian linear combinations $\hat{X} = \frac{1}{2}(\hat{c}+\hat{c}^\dagger)$ and $\hat{Y} = \frac{1}{2i}(\hat{c}-\hat{c}^\dagger)$, commonly known as "quadrature operators." Since $\left[\hat{X}, \hat{Y}\right] = i/2$, the operators obey the uncertainty relation $\langle\delta^2\hat{X}\rangle^{1/2}\langle\delta^2\hat{Y}\rangle^{1/2} \geq 1/4$.

2.1.1 Coherent states

Coherent states, first introduced by Glauber [132, 133] in 1963, are the quantum states whose properties most closely resemble those of a classical electromagnetic wave. They are well-approximated in the laboratory because a single-mode laser generates a coherent state excitation when operated well above threshold. Other than their utility in conceptualizing quantum optics experiments, they are also used to define path integrals [134].

Most modern quantum optics texts [126–130] discuss coherent photonic states and their utility in laser studies. A more mathematical presentation of coherent states and their applications is Zhang *et al*'s 1990 review [135].

Coherent states are normalized eigenstates of the creation operator:

$$\hat{c}\,|\mathrm{coh}\rangle = \alpha\,|\mathrm{coh}\rangle\,, \quad \text{or} \quad \hat{c}\,|\alpha\rangle = \alpha\,|\alpha\rangle$$

Here $\alpha = |\alpha|\,e^{i\theta}$ is the complex coherence parameter. Using this definition, one can express coherent states in a Fock basis:

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,.$$

This can also be written in terms of operators acting on the vacuum:

$$|\alpha\rangle = \exp\left[\alpha\hat{c}^\dagger - \tfrac{1}{2}|\alpha|^2\right]|\mathrm{vac}\rangle = \exp\left[\alpha\hat{c}^\dagger - \alpha^*\hat{c}\right]|\mathrm{vac}\rangle\,.$$

Any one of the above equations can be equivalently used as the definition of a coherent state, and the other equations can then be derived from the definition.

Coherent states are minimum-uncertainty states, as are the vacuum and the squeezed states to be discussed later. Different coherent states are not orthogonal, $\langle\alpha|\beta\rangle = e^{\alpha^*\beta - \frac{1}{2}|\alpha|^2 - \frac{1}{2}|\beta|^2}$, but have small overlap for large $|\alpha - \beta|$.

The coherent-state displacement operator $\hat{D}(\alpha)$ is defined as

$$\hat{D}(\alpha) = \exp\left[\alpha\hat{c}^\dagger - \alpha^*\hat{c}\right]\,.$$

The displacement operator converts the vacuum state into a coherent state:

$$|\mathrm{coh}\rangle = \hat{D}\,|\mathrm{vac}\rangle\,.$$

The reason for the name "displacement operator" is that $\hat{D}$ represents a unitary operation that displaces the mode operators $\hat{c}$, $\hat{c}^\dagger$:

$$\hat{D}^\dagger(\alpha)\hat{c}\hat{D}(\alpha) = \hat{c} + \alpha\,, \qquad \hat{D}^\dagger(\alpha)\hat{c}\hat{D}(\alpha) = \hat{c} + \alpha^*\,.$$

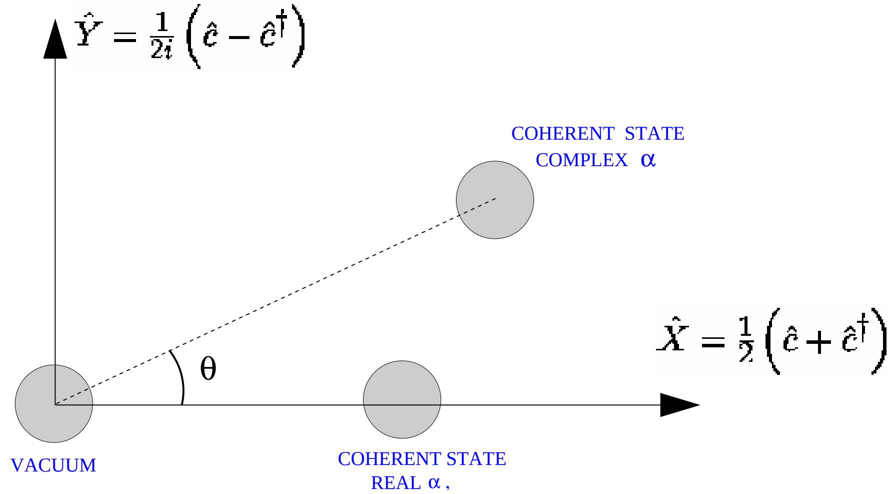


Figure 2.1: The coherent state as a "displaced" vacuum in quadrature space: variance contours of the vacuum and two coherent states.

A visual representation of the effect of the displacement operator is obtained by comparing variance contours of the vacuum and the coherent state in the plane of the quadratures $\hat{X}$, $\hat{Y}$ (Fig. 2.1). In the vacuum, these operators have zero expectation value and equal fluctuations $\langle \delta^2 \hat{X} \rangle = \langle \delta^2 \hat{Y} \rangle = 1/4$. The representation for the vacuum is therefore a circle of minimal radius $1/2$ centered at the origin. The coherent state is also a minimum-uncertainty state with equal variances $\langle \delta^2 \hat{X} \rangle = \langle \delta^2 \hat{Y} \rangle = 1/4$. However, the expectation values are now

$$\langle \hat{X} \rangle = \mathcal{R}e[\alpha] = |\alpha| \cos\theta \, , \qquad \langle \hat{Y} \rangle = \mathcal{I}m[\alpha] = |\alpha| \sin\theta \, .$$

The representation for the coherent state is therefore a minimal-radius circle, centered at coordinates $(|\alpha| \cos\theta, \ |\alpha| \sin\theta)$. The displacement operator $\hat{D}(\alpha)$ thus displaces the variance contour by α in the $\hat{X}$-$\hat{Y}$ plane, without modifying its circular shape.

2.1.2 Squeezed coherent states

The single-mode squeeze operator is defined as

$$\hat{S} = \exp\left[\gamma \hat{c}^\dagger \hat{c}^\dagger - \gamma^* \hat{c}\hat{c}\right], \qquad \gamma = s\, e^{i\phi}.$$

This operator, when applied on a coherent state, produces a squeezed coherent state:

$$|\text{sq\_coh}\rangle = \hat{S}\,|\text{coh}\rangle = \hat{S}\hat{D}\,|\text{vac}\rangle\,.$$

In the quantum optics literature, it has become more usual to use the inverted order of operators, $\hat{D}\hat{S}\,|\text{vac}\rangle$, but we prefer the $\hat{S}\hat{D}$ form. The fluctuations of $\hat{X}$, $\hat{Y}$ are the same in this alternate form, but the expectation values differ by the factor $F = \cosh(2s) + \sinh(2s)$.

The $\hat{D}\hat{S}$ state has expectation values $\langle\hat{X}\rangle_{DS} = |\alpha|\cos\theta$, $\langle\hat{Y}\rangle_{DS} = |\alpha|\sin\theta$, i.e., the fluctuation profile in the $\hat{X}$-$\hat{Y}$ plane is centered at $(|\alpha|\cos\theta,\, |\alpha|\sin\theta)$ just like the coherent state. For the $\hat{S}\hat{D}$ state, the center is at $(|\alpha|F\cos\theta,\, |\alpha|F\sin\theta)$.

As mentioned already, squeezed coherent states are minimum-variance states. However, the uncertainties are unequal in different directions. For both $\hat{D}\hat{S}$ and $\hat{S}\hat{D}$ states, the fluctuations are

$$\begin{aligned}
\langle\delta^2\hat{X}\rangle &= \frac{1}{4}\left[e^{2s}\cos^2(\phi/2) \;+\; e^{-2s}\sin^2(\phi/2)\right], \\
\langle\delta^2\hat{Y}\rangle &= \frac{1}{4}\left[e^{2s}\sin^2(\phi/2) \;+\; e^{-2s}\cos^2(\phi/2)\right].
\end{aligned}$$

(For $\phi = 0$ the fluctuations are $e^{\pm 2s}$.) The uncertainty contour is thus elliptical rather than circular, centered at a displaced position (α or αF). For positive s, the major axis of the ellipse makes an angle $\phi/2$ with the horizontal.

The effect of the $\hat{S}$ operator is thus to flatten (squeeze) the variance contour. This explains the names ‘squeeze operator’ and ‘squeezed state’.

It is common to lock the phase of the squeeze parameter, $\gamma = s\, e^{i\phi}$, to twice the phase of the coherence parameter, $\alpha = |\alpha|\, e^{i\theta}$, i.e., to set $\phi = 2\theta$. In that

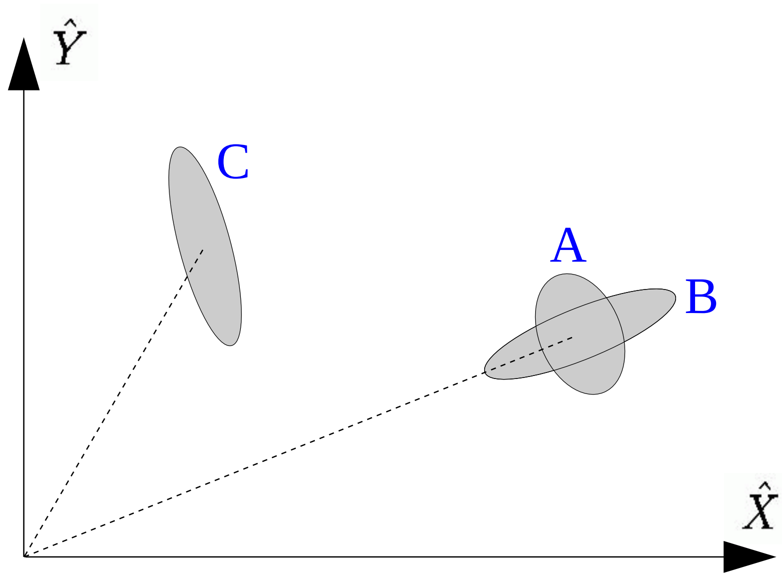


Figure 2.2: Squeezed coherent states (variance contours). States A and B have their squeeze phases locked to twice the coherence phase ($\phi = 2\theta$), and are therefore respectively amplitude-squeezed ($s < 0$) and phase-squeezed ($s > 0$) states. State C has $\phi \neq 2\theta$.

case, the elongation of the variance profile is in the radial direction for $s > 0$ (phase-squeezed state), and in the polar direction for $s < 0$ (amplitude-squeezed state).

The literature of squeezed coherent states can be traced back to Stoler [136, 137] and to Yuen [138]. Stoler discussed minimum-uncertainty wave packets whose variance profiles are not necessarily circular (in modern language: squeezed states). Yuen in 1976 recognized that these states are produced by $\hat{c}^\dagger\hat{c}^\dagger$-type operators, and so called them "two-photon coherent states." He used the $\hat{S}\hat{D}$ form, and derived many of the properties of these states. Discussion of the $\hat{D}\hat{S}$ form was first given by Caves [139]. Loudon and Knight's review [131] summarizes much of the literature of squeezed optical states, and more recent accounts are found in several texts [126–130].

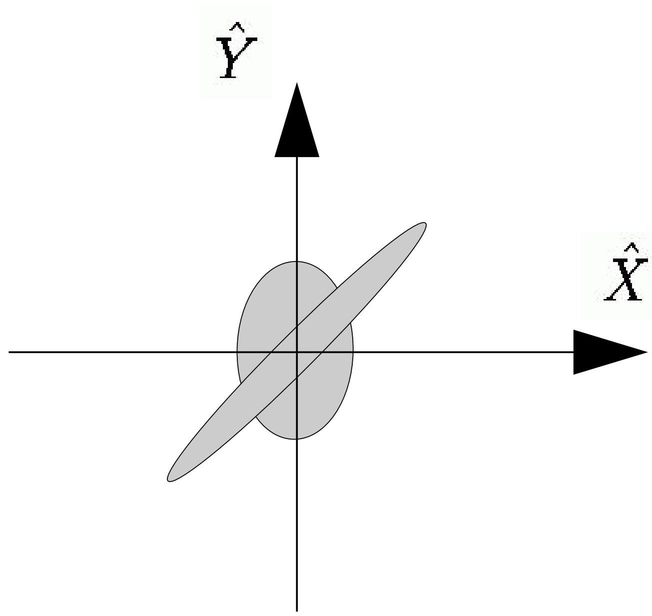


Figure 2.3: Squeezed vacua. The moderately squeezed state has $\phi = 0$, $s < 0$. The highly squeezed state corresponds to large and positive s, and $\phi = \pi/2$, so that the tilt is $\pi/4$.

2.1.3 Squeezed vacuum

A squeezed vacuum is produced by applying the squeeze operator $\hat{S}$ on the vacuum. We first consider the single-mode squeezed vacuum:

$$|\mathrm{sq\_vac(single)}\rangle = \exp\left[\gamma \hat{c}^\dagger \hat{c}^\dagger - \gamma^* \hat{c}\hat{c}\right] |\mathrm{vac}\rangle$$

In this state, the quadrature operators have zero expectation values, while the variances are the same as in the squeezed coherent state: $\langle \delta^2 \hat{X} \rangle = \frac{1}{4}\left[e^{2s}\cos^2\frac{\phi}{2} + e^{-2s}\sin^2\frac{\phi}{2}\right]$ and $\langle \delta^2 \hat{Y} \rangle = \frac{1}{4}\left[e^{2s}\sin^2\frac{\phi}{2} + e^{-2s}\cos^2\frac{\phi}{2}\right]$. The variance profiles are thus elliptical and centered at the origin, with the ellipse major axis inclined at angle $\phi/2$ to the horizontal.

It is also useful to consider mixed-mode squeezed vacua,

$$|\mathrm{sq\_vac(mixed)}\rangle = \exp\left[\gamma \hat{c}_m^\dagger \hat{c}_n^\dagger - \gamma^* \hat{c}_n \hat{c}_m\right] |\mathrm{vac}\rangle\,,$$

which will be the fate of $\mathbf{k} \neq 0$ states in our variational treatment of the WIBG, in chapter 3.

For this kind of state, the variances of quadrature variables $\hat{X}_m = \frac{1}{2}(\hat{c}_m + \hat{c}_m^\dagger)$ and $\hat{Y}_m = \frac{1}{2i}(\hat{c}_m - \hat{c}_m^\dagger)$ are equal, as are those of $\hat{X}_n$, $\hat{Y}_n$. Flattening of variance contours is seen in the *mixed* quadrature variables $\hat{X} = \frac{1}{\sqrt{2}}(\hat{X}_m + \hat{X}_n)$, $\hat{Y} = \frac{1}{\sqrt{2}}(\hat{Y}_m + \hat{Y}_n)$.

2.1.4 Coherence and squeezing in the WIBG

At this juncture, we briefly state why the ideas of coherence and squeezing are relevant to the study of the Bose-condensed WIBG.

The essence of Bose-Einstein condensation is that the field operators choose a particular phase. This violates number conservation, but allows $\langle\psi(\mathbf{r})\rangle$ or $\langle\hat{c}_0\rangle$ to attain a macroscopic value.

A coherent state embodies exactly the same idea: it has a definite phase (the θ of Sec. 2.1.1), but is a linear combination of many Fock (number) states. A coherent state thus seems to be a natural description of the $\mathbf{k} = 0$ mode of a uniform BEC, or for the lowest single-particle mode in the non-uniform case.

For an *interacting* Bose gas, one effect of interactions is to deplete the condensate. In the uniform case, this predominantly leads to the population of $\pm\mathbf{k}$ pair modes. (The Bogoliubov approximation takes into account only this type of depletion.) An operator of the form

$$\exp\left[\gamma_{\mathbf{k}}\hat{c}_{\mathbf{k}}^\dagger\hat{c}_{-\mathbf{k}}^\dagger - \gamma_{\mathbf{k}}^*\hat{c}_{\mathbf{k}}\hat{c}_{-\mathbf{k}}\right]$$

would populate modes in opposite-momenta pairs. Of course this is just a mixed-mode squeeze operator.

Thus, a coherent state at the lowest one-particle mode (condensate mode), and mixed-mode squeezing in the other modes, is a natural way to describe the WIBG.

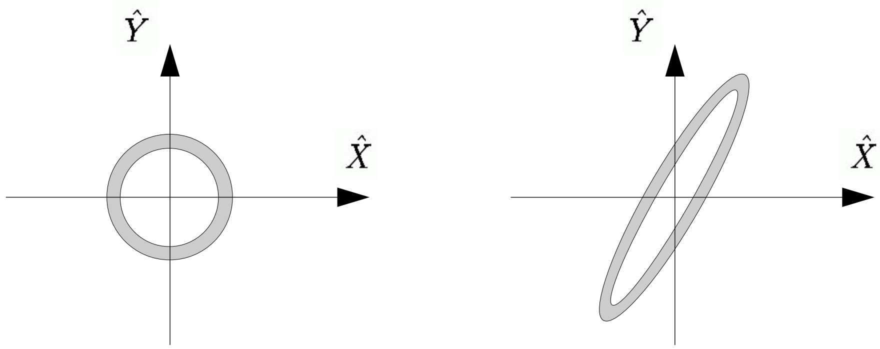


Figure 2.4: A Fock state is represented by a circular annulus, and a squeezed Fock state by an elliptical annulus.

A coherent state formulation describes only the condensate dynamics. In real space, such a description leads to the Gross-Pitaevskii equation [140–142]. One would expect that incorporation of squeezing would allow one to include excitations around the Gross-Pitaevskii ground state, e.g., the Bogoliubov or HFB equations for excitation fields found by linearizing around the Gross-Pitaevskii ground state [32, 143]. The program of deriving the non-uniform Bogoliubov or HFB equations, starting from a squeezed coherent description, has not been carried out, except for the calculation in [79] using "generalized coherent states." These investigations will be reviewed in Secs. 2.2 and 2.3.

2.1.5 Other Quantum States

Of course squeezed and coherent states are not the only types of bosonic states used in quantum optics. We have already met Fock states $|N\rangle = (\hat{c}^\dagger)^N |0\rangle$. The appropriate phase space representation of such a state is an annulus, the inner and outer circles having radius $N \pm \frac{1}{2}$. Unfortunately, this representation cannot be obtained by calculating variances of $\hat{X}$ and $\hat{Y}$; more sophisticated tools (Wigner function or Q function) are required [126–130].

One can also apply a squeeze operator to a number state: the resulting squeezed Fock state is represented by an elliptical ring [144].

Thermal states are linear combinations of number states with Boltzmann or Bose-Einstein weights, while squeezed thermal states are properly weighted sums of squeezed Fock states [144].

2.2 Coherent States in Condensed Matter

Although coherent states were initially used to describe coherence properties of optical systems [132, 133], they have also been applied to describe condensed-matter phenomena, ever since they were introduced. We survey some examples, focusing (Sec. 2.2.1) on their use in describing the WIBG.

Other than their use to describe lasing and Bose condensation, coherent states have also been used extensively in the analysis of superradiance from many-atom systems (e.g., [145–147]). As a combined example, in the work of Littlewood's group on cavity polariton systems [148, 149], a coherent state $e^{\lambda \hat{c}^\dagger} |\mathrm{vac}\rangle$ is used for a variational treatment of a generalized Dicke model, and coherent states are found to describe superradiance, lasing and polariton BEC in different regimes of this model.

Coherent states have been used to study Jahn-Teller phenomena [150], Landau diamagnetism (electron in magnetic field) [151], anharmonic crystals [152] and N-spin systems [145]. They form zeroth-order solutions to the spin-boson model [153], described in more detail in Sec. 2.3.

In an early study of the 2D Bose gas, Lasher used a coherent state of phonons to investigate the presence of superfluidity despite the absence of long-range order [42].

Coherent states are fundamental to the development of the path integral formulation of many-particle physics [134]. A variant, spin coherent states [154], is widely used in the study of quantum magnetism.

2.2.1 Condensate as a coherent state

The idea of using a coherent state to describe a Bose-Einstein condensate is a natural and attractive one. Gross actually attempted a coherent-state description of a condensed WIBG [155] even before Glauber [132, 133] formally introduced the idea.

Cummings and Johnston in 1966 [156,157] developed a theory of superfluidity for liquid helium using the postulate that a coherent state of the quasiparticles of normal helium is appropriate in describing the superfluid ground state. When expressed in terms of the bare bosons $\hat{c}$, $\hat{c}^\dagger$ (helium particle operators), the wavefunction takes the form

$$\prod_{\mathbf{p}} \exp\left(\alpha_{\mathbf{p}} \hat{c}^\dagger_{\mathbf{p}} - \alpha^*_{\mathbf{p}} \hat{c}_{\mathbf{p}}\right) \prod_{\mathbf{k}} \exp\left(\gamma_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}} \hat{c}^\dagger_{-\mathbf{k}} - \gamma^*_{\mathbf{k}} \hat{c}_{\mathbf{k}} \hat{c}_{-\mathbf{k}}\right) |\mathrm{vac}\rangle \,.$$

Gross's work [155] had already resulted in the same form. In modern language, one would call this a squeezed coherent state of $\hat{D}\hat{S}$ form for *each* momentum mode.

Langer [140,141] used a coherent-state description to derive a Landau-Ginzburg functional and the Gross-Pitaevskii equation for the non-uniform WIBG. His formulation involves a coherent occupation of each momentum mode, $\prod_{\mathbf{k}} \hat{D}(\alpha_{\mathbf{k}}) |\mathrm{vac}\rangle$. The Landau-Ginzburg order parameter (or the classical wavefunction of the Gross-Pitaevskii equation) $\psi(\mathbf{r})$ appears as the Fourier transform of the coherence parameter function $\alpha_{\mathbf{k}}$.

Glassgold and Sauerman [158] also attempted a description of a WIBG using a coherent state. They found that the coherent state needs to be corrected with an operator involving $\exp[\lambda(\hat{c}^\dagger_0)^2]$; i.e., zero-momentum squeezing.

The basis of the idea of using a coherent state to describe a BEC is that a condensate can be ascribed a specific phase, at the expense of number conservation. The concept of a macroscopic system with a definite phase, and the consequences of a macroscopic phase, have long intrigued many-body physicists.

Important discussions have been given by Anderson in 1966 [159] and 1986 [160], by Feynman in his Lectures [161], and by Leggett & Sols in 1991 [162].

A more relevant analysis was presented in 1996 by Barnett, Burnett and Vaccarro, who used the criterion of "robustness" to determine the quantum state of the condensate [142]. It was found that the coherent state is the description of the condensate which makes it most robust against interactions with the environment. Barnett *et al.* proposed this to be the argument for regarding a BEC as having a well-defined phase.

2.2.2 Avoiding a coherent state: fixed-N descriptions

Sacrificing the comfort of a well-defined particle number provoked resistance from the atomic-physics and quantum-optics community, since the magnetically trapped atomic-gas condensates are well-insulated from their environments and thus "obviously" have fixed numbers of atoms. There have been several recent attempts to formulate "fixed-N" descriptions of BECs [71–75]. In addition, Girardeau has pointed out [163] that the 1959 Girardeau & Arnowitt paper [27] is also number-conserving.

The idea is to use a Fock state instead of a coherent state, and re-formulate the Gross-Pitaevskii equations and the Bogoliubov-deGennes equations in a $U(1)$-symmetric form. These schemes tend to be awkward compared to conventional many-body techniques based on spontaneous symmetry breaking.

Some of the number-conserving formulations introduce phonon operators somewhat different from the Bogoliubov phonons. In Sec. 3.3, we utilize the phonon operators appearing in Ruckenstein's number-conserving formulation [71], to find a physical interpretation of the mixed-mode $\pm\mathbf{k}$ pair squeezing in a uniform WIBG.

2.3 Squeezed States in Condensed Matter

After squeezed states were popularized in quantum optics in the 1980s, the concept was utilized in the analysis of several condensed matter systems, mainly in the 1990s. We give brief examples of squeezed states in the analysis of other condensed-matter systems, before moving to a more detailed review (in 2.3.1) of squeezing in a Bose condensate.

Squeezed phonon states have been used as a variational framework for describing correlated electron-phonon problems. A generic simple model is a dissipative two-state system,

$$\hat{H} = -\Delta\sigma_x + \omega\hat{b}^\dagger\hat{b} + g_i(\hat{b}^\dagger + \hat{b})\sigma_z \,,$$

where $\hat{b}$ is the phonon mode, and the Pauli matrix describes the two-state system. This "spin-boson" model, and its extensions, are useful for describing dissipative tunneling [164], defect tunneling in solids, the polaron problem, etc.

The lowest-order solution for the ground state is a phonon coherent state, created with the displacement operator $\exp\left[-\lambda\sigma_z(\hat{b}^\dagger - \hat{b})\right]$; this solution was first studied in the context of Kondo physics [153]. Since then, a squeezed coherent state of the $\hat{D}\hat{S}$ form, commonly called a "displaced squeezed" state in this literature, has been extensively discussed in connection with this model and its extensions (representative Refs. [165–168]).

Ref. [169] gives a general discussion of squeezing and coherence in phonon systems. Phonon squeezing can be induced either by phonon-phonon or by phonon-photon interactions. Creation of squeezed phonon states in a solid via phonon-photon interactions has been reported [169].

Squeezed states have also been used for polaritons (coupled exciton-photon system, Hopfield model and generalizations) [170], exciton-phonon systems [171], and bilayer quantum Hall systems [172].

Somewhat closer to our topic is the work of Sá de Melo [173], who studied a

Bose gas with *attractive* interactions. The focus is on the transition between pair condensation and single-boson condensation, a consideration motivated by exciton systems where bi-exciton condensation may compete with exciton condensation.

2.3.1 Squeezing in the condensed WIBG

A common trend emerges from the early studies of the Bose-condensed WIBG using coherent states [155, 156, 158], discussed in 2.3.1. It seems that coherent states are not by themselves enough to describe interaction effects, and need to be modified by a squeeze-type operator.

Depletion effects are described by nonzero $\pm\mathbf{k}$ pair-mode squeezing [155,156]. On the other hand, analysis of the fluctuations of the condensate mode operators $\hat{c}_0$, $\hat{c}_0^\dagger$ leads to a description of the $\mathbf{k} = 0$ mode involving an $\exp\left[\lambda(\hat{c}_0^\dagger)^2\right]$ -type operator, i.e., single-mode squeezing [158]. Thus both single-mode $\mathbf{k} = 0$ squeezing, and nonzero $\pm\mathbf{k}$ pair-mode squeezing, seem to be present in the interacting Bose condensate.

The description of the WIBG using states created by operators of the type $\exp\left[\gamma\hat{c}^\dagger\hat{c}^\dagger - \gamma^*\hat{c}\hat{c}\right]$ was attempted even before ideas of squeezing were developed. It is only in the new (trapped-gas) era of WIBG research that the language of quantum optics has been available for discussing squeezing phenomena in the interacting Bose gas. Here we review some such descriptions, from both eras. Our own efforts in this direction are described in chapter 3.

As early as 1958, Valatin and Butler suggested [104] for the interacting bosonic system a variational ground state which, in today's language, would be described as a squeezed vacuum (single-mode for $\mathbf{k} = 0$ mode and mixed-mode for nonzero $\pm\mathbf{k}$ pair modes). The work was strongly motivated by the idea that superfluids and superconductors should have similar descriptions. A squeeze operator looks superficially like a Bardeen-Cooper-Schrieffer (BCS) state and hence was attractive to these researchers, shortly after the invention of BCS theory.

Later, Valatin [105] modified their earlier wavefunction to enforce a nonzero occupation of the different modes. By a different route, Valatin thus arrived at the same conclusion as the workers who had started with a coherent state [155, 156, 158], that both coherence and squeezing is necessary to describe an interacting Bose fluid.

The 1959 paper by Girardeau and Arnowitt [27] is also a variational calculation for the WIBG. The variational wavefunction is obtained by using the operator

$$\hat{U} = \exp\left\{ \tfrac{1}{2} \sum_{\mathbf{k}\neq 0} \psi_{\mathbf{k}} \left[(\hat{\beta}_0^\dagger)^2 \ \hat{c}_{-\mathbf{k}}\hat{c}_{\mathbf{k}} \ - \ \hat{c}_{\mathbf{k}}^\dagger \hat{c}_{-\mathbf{k}}^\dagger \ (\hat{\beta}_0)^2 \right] \right\}$$

on the number (Fock) state $|N\rangle = (N!)^{-1/2} (\hat{c}_0^\dagger)^N \, |0\rangle$. The quantity $\hat{\beta}_0 = \hat{N}_0^{-1/2} \hat{c}_0$ is itself an operator, so $\hat{U}$ is slightly different from the usual squeeze operator. Other than this distinction, the structure of the theory seems to be that of a squeezed number state for the $\mathbf{k} = 0$ mode, and mixed-mode squeezed vacua for the nonzero $\pm\mathbf{k}$ pair modes.

To the best of our knowledge, the Girardeau-Arnowitt variational ansatz has not been re-analyzed in terms of modern quantum-optics concepts, and it is not known what additional squeezing physics it might contain in comparison to the squeezed-coherent formulation that is explored in chapter 3 and in Refs. [76, 77].

Dunningham *et al.* studied the quantum state of a *trapped* BEC, focusing on the condensate mode only [78]. (The equivalent in the uniform case would be to focus on the $\mathbf{k} = 0$ mode only.) They find, first of all, an amplitude-squeezed coherent state. In attempting to account for the fixed number of atoms in the trap, the variance contour is found to be distorted from the elliptical shape to a “banana” shape, i.e., the contour becomes more annulus-like (the state becomes more Fock-like).

Roger-Salazar *et al.* have also studied the quantum state of a condensate, but they focus on the evolution of the quantum state as the condensate expands after being released from a trap [80]. Their description of a trapped BEC is as a squeezed coherent state, which evolves into a thermal state after release. This calculation also focuses on the condensate mode only.

Chernyak, Choi and Mukamel have used [79] a generalized coherent state,

$$|\Psi(\tau)\rangle \;=\; \exp\left(\sum_i \alpha_i(\tau)\hat{c}_i^\dagger + \sum_{i,j}\beta_{ij}(\tau)\hat{c}_i^\dagger\hat{c}_j^\dagger\right)|\mathrm{vac}\rangle\,,$$

in an arbitrary basis indexed by i, j, to describe a Bose condensate in any trap. Such a state contains squeezing (β_{ij}), both single-mode and two-mode, as well as coherent populations α_i. Starting from the generalized coherent state, Chernyak *et. al.* derived the time-dependent HFB equations [143]. As discussed in 2.1.4, this is an extension of using a pure coherent state to derive the Gross-Pitaevskii equation.

Navez [76] and Solomon, Feng and Penna [77] have both explored a state similar to our formulation in the next chapter. The ket vector used in these calculations involve a squeezed coherent state of the $\hat{D}\hat{S}$ type for the $\mathbf{k}=0$ mode, and the usual $\pm\mathbf{k}$ pair squeezed vacua for the $\mathbf{k}\neq 0$ modes. Navez's focus is on regulating anomalous fluctuations by using linear combinations of the states with different phases. Solomon *et. al.* use the formalism to calculate second- and third-order correlation functions (also known as degrees of coherence [128,129,132]) and compare with experimental determinations of the same quantities.

Our focus in the next chapter is to investigate squeezing properties of the Bose-condensed WIBG. We seek to make a specific statement about the existence of squeezing ($\mathbf{k}=0$ mode and nonzero mixed-mode) at mean field level. In addition, we derive an alternate formulation (Sec. 3.3) that allows a more physical understanding of the variables involved in $\mathbf{k}\neq 0$ squeezing.

In contrast to the $\hat{D}\hat{S}$ state of Refs. [76, 77], we will be using a $\hat{S}\hat{D}$ form of squeezed coherent state for the $\mathbf{k}=0$ mode. It seems sensible to *first* apply the coherent state operator to produce a macroscopic occupation, which is the basic effect of Bose-Einstein condensation, and only after that apply a squeeze operator to take into account some effects of the interaction. The $\hat{D}\hat{S}$ and $\hat{S}\hat{D}$ forms seem to yield similar results, except for factors of $\cosh(2s_0) + \sinh(2s_0)$ (Sec. 2.1.2). It has not been investigated in detail whether there are any differences in the physical predictions of the two approaches.

2.4 Relevant Experiments with Trapped Atomic Gases

With the advent of the trapped-gas WIBGs which are adaptable to diverse types of manipulations and measurements, some aspects of coherence have been investigated in the laboratory. Also, schemes have been suggested for creating condensates with enhanced squeezing.

2.4.1 Coherence

The macroscopic coherence of a Bose condensate has been demonstrated dramatically in 1997 by an interference experiment at MIT [174]. Two condensates, originally separated in a double-well trap, were allowed to expand freely. In the region where they overlapped, interference fringes could be observed.

Since then, many other interference experiments have been performed. We mention, as an example, the experiment by Bloch *et al.* [175] that demonstrates spatial coherence in a single Bose condensate, by probing the relative phase between two separated regions of a single condensate. Segments taken from spatially separated regions, when brought together, were seen to form interference fringes.

Another demonstration of macroscopic coherence would have been Josephson tunneling between two weakly-coupled BECs. There has been extensive theoretical discussion of Josephson tunneling, but the only experiment, as far as we know, has been performed in the context of an optical lattice, i.e., with many condensates instead of two [176].

There has also been a measurement [177] of the third-order correlation function, $g^{(3)}(0) \propto \langle (\hat{c}^{\dagger})^3 (\hat{c})^3 \rangle$, via observation of the decay of the condensate by three-body recombination. As predicted [178], three-atom-bunching is reduced in a coherent state, and $g^{(3)}(0)$ is found to be indeed smaller in the BEC than in the un-condensed gas.

2.4.2 Squeezing

The type of squeezing we investigate in the next chapter is that intrinsically present in the WIBG (either due to Bogoliubov $\pm\mathbf{k}$ correlations or due to quantum fluctuations in the condensed mode). As of now, this type of squeezing has not been probed or measured in any methodical experiment.

In contrast, there have been several proposals for *producing* highly squeezed states. We mention two schemes here.

First, there is the idea that four-wave mixing, or parametric amplification, would create two matter waves whose relative numbers are squeezed [179–181]. In these experiments, three coherent matter waves (condensates) are allowed to overlap in space. Two of them act as source and the third (smaller) one functions as the seed. The result is the creation of a fourth wave in the conjugate momentum state, resulting in an amplified seed-conjugate wave pair. The seed-conjugate pair is squeezed in relative number.

This procedure is analogous to producing squeezed light, in quantum optics, via four-wave mixing. In the photonic case, a nonlinear ("Kerr") medium is required. In the case of colliding WIBGs, the nonlinearity is provided by the interaction term of the WIBG Hamiltonian.

The second idea is to use the analog of what is known as "parametric down-conversion" in quantum optics. Squeezed light can be produced by the process of breaking a photon into two, each of half the frequency and opposite polarization. The process is the inverse of second harmonic generation. The two half-frequency beams are entangled, and the resulting state is squeezed.

The corresponding scheme for Bose condensates is to use the dissociation of di-atomic molecules [182–186]. Starting with a molecular condensate, photo-dissociation (or magnetic field-induced dissociation) would create correlated atomic pairs in correlated (mixed-mode squeezed) states.

2.5 Perspective

In this chapter, we have provided background on some concepts developed in the context of photonic systems. We also tried to indicate the relevance of these ideas, in the study of condensed matter systems in general and for the Bose-condensed fluid in particular. The application of quantum-optics tools, to bosonic systems appearing in condensed matter, is an emerging paradigm. In particular, the studies of the "quantum state" of single BECs are just beginning to appear.

In the next chapter, we present a systematic investigation of squeezing in the condensed uniform $T = 0$ WIBG, via a variational approach. This research is complementary to two other recent studies of the quantum state of the uniform BEC (Refs. [76, 77]), discussed in Sec. 2.3.1.

Our choice of quantum-optics material mentioned in this chapter was quite restricted. We omitted, for example, a discussion of phase space functions (Wigner, Q, P functions) which have become common tools in the study of photonic states. For our purposes, variance contours will be a sufficient tool for characterizing quantum states. However, it is worth noting that a sophisticated set of concepts for describing bosonic states exist in the quantum optics literature, waiting to be exploited in the study of condensed matter systems.

Chapter 3

Squeezing in the Zero-Temperature Condensate

One motivation for studying squeezing phenomena in the interacting Bose gas is to describe aspects of WIBG physics in a new language, since interaction effects can be re-formulated in terms of squeezing behavior of a number of variables. The concepts of squeezing allows one to ask a new class of questions (involving uncertainty and squeezing in different modes) that have not been considered before in the Bose-gas context, simply because this language did not exist until recently.

Another reason to look at squeezing phenomena in WIBG physics is that, in this context, it becomes natural to investigate squeezing in variable pairs other than the usual quadrature operators used in quantum optics, such as density-phase or density-current pairs. One thus hopes to learn something new about squeezing phenomena as well as something new about Bose-gas physics.

After the survey in chapter 2 of research on squeezing in the Bose-condensed system, in this chapter we present our own studies in this direction.

The main part of the work is a varational treatment of the zero-temperature Bose gas (Sec. 3.1) using a squeezed coherent state for the zero-momentum mode and a mixed-mode squeezed state for each opposite-momenta pair of modes. The manifestation of $U(1)$ symmetry breaking within this formalism is explored (Sec. 3.2). The resulting form of the Pines-Hugenholz theorem, when compared with the Hartree-Fock-Bogoliubov form, gives information about the presence of $\mathbf{k} = 0$ mode squeezing at mean field.

While this formalism also provides information about squeezing in the $\mathbf{k}\neq 0$

modes, in Sec. 3.3 we seek a more physical understanding of $\mathbf{k}\neq 0$ squeezing by using an alternate variational state. This state is squeezed in operators for density-oscillation modes, which were introduced in Ref. [71] by A. E. Ruckenstein, in the course of a number-conserving description of the Bose-condensed state. The alternate formulation allows an interpretation of $\mathbf{k}\neq 0$ squeezing in terms of variances of density and phase modes.

3.1 Formalism

In this section we introduce and justify our variational wavefunction (3.1.1), determine the variational parameters by minimization (3.1.2), and discuss variances and squeezing properties (3.1.3 and 3.1.4).

3.1.1 Variational wave-function for WIBG

We choose a squeezed coherent wavefunction as the variational state for a weakly-interacting bose gas. For a uniform condensate, the macroscopic occupation is in the zero-momentum state, so we will have a coherent occupation of the $\mathbf{k} = 0$ mode only: $\hat{D} = \hat{D}_0 = \exp\left[\alpha \hat{c}_0^\dagger - \alpha^* \hat{c}_0\right]$, with the complex coherence parameter $\alpha = f\ e^{i\theta}$.

We will apply a mixed-mode squeeze operator for each opposite-momenta mode pair. Thus the variational ground state is $|\mathrm{sq\_gr}\rangle = \hat{S}\,|\mathrm{coh}\rangle = \hat{S}\hat{D}\,|\mathrm{vac}\rangle$, with

$$\hat{S} = \prod_{\mathbf{k}} \hat{S}_{\mathbf{k}} = \prod_{\mathbf{k}} \exp\left[\frac{1}{2}\sum_{\mathbf{k}} \left(\gamma_{\mathbf{k}} \hat{c}_{\mathbf{k}}^\dagger \hat{c}_{-\mathbf{k}}^\dagger - \gamma_{\mathbf{k}}^* \hat{c}_{\mathbf{k}} \hat{c}_{-\mathbf{k}}\right)\right] , \qquad \gamma_{\mathbf{k}} + \gamma_{-\mathbf{k}} = 2 s_{\mathbf{k}} e^{i\phi_{\mathbf{k}}} .$$

(Note that the $\mathbf{k} = 0$ mode squeezing is single-mode.) Our variational wavefunction for the uniform WIBG is thus a squeezed coherent state for the $\mathbf{k} = 0$ mode and a mixed-mode squeezed vacuum for each $\mathbf{k}\neq 0$ mode pair.

The motivation for using $\hat{c}_{\mathbf{k}}^\dagger \hat{c}_{-\mathbf{k}}^\dagger$-type squeezing is clear if we consider the

Bogoliubov Hamiltonian for the Bose gas, where the anomalous (number-non-conserving) terms indicate the importance of opposite-momenta pair excitations.

(The need to capture the physics of $\pm\mathbf{k}$ pair excitations has been appreciated from the early days of WIBG research, e.g., in the 1959 variational calculation of Girardeau and Arnowitt [27].)

For squeezed coherent states, it is common in the quantum-optics literature to lock the phases of the squeezing and coherence parameters as $\phi = 2\theta$. In addition to the $\mathbf{k} = 0$ squeeze-parameter phase, we find it necessary to lock the squeeze phases of *each* momentum-pair mode to twice the phase of the $\mathbf{k} = 0$ coherence parameter, i.e., $\phi_{\mathbf{k}} = 2\theta$ for *all* $\mathbf{k}$. We tried relaxing this condition, in which case one has the extra set of parameters $\phi_{\mathbf{k}}$ to minimize against. Minimization of variational energy $\langle \hat{K} \rangle$ imposes the condition $\theta = \frac{1}{2}\phi_{\mathbf{k}}$. Therefore, in the following description, we will simply start with $\theta = \frac{1}{2}\phi_{\mathbf{k}}$, to avoid cumbersome typing of equations involving $\phi_{\mathbf{k}}$ and θ.

3.1.2 Minimization

To determine the variational parameters, we need to minimize the expectation value of the grand-canonical Hamiltonian $\hat{K} = \hat{H} - \mu\hat{N}$.

Expectation values of operators in our variational state can be calculated using the relations $\hat{D}^{\dagger}\hat{c}_{\mathbf{k}}\hat{D} = \hat{c}_{\mathbf{k}} + \alpha\delta_{\mathbf{k},0}$ and $\hat{S}^{\dagger}\hat{c}_{\mathbf{k}}\hat{S} = \cosh(s_{\mathbf{k}})\hat{c}_{\mathbf{k}} + \sinh(s_{\mathbf{k}})e^{i\phi_{\mathbf{k}}}\hat{c}^{\dagger}_{-\mathbf{k}}$. The required quantities are $\langle \hat{N}_{\mathbf{k}} \rangle = \langle \hat{c}^{\dagger}_{\mathbf{k}}\hat{c}_{\mathbf{k}} \rangle$ and $\langle \hat{H}_{\text{int}} \rangle$, calculated in the variational ground state $|\text{sq\_gr}\rangle$:

$$\langle \hat{N}_{\mathbf{k}} \rangle = \sinh^2(s_{\mathbf{k}}) + N_{\mathrm{c}}\delta_{\mathbf{k},0}\,,$$

$$\begin{aligned}\langle \hat{H}_{\text{int}} \rangle &= \frac{U}{2V}N_{\mathrm{c}}^2 + \frac{U}{2V}N_{\mathrm{c}}\left\{\sum_{\mathbf{q}}\cosh(s_{\mathbf{q}})\sinh(s_{\mathbf{q}}) + 2\sum_{\mathbf{q}}\sinh^2(s_{\mathbf{q}})\right\} \\ &\quad + \frac{U}{2V}\left\{\left[\sum_{\mathbf{q}}\cosh(s_{\mathbf{q}})\sinh(s_{\mathbf{q}})\right]^2 + 2\left[\sum_{\mathbf{q}}\sinh^2(s_{\mathbf{q}})\right]^2\right\}.\end{aligned}$$

Here we have defined

$$N_{\rm c} \;=\; f^2\left[\cosh(2s_0)+\sinh(2s_0)\right] \;=\; f^2\left[\cosh(s_0)+\sinh(s_0)\right]^2 ,$$

The $\mathbf{k}=0$ occupancy and the total number of particles are respectively

$$N_0 = N_{\rm c} + \sinh^2(s_0) \qquad \text{and} \qquad N = N_{\rm c} + \sum_{\mathbf{k}} \sinh^2(s_{\mathbf{k}})\,.$$

We now minimize $\langle \hat{K}\rangle = \sum_{\mathbf{k}}(\epsilon_{\mathbf{k}}-\mu)\langle \hat{N}_{\mathbf{k}}\rangle + \langle \hat{H}_{\rm int}\rangle$ with respect to $N_{\rm c}$ and $s_{\mathbf{k}}$. (This is equivalent to using α and $s_{\mathbf{k}}$.) Setting $\delta\hat{K}/\delta N_{\rm c}$ and $\delta\hat{K}/\delta s_{\mathbf{k}}$ to zero, we get respectively

$$\mu \;=\; 2Un + Um - 2Un_{\rm c} \tag{3.1}$$

and

$$\tanh(2s_{\mathbf{k}}) = \frac{-Um}{\epsilon_{\mathbf{k}}-\mu+2Un}\,, \quad \sinh(2s_{\mathbf{k}}) = \frac{-Um}{\sqrt{(\epsilon_{\mathbf{k}}-\mu+2Un)^2-(Um)^2}}\,.$$

Here we have defined

$$M = \sum_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}}\hat{c}^\dagger_{-\mathbf{q}}\rangle = N_{\rm c} + \sum_{\mathbf{q}} \frac{1}{2}\sinh(2s_{\mathbf{q}})\,,$$

and $m = M/V$.

In 3.2.2, we will find $\mu - 2Un = -Um$. Therefore

$$\sinh(2s_{\mathbf{k}}) = \frac{-Um}{E_{\mathbf{k}}} = \frac{-Um}{\sqrt{\epsilon_{\mathbf{k}}^2 + 2(Um)\epsilon_{\mathbf{k}}}}\,. \tag{3.2}$$

3.1.3 Squeezing in the zero-momentum state

Eqn. (3.2) shows that the squeezing parameter $s_{\mathbf{k}}$ is negative, and becomes large for small $\mathbf{k}$, and seems to diverge for $\mathbf{k}=0$. This infrared divergence indicates that $\sinh(2s_0)$ scales as a positive power of the volume (or particle number): i.e.,

$$\sinh(2s_0) = -\alpha N^\beta \quad \text{and} \quad |s_0| \sim \mathcal{O}(\ln N)\,.$$

However, it is not clear whether it is meaningful to try to extract β and α from Eqn. (3.2).

3.1.4 Expectation values and fluctuations

<u>$\mathbf{k}=0$ *mode.*</u>

Expectation values of the bosonic operators are:

$$\langle \text{sq\_gr}|\hat{c}_0|\text{sq\_gr}\rangle \;=\; \sqrt{N_c}e^{i\theta}\,, \qquad \langle \text{sq\_gr}|\hat{c}_0^\dagger|\text{sq\_gr}\rangle \;=\; \sqrt{N_c}e^{-i\theta}\,.$$

It is interesting to contrast this with Bogoliubov's mean-field prescription, $\langle\hat{c}_0\rangle = \langle\hat{c}_0^\dagger\rangle = \sqrt{N_0}$. Since $N_0 = N_c + \sinh^2(s_0)$, the $\mathbf{k}=0$ squeezing parameter measures the deviation of our model from mean-field physics. This will be discussed further in 3.2.3.

The hermitian quadrature operators $\hat{X}_0 = \frac{1}{2}(\hat{c}_0+\hat{c}_0^\dagger)$ and $\hat{Y}_0 = \frac{1}{2i}(\hat{c}_0-\hat{c}_0^\dagger)$ have expectation values $\langle\hat{X}_0\rangle = N_c\cos\theta$, $\langle\hat{Y}_0\rangle = N_c\sin\theta$, and fluctuations

$$\begin{aligned}\langle\delta^2\hat{X}_0\rangle &= \frac{1}{4}\left[e^{2s_0}\cos^2\theta \;+\; e^{-2s_0}\sin^2\theta\right]\,,\\ \langle\delta^2\hat{Y}_0\rangle &= \frac{1}{4}\left[e^{2s_0}\sin^2\theta \;+\; e^{-2s_0}\cos^2\theta\right]\,.\end{aligned}$$

For $\theta=0$, the fluctuations are $\frac{1}{4}e^{\pm s_0}$. These results are identical to those for a squeezed coherent state in quantum optics. Since s_0 is negative, as indicated by Eqn. (3.2), the $\mathbf{k}=0$ state is squeezed in the radial direction in the $\hat{X}_0$, $\hat{Y}_0$ plane.

<u>$\mathbf{k}\neq 0$ *modes.*</u>

Because the nonzero-momentum modes have the structure of two-mode squeezed vacua, the operators have zero expectation values, $\langle\hat{c}_\mathbf{k}\rangle = \langle\hat{c}_\mathbf{k}^\dagger\rangle = 0$. There is also no squeezing in quadrature operators defined within a single mode: the usual $\hat{X}_\mathbf{k}$, $\hat{Y}_\mathbf{k}$ have zero expectation values and equal fluctuations.

Squeezing can be seen if one defines the mixed-mode operators

$$\begin{aligned}\hat{X}_{\pm\mathbf{k}} &= \frac{1}{\sqrt{2}}(\hat{X}_\mathbf{k}+\hat{X}_{-\mathbf{k}}) \;=\; \frac{1}{2\sqrt{2}}(\hat{c}_\mathbf{k}+\hat{c}_{-\mathbf{k}}+\hat{c}_\mathbf{k}^\dagger+\hat{c}_{-\mathbf{k}}^\dagger)\,,\\ \hat{Y}_{\pm\mathbf{k}} &= \frac{1}{\sqrt{2}}(\hat{Y}_\mathbf{k}+\hat{Y}_{-\mathbf{k}}) \;=\; \frac{1}{2\sqrt{2}}(\hat{c}_\mathbf{k}+\hat{c}_{-\mathbf{k}}-\hat{c}_\mathbf{k}^\dagger-\hat{c}_{-\mathbf{k}}^\dagger)\,.\end{aligned}$$

These quadrature operators have zero expectation values and unequal (squeezed) variances $\frac{1}{2}\,e^{\pm s_\mathbf{k}}$.

3.2 Symmetry Breaking and Goldstone's Theorem

We now investigate the symmetry-broken nature of the ground state of the Bose gas. The ground state has a particular phase, and thus spontaneously breaks the $U(1)$ symmetry present in the Hamiltonian. In our formalism, the phase of the ground state is given by the phase of the coherence parameter, $\alpha = |\alpha|\, e^{i\theta}$.

According to Goldstone's theorem, a symmetry-broken phase should have a gapless mode. Therefore the excitation spectrum of the weakly-interacting Bose gas is gapless. This condition imposes one more constraint on the variational parameters of our theory.

For the weakly-interacting Bose gas, the condition for a gapless spectrum is the Hugenholtz-Pines (H-P) theorem [26], $\mu = \Sigma_{11}(0,0) - \Sigma_{12}(0,0)$, where $\Sigma_{11}(\mathbf{k},\omega)$ and $\Sigma_{12}(\mathbf{k},\omega)$ are the normal and anomalous self-energies. As discussed in 1.2.2 (chapter 1), the H-P theorem can be deduced from the existence of a symmetry-broken ground state, as well as from the requirement of a gapless excitation spectrum.

In 3.2.1 we use the fact that the Hamiltonian has a $U(1)$ symmetry while the ground state (and hence our variational wavefunction) does not. In 3.2.2 we calculate the excitation spectrum by constructing a single-quasiparticle wavefunction, and impose the requirement of gaplessness. The two considerations lead to the same condition for the variational parameters, which is comforting in light of Goldstone's theorem. The condition should be equivalent to the H-P theorem. In 3.2.3 we compare our condition with the Hartree-Fock-Bogoliubov (mean-field) form of the H-P theorem. The comparison provides a conclusion concerning the presence of squeezing at the mean field level.

3.2.1 $U(1)$ symmetry breaking

In our variational wavefunction, the symmetry-broken nature of the ground state appears as the definite phase of the coherence parameter $\alpha = fe^{i\theta}$. A shift of this phase would obviously change the wavefunction, but should not affect the ground-state energy, since the Hamiltonian is $U(1)$-invariant. This requirement gives us the H-P theorem for our formalism.

We examine the transformation

$$\alpha \to \widetilde{\alpha} = \alpha e^{i\lambda} = fe^{i(\theta+\lambda)}, \quad \hat{D} \to \widetilde{\hat{D}}\,,$$

so that the ground state is shifted, $|\text{sq\_gr}\rangle \to \left|\widetilde{\text{sq\_gr}}\right\rangle = \hat{S}^\dagger\,(\widetilde{\hat{D}})^\dagger\,|\text{vac}\rangle$. We will consider infinitesimal λ, so that α and N_c transform respectively to $\widetilde{\alpha} \approx \alpha(1+i\lambda)$, and $\widetilde{N_\text{c}} = N_\text{c}(1+\lambda^2)$.

The shift in the grand-canonical potential is

$$\delta\langle\hat{K}\rangle = \langle\widetilde{\text{sq\_gr}}|\,\hat{K}\,|\widetilde{\text{sq\_gr}}\rangle - \left\langle \text{sq\_gr}|\hat{K}|\text{sq\_gr}\right\rangle = \lambda^2 N_\text{c}\left\{-\mu + 2Un - Un_\text{c}\right\}\,.$$

We now use the requirement that the grand-canonical energy should not be changed by a shift of the ground state phase, i.e., $\delta\langle\hat{K}\rangle = 0$. Thus we get

$$\mu = 2Un - Un_\text{c}\,. \tag{3.3}$$

Within the variational formalism, this is our equivalent of the H-P relation.

3.2.2 Excitation spectrum

We first construct a wavefunction for a Bose-gas ground state with a single-quasiparticle excitation added on top of it:

$$|\text{sq\_exc}(\mathbf{k}_1)\rangle = \hat{S}\hat{D}\,\hat{c}^\dagger_{\mathbf{k}_1}\,|\text{vac}\rangle$$

The idea is that, since the squeeze operator represents interaction effects in the present formalism, the particle creation operator $\hat{c}^\dagger_{\mathbf{k}_1}$ should produce a Bogoliubov quasiparticle when used in conjunction with the $\hat{S}$ operator.

One can evaluate matrix elements of $\hat{N}_{\mathbf{k}}$ and $\hat{H}_{\rm int}$ in the state $|\text{sq\_exc}(\mathbf{k}_1)\rangle$ just as was done in the ground state $|\text{sq\_gr}\rangle$. The calculation is lengthier but straightforward. Calculating $\langle\hat{K}\rangle$, one now obtains the excitation spectrum as the (grand-canonical) energy of the new state with respect to the ground state.

$$\begin{aligned} E_{\mathbf{k}} &= \langle \text{sq\_exc}(\mathbf{k}_1)|\,\hat{K}\,|\text{sq\_exc}(\mathbf{k}_1)\rangle \;-\; \langle \text{sq\_gr}|\,\hat{K}\,|\text{sq\_gr}\rangle \\ &= \cosh(2s_{\mathbf{k}_1})\,[\epsilon_{\mathbf{k}_1}-\mu+2Un] + \;\sinh(2s_{\mathbf{k}_1})\cdot Um \\ &= \left[(\epsilon_{\mathbf{k}_1}-\mu+2Un)^2-(Um)^2\right]^{1/2} \end{aligned}$$

For the spectrum to be gapless, one requires $\mu-2Un=\pm Um$. The positive sign is inconsistent (resulting in $n_{\rm c}=0$). Therefore, we have

$$\mu = 2Un - Um \tag{3.4}$$

Taken together with Eqn. (3.1), this is indeed identical to the condition (3.3) obtained from consideration of $U(1)$ symmetry breaking, as expected.

The spectrum we have is thus $E_{\mathbf{k}}=\sqrt{\epsilon_{\mathbf{k}}^2+2(Un_{\rm c})\epsilon_{\mathbf{k}}}=\sqrt{\epsilon_{\mathbf{k}}^2+2(Um)\epsilon_{\mathbf{k}}}\,$, which may be contrasted with the Bogoliubov spectrum $E_{\mathbf{k}}=\sqrt{\epsilon_{\mathbf{k}}^2+2(Un_0)\epsilon_{\mathbf{k}}}\,$.

3.2.3 Mean field and beyond

At the Hartree-Fock-Bogoliubov (HFB) level, $\Sigma_{11}=2Un$ and $\Sigma_{12}=Un_0$, so that the H-P theorem is $\mu=2Un-Un_0$. Comparing with our form $\mu=2Un-n_{\rm c}=2Un-m$, we conclude that the $\hat{S}\hat{D}\,|\text{vac}\rangle$ formalism reduces to the mean-field Hartree-Fock-Boholiubov (HFB) results if $N_0=N_{\rm c}=M$. Since

$$N_0=\langle\hat{c}_0^\dagger\hat{c}_0\rangle = N_{\rm c}+\sinh^2(s_0)\,, \tag{3.5}$$

the condition for our formalism to be restricted to mean-field physics is $s_0=0$. The $\mathbf{k}=0$ squeezing parameter is thus a measure of the deviation of the formalism from HFB physics.

The point is further emphasized by rewriting (3.5) as

$$\langle \hat{c}_0^\dagger \hat{c}_0 \rangle = \langle \hat{c}_0^\dagger \rangle \langle \hat{c}_0 \rangle + \sinh^2(s_0)\,.$$

Again, $\sinh^2(s_0)$ acts as a correction to mean-field type decomposition.

The argument can be inverted to state that, at mean field level, the weakly interacting $T = 0$ Bose gas has no squeezing in the zero-momentum mode. Since $s_0 = 0$ at mean field level, the $\mathbf{k} = 0$ squeezing must come from beyond mean field.

While s_0 is not strictly determined in our formalism, Eqn. (3.2) indicates that it is nonzero and large (Sec. 3.1.3). We therefore conclude that our $\hat{S}\hat{D}\,|\text{vac}\rangle$ description includes contributions from diagrams other than the mean-field (HFB) ones.

In contrast to the $\mathbf{k} = 0$ mode squeezing, the nonzero $\pm\mathbf{k}$ mixed-mode squeezing in the $\mathbf{k}\neq 0$ modes (mixed mode squeezing) is present at mean field level already.

3.3 Squeezing in "Fixed-N" Excitations

The squeezing of the nonzero-momentum modes can be given a more physical interpretation by using a different variational formulation. For this purpose, we will use the bosonic operators introduced by A.E. Ruckenstein in [71].

3.3.1 Bosonic fields and excitation Hamiltonian

Ref. [71] presents a current algebra approach to formulating a number-conserving description of the WIBG. The Hamiltonian is written in terms of density and current operators $\hat{n}(\mathbf{r}) = \hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r})$ and $\hat{\mathbf{j}} = -\frac{i}{2m}[\hat{\psi}^\dagger(\mathbf{r})\nabla\hat{\psi}(\mathbf{r}) - \nabla\hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r})]$.

The density fluctuation operator $\hat{\eta}$, defined as $\hat{\eta}(\mathbf{r}) = \hat{n}(\mathbf{r}) - \langle \hat{n}(\mathbf{r})\rangle = \hat{n}(\mathbf{r}) - n_{\mathrm{G}}(\mathbf{r})$, and the phase operator $\hat{\phi}$, defined by $\hat{\mathbf{j}} = \frac{n_{\mathrm{G}}(\mathbf{r})}{m}\nabla\hat{\phi}(\mathbf{r})$, are canonically conjugate. It is convenient to define linear combinations

$$\hat{b}(\mathbf{r}) = \frac{1}{2\sqrt{n_{\rm G}(\mathbf{r})}}\left[\hat{\eta}(\mathbf{r}) + 2in_{\rm G}(\mathbf{r})\hat{\phi}(\mathbf{r})\right] ,$$
$$\hat{b}^\dagger(\mathbf{r}) = \frac{1}{2\sqrt{n_{\rm G}(\mathbf{r})}}\left[\hat{\eta}(\mathbf{r}) - 2in_{\rm G}(\mathbf{r})\hat{\phi}(\mathbf{r})\right] .$$

The Hamiltonian in [71] takes the form

$$\hat{H} \;=\; E_{\rm GS}\left[n_{\rm G}(\mathbf{r})\right] \;+\; \hat{H}_X\left[n_{\rm G}(\mathbf{r}), \hat{b}(\mathbf{r}), \hat{b}^\dagger(\mathbf{r})\right] .$$

$E_{\rm GS}$ describes the mean-field ground state, and minimizing this functional determines $n_{\rm G}(\mathbf{r})$. We are interested in $\hat{H}_X$ which gives the low-lying, large-lengthscale excitations:

$$\hat{H}_X \;\approx\; \int d^3\mathbf{r}\left\{\frac{1}{2m}\left[\nabla\hat{b}^\dagger(\mathbf{r})\right]\cdot\left[\nabla\hat{b}^\dagger(\mathbf{r})\right] \;+\; \tfrac{1}{2}Un_{\rm G}(\mathbf{r})\left[\hat{b}(\mathbf{r}) + \hat{b}^\dagger(\mathbf{r})\right]^2\right\} .$$

We are interested in the uniform WIBG, $n_{\rm G}(\mathbf{r}) = n_{\rm G}$; so we define Fourier space quantities $\hat{b}_\mathbf{k} = \int_\mathbf{r} \hat{b}(\mathbf{r})e^{-i\mathbf{k}\cdot\mathbf{r}}$, $\hat{\eta}_\mathbf{k} = \int_\mathbf{r} \hat{\eta}(\mathbf{r})e^{-i\mathbf{k}\cdot\mathbf{r}}$, and $\hat{\phi}_\mathbf{k} = \int_\mathbf{r} \hat{\phi}(\mathbf{r})e^{-i\mathbf{k}\cdot\mathbf{r}}$. Unfortunately, $\hat{\eta}_\mathbf{k}$ and $\hat{\phi}_\mathbf{k}$ are not hermitian: $\hat{\eta}_\mathbf{k}^\dagger = \hat{\eta}_{-\mathbf{k}}$ and $\hat{\phi}_\mathbf{k}^\dagger = \hat{\phi}_{-\mathbf{k}}$.

We also define the mixed-mode hermitian quadrature operators

$$\hat{X}_{\pm\mathbf{k}} \;=\; \frac{1}{4}\left(\hat{b}_\mathbf{k} + \hat{b}_{-\mathbf{k}} + \hat{b}_\mathbf{k}^\dagger + \hat{b}_{-\mathbf{k}}^\dagger\right) \;=\; \frac{1}{4n_{\rm G}}\left(\hat{\eta}_\mathbf{k} + \hat{\eta}_\mathbf{k}^\dagger\right) , \tag{3.6}$$
$$\hat{Y}_{\pm\mathbf{k}} \;=\; \frac{1}{4i}\left(\hat{b}_\mathbf{k} + \hat{b}_{-\mathbf{k}} - \hat{b}_\mathbf{k}^\dagger - \hat{b}_{-\mathbf{k}}^\dagger\right) \;=\; \frac{n_{\rm G}}{2}\left(\hat{\phi}_\mathbf{k} + \hat{\phi}_\mathbf{k}^\dagger\right) . \tag{3.7}$$

In momentum space, the excitation Hamiltonian reads:

$$\hat{H}_X \;=\; \sum_{\mathbf{k}\neq 0}\epsilon_\mathbf{k}\;\hat{b}_\mathbf{k}^\dagger\hat{b}_\mathbf{k} \;+\; \tfrac{1}{2}Un_{\rm G}\sum_{\mathbf{k}\neq 0}\left(\hat{b}_\mathbf{k}^\dagger\hat{b}_{-\mathbf{k}}^\dagger + \hat{b}_\mathbf{k}\hat{b}_{-\mathbf{k}} + 2\;\hat{b}_\mathbf{k}^\dagger\hat{b}_\mathbf{k}\right) . \tag{3.8}$$

I am ignoring an additive constant that appears while taking the Fourier transform.

The Hamiltonian (3.8) looks identical to that derived by Bogoliubov. However, the operators in the Bogoliubov Hamiltonian are the original bosonic operators $\hat{c}$, $\hat{c}^\dagger$, rather than the peculiar bosons $\hat{b}$, $\hat{b}^\dagger$, that we have here. The interpretation is very different; the Bogoliubov picture involves an order parameter and $\pm\mathbf{k}$ pairs can appear from or disappear into the condensate. In the fixed-N picture, there is no order parameter. $\hat{H}_X$ should not be regarded as a quasiparticle Hamiltonian, but rather as the Hamiltonian describing low-lying *density and current oscillations* of the system at a fixed total particle number. It is then no surprise that Eqn. (3.8) does not conserve the number of bosons, $\int_{\mathbf{r}} \hat{b}^\dagger(\mathbf{r})\hat{b}(\mathbf{r})$. Our reason for using this formalism is that the $\hat{b}$ bosons can be interpreted in terms of density fluctuation and phase operators.

3.3.2 Variational treatment

Let us define the reference state $|\mathrm{ref}\,\rangle$ as the vacuum for the $\hat{b}_{\mathbf{k}}$ bosons. We now introduce the following wavefunction as a variational state for the system:

$$|\mathrm{sq2}\rangle = \prod_{\mathbf{k}\neq 0} \exp\left[\gamma_{\mathbf{k}}\hat{b}^\dagger_{\mathbf{k}}\hat{b}^\dagger_{-\mathbf{k}} - \gamma^*_{\mathbf{k}}\hat{b}_{\mathbf{k}}\hat{b}_{-\mathbf{k}}\right] |\mathrm{ref}\,\rangle \;=\; \prod_{\mathbf{k}\neq 0} \hat{S}_{\mathbf{k}}\,|\mathrm{ref}\,\rangle \;=\; \hat{S}\,|\mathrm{ref}\,\rangle\,.$$

with the usual $\gamma_{\mathbf{k}} + \gamma_{-\mathbf{k}} = 2s_{\mathbf{k}}\, e^{i\phi_{\mathbf{k}}}$.

The state $|\mathrm{ref}\,\rangle$ itself is determined from the E_{GS} part of the theory, i.e., using $\hat{H} \approx E_{\mathrm{GS}}[n_{\mathrm{G}}(\mathbf{r})]$.

The expectation values in our variational state $|\mathrm{sq2}\rangle$ are $\langle\hat{b}^\dagger_{\mathbf{k}}\hat{b}_{\mathbf{k}}\rangle = \sinh^2(s_{\mathbf{k}})$ and $\langle\hat{b}^\dagger_{\mathbf{k}}\hat{b}^\dagger_{-\mathbf{k}}\rangle = \frac{1}{2}\sinh(2s_{\mathbf{k}})e^{-i\phi_{\mathbf{k}}}$.

We will minimize $\langle H_X\rangle$, not $\langle H_X - \mu\hat{N}\rangle$. The number of excitations are not conserved, and μ is determined in the E_{GS} part of the theory.

$$\langle H_X\rangle = \sum_{\mathbf{k}\neq 0} (\epsilon_{\mathbf{k}} + Un_{\mathrm{G}})\sinh^2(s_{\mathbf{k}}) + Un_{\mathrm{G}}\sum_{\mathbf{k}\neq 0}\sinh(2s_{\mathbf{k}})\cos\phi_{\mathbf{k}}\,.$$

Minimization leads to: $\phi_{\mathbf{k}} = 0$, and

$$\tanh(2s_{\mathbf{k}}) = \frac{-Un_{\mathrm{G}}}{\epsilon_{\mathbf{k}} + Un_{\mathrm{G}}}\,, \qquad \sinh(2s_{\mathbf{k}}) = \frac{-Un_{\mathrm{G}}}{\sqrt{\epsilon^2_{\mathbf{k}} + 2Un_{\mathrm{G}}\epsilon_{\mathbf{k}}}}\,.$$

Finally, we will calculate the excitation spectrum from this alternate variational procedure. Using the same idea described in 3.2.2, we construct the excited state $|\mathrm{sq2.exc}(\mathbf{p})\rangle = \hat{S}\ \hat{b}^{\dagger}_{\mathbf{p}}\,|\mathrm{ref}\rangle$. The excitation spectrum is found to be

$$\begin{aligned} E(\mathbf{p}) &= \langle \mathrm{sq2.exc}(\mathbf{p})|\,\hat{H}_X\,|\mathrm{sq2.exc}(\mathbf{p})\rangle - \langle \mathrm{sq2.gr}|\,\hat{H}_X\,|\mathrm{sq2.gr}\rangle \\ &= \sqrt{(\epsilon_{\mathbf{p}} + Un_{\mathrm{G}})^2 - (Un_{\mathrm{G}})^2} \;=\; \sqrt{\epsilon_{\mathbf{p}}^2 + 2\,(Un_{\mathrm{G}})\,\epsilon_{\mathbf{p}}}\ , \end{aligned} \tag{3.9}$$

which is the Bogoliubov spectrum, assuming $n_{\mathrm{G}} = n_0$.

3.3.3 Squeezing of variables

Having obtained the spectrum (3.9) from the variational state, we conclude that the new variational formulation reflects the physics of the WIBG at least up to mean-field level, and hence we can use it to draw conclusions about the $\mathbf{k}\neq 0$ mixed-mode squeezing.

The two-mode squeezed $\pm\mathbf{k}$ modes will display squeezing in the plane of mixed-mode quadrature operators $\hat{X}_{\pm\mathbf{k}} = \frac{1}{\sqrt{2}}(\hat{X}_{\mathbf{k}} + \hat{X}_{-\mathbf{k}})$ and $\hat{Y}_{\pm\mathbf{k}} = \frac{1}{\sqrt{2}}(\hat{Y}_{\mathbf{k}} + \hat{Y}_{-\mathbf{k}})$. This is analogous to our discussion in 3.1.4 for our first variational wavefunction, and in 2.1.3 for a general two-mode squeezed vacuum.

However, at this stage the relevant quadrature operators are physically meaningful: $\hat{X}_{\pm\mathbf{k}} = \frac{1}{4n_{\mathrm{G}}}\left(\hat{\eta}_{\mathbf{k}} + \hat{\eta}^{\dagger}_{\mathbf{k}}\right)$ and $\hat{Y}_{\pm\mathbf{k}} = \frac{n_{\mathrm{G}}}{2}\left(\hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}}\right)$, as defined in Eqn. (3.6).

Since $s_{\mathbf{k}}$ is negative, the $\hat{X}_{\mathbf{k}}$ variable is squeezed. Thus the density fluctuation operators are squeezed while phase operators have dilated variance. The squeezing is larger for lower momentum. An example is shown in Fig. 3.1.

Thus our study of the alternate variational formulation, in terms of the bosons of the "fixed-N" theory [71], has led to two noteworthy consequences. First, we have managed to look at a squeezing phenomenon in terms of variables that have the very physical meaning of density fluctuations and phases, albeit in momentum space. Second, we have found a new formulation of the old idea, attributed to Feynman [11,187], that in a repulsive Bose condensate, density fluctuations should be suppressed. (In our new language: squeezed.)

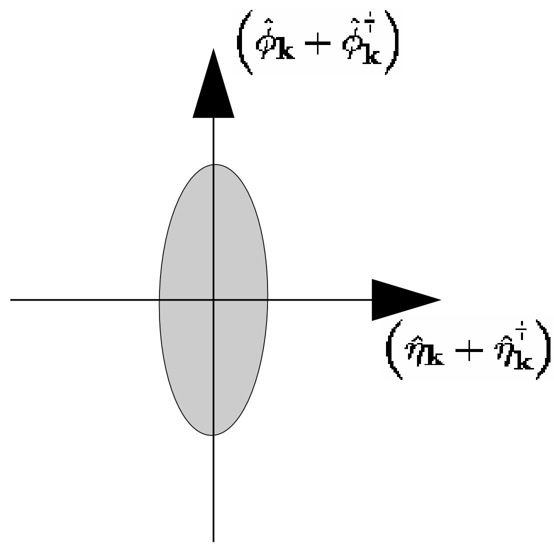


Figure 3.1: Squeezing of nonzero-momentum modes, in terms of "physical" variables. The variables are Fourier transforms of density fluctuation and phase, in units of n_{G} and n_{G}^{-1} respectively. The density fluctuation variables are squeezed. The squeezing shown here is very moderate, i.e., a large-momentum mode or a condensate with small number of particles.

3.4 Closing Remarks

Squeezing in various modes of a WIBG ground state has been explored in this chapter, using two different variational wavefunctions. We have shown that the squeezing of the zero-momentum mode does not show up at mean field or Hartree-Fock-Bogoliubov (HFB) level, but comes from higher-order diagrams. On the other hand, mixed-mode squeezing for $\pm\mathbf{k}$ pair modes already shows up at HFB order. With our second variational wavefunction, we have managed to interpret the $\mathbf{k}\neq 0$ pair-mode squeezing in terms of density-phase variables, and we see that the density fluctuations are squeezed.

These results open up a number of new questions which have not been yet addressed.

For example, one would like to compare our treatment to diagrammatic methods for the Bose-condensed WIBG. Since our method recovers Σ_{11} up to HFB level and incorporates some extra-HFB physics into Σ_{12} , one would like to know exactly what diagrams are incorporated into the anomalous self-energy Σ_{12} obtained from the variational procedure.

A related question is: which diagrams provide zero-momentum squeezing, i.e., how would one perform a perturbative calculation of s_0? A hint for both these questions is provided by considering the physical source of the squeezing in the condensate mode. This squeezing represents quantum fluctuations of the $\mathbf{k} = 0$ mode operators ($\hat{c}_0$, $\hat{c}_0^\dagger$). The correct way to include such fluctuations, in a condensed-phase perturbative description, has not been developed, to the best of our knowledge.

In addition, a whole number of other quantum states are studied in quantum optics which have not been tried in the description of WIBG physics. For example, it is not known what further information can be obtained by using, e.g., a squeezed Fock state instead of a squeezed coherent state for the $\mathbf{k} = 0$ mode.

Extending the questions discussed here, to the case of non-uniform (e.g., magnetically trapped, or rotating) WIBGs, also remains an open task. It would also be interesting to develop a similar description for the 2D WIBG, where there is no real condensate, and for the 3D WIBG at finite temperatures, where the depletion is much higher.

Other than the present class of variational wavefunctions based on squeezing and coherence ideas, a *real-space* variational procedure has also been widely used for the Bose condensate (discussion in Sec. 1.3.4, Refs. [51, 107, 113]). A natural question is the relation between the two descriptions. Do the Bijl-Dingle-Jastrow wavefunctions have $\pm\mathbf{k}$ pair squeezing built into them? Do they contain fluctuations of the condensate mode, i.e., $\mathbf{k} = 0$ squeezing? What is the overlap between the higher-order diagrams included in the Jastrow-type wavefunctions and those included in our $\hat{S}\hat{D}$ states? These problems remain unaddressed at this point.

Finally, we should note that, while we have obtained a "physical" interpretation for the $\pm\mathbf{k}$ pair squeezing in Sec. 3.3, the interpretation is not necessarily unique. Further elucidation of squeezing, in terms of other variables, would be welcome.

Chapter 4

The T_c Problem: Overview

The $T_{\rm c}$ problem was introduced briefly in section 1.4. In this chapter, we provide a short survey of previous work on this topic, and analyze several issues related to these calculations. Such an overview seems appropriate considering the diverse approaches to the problem, and the variety of issues that have arisen in this connection.

The main part of our work on the $T_{\rm c}$ problem is the development and exploration of a quasiparticle formulation of the problem; this is deferred to chapter 5.

We are concerned with the Bose condensation temperature ($T_{\rm c}$) of the uniform WIBG model, given by the Hamiltonian of Eqn. (1.1) in chapter 1. The transition temperature for the ideal gas is $T_{\rm c}^0 = \left(2\pi/m[\zeta(\frac{3}{2})]^{2/3}\right) n^{2/3}$. The program is to calculate the shift in $T_{\rm c}$, as compared to a noninteracting gas of the same density, as a function of the dimensionless parameter $an^{1/3}$, for small $an^{1/3}$.

Other than $an^{1/3}$, it is also common to use the ratio a/λ to measure the interaction strength; here $\lambda = (2\pi\hbar^2/mk_{\rm B}T)^{1/2}$ is the thermal wavelength. At free-boson criticality, the two dimensionless quantities differ by a factor of $[\zeta(\frac{3}{2})]^{1/3}$. (Of course this needs to be corrected when $\lambda_{\rm c} \propto T_{\rm c}^{-1/2}$ gets shifted. But we will be interested only in the leading correction, so $an^{1/3}$ and $a/\lambda_{\rm c}$ can be thought of as being simply proportional.)

In section 4.1, the background is set by summarizing different calculations and basic results. Section 4.2 contains comments and analysis on various approaches to the T_c problem.

Kanno's research [125, 188, 189] is particularly relevant to our calculations in the next chapter; his calculations motivated our non-selfconsistent regularization procedure in Sec. 5.4. We provide a more detailed discussion, and a diagrammatic re-formulation, of Kanno's work, in Sec. 4.3.

In the concluding section 4.4, some directions for additional calculations are suggested.

4.1 Background and Perspective

Early predictions for the shift in transition temperature appear in the work of Lee, Yang and Huang, around forty years ago, as summarized in [190]. Recently, Huang has published [191] a prediction for an increase in T_c due to the inter-particle repulsion, proportional to the square root of the interaction strength, $T_c - T_c^{(0)} \sim \sqrt{an^{1/3}}$. Schakel in a short note [192], and Baym *et al.* in [190], have both discussed the problem with this work; in particular [190] contains an analysis showing how Huang's result is generated spuriously from a truncation of the virial expansion used at mean field level.

A similar spurious result appearing in the literature is that of Toyoda [116], who in 1982 performed mean-field-like calculations deriving a Landau-Ginzburg type thermodynamic function. His prediction was a *decrease* of the transition temperature, $\Delta T_c \sim -\sqrt{an^{1/3}}$. Again, this is believed to be an artifact of mean field approximations. The present understanding is that there is no mean field effect on T_c, and that higher-order correlations are required to find the leading shift in T_c. More detailed comments on mean-field effects appear in 4.1.1.

A more recent work that also predicts a negative shift in T_c is the canonical-ensemble work of Wilkens *et al.* [193]. This work has been critiqued in [194]. We discuss some related issues in section 4.2.5.

In addition, in view of the continuing appearance of negative-shift predictions, we give in section 4.2.2 a simple argument for the *increase* of the transition temperature due to the addition of a repulsive interaction. This argument makes use of Eqn. (4.2), which is derived independent of any particular method of treating interaction effects.

Baym, Laloë and collaborators have published a large body of work on the T_c problem during the last several years [190, 194–200]. A number of authors have also produced followup calculations [201–203]. These investigations predict a positive shift in T_c that to leading order is *linear* in the scattering length; $\Delta T_c = c_1 a n^{1/3} + \mathcal{O}(a^2 n^{2/3})$. Attempts to calculate the coefficient c_1, however, have continued to give fluctuating results, and this has prompted the Baym group to predict [197] a logarithmic contribution at the *next* order, i.e., a contribution $\propto -a^2 \ln a$. The presence of such a non-analyticity might explain the difficulty in any numerical estimate of c_1.

Stoof's renormalization-group analysis [16, 68] of the three-dimensional bose gas has led to a prediction [16, 204] of an increase of the transition temperature proportional to $a|\ln a|$, i.e., a non-analyticity at *leading* order. A leading non-analytic behavior would actually explain even better the uncertain results obtained in attempts to calculate the coefficient c_1, based on the assumption of linear shift $\Delta T_c \sim c_1 a n^{1/3}$.

An older attack on the T_c problem is that of Kanno [125, 188, 189]. In a series of papers, Kanno introduced a "quasilinear" canonical transformation approach to the many-body problem, which he used to calculate, among other things, the free energy and hence transition temperature of the bose gas. Since Kanno's work has mostly been ignored by the current T_c community, we give a description of his work in section 4.3. We also show how some of his derivations can be performed in a modern, far more transparent, manner.

Schakel's work on developing an effective theory [102] of the weakly non-ideal

Bose gas has led to the prediction [103] of a shift in the transition temperature that is strictly linear in the interaction. The connection between this calculation and other approaches has not yet been explored in detail. Some issues are pointed out in section 4.2.4.

Our recent report (Ref. [2] and chapter 5 of this thesis) addresses the question of T_c within a quasiparticle framework. Other than working out details of a quasiparticle description (e.g., ultraviolet divergence, effect of quasiparticle lifetime), we also provide a non-selfconsistent method of regularizing the infrared (IR) divergences appearing in the description of the transition point. This results in a shift of T_c approximately $\propto a\sqrt{|\log a|}$. The procedure is uncontrolled. Nevertheless, this result points strongly to a non-analytic contribution in the dependence of T_c on the interaction.

Other relevant calcultaions include Kleinert's "five-loop" determination [205] of the coefficient c_1 mentioned above, assuming a purely linear leading shift, and the "linear δ-expansion" calculations [206–209] of de Souza Cruz *et al.* Kleinert has questioned this method [205]. However, a detailed analysis of the relationship of the "δ-expansion" approach to others seems to be lacking in the literature.

4.1.1 Mean field

For a momentum-independent potential, the two mean field contributions to the self-energy (Hartree and Fock) are equal, $\Sigma^{(1)}(\mathbf{k},\omega) = Un + Un = 2Un$. Since the self-energy is momentum-independent, it can simply be absorbed into a redefinition of the chemical potential.

$$E_{\mathbf{k}}^{(1)} = \frac{k^2}{2m} - \mu + \Sigma^{(1)}(\mathbf{k}, E_{\mathbf{k}}^{(1)}) = \frac{k^2}{2m} - \mu_1 \tag{4.1}$$
$$N = N_0 + \frac{V}{\lambda^3} g_{3/2}(e^{\beta\mu_1})$$

Here $\mu_1 = \mu - \Sigma^{(1)} = \mu - 2Un$, and $g_\nu(x)$ represents a Bose-Einstein integral function [115]. Comparison with the ideal-gas relation $N = N_0 + \frac{V}{\lambda^3} g_{3/2}(e^{\beta\mu})$

shows that the transition now simply corresponds to $\mu_1 = 0$ instead of $\mu = 0$, and T_c remains unchanged. The lesson here is that a momentum-dependent self-energy is required for a shift in T_c.

In second-order calculations, (e.g. in 4.3.3 and in chapter 5) it is usual to have the 1st-order self-energy already absorbed in the chemical potential, i.e., to use μ_1 as the chemical potential. This allows one to exclude those second-order diagrams which can be obtained by 1st-order self-energy insertions into 1st-order diagrams.

Spatially extended interaction. In a real system, of course, the description of the interaction in terms of a single parameter (U_0 or a) is valid only for very weak interactions or at very low temperatures/densities; if these conditions are not satisfied one has to take into account the detailed shape of the potential $V(x) = V_0\sigma(x)$ which is not a delta function in real space (not a constant in momentum space) any more. In such a case we get a momentum-dependent Fock term, and hence a contribution to T_c from the first-order self-energy. Assuming a Gaussian $V(x)$ of width a and height V_0, one gets a correction to the spectrum that can be incorporated as an effective mass $m^* = m[1 - V_0 a^2 nm/3]^{-1}$, resulting in a negative shift [6]:

$$\frac{\Delta T_c}{T_c^{(0)}} = -\frac{1}{3} mnV_0 a^2 \ .$$

The negative shift at higher densities is present in liquid helium and presumably in other Bose fluids. However, for weak interactions, this shift is less important than the leading $\mathcal{O}(a)$ effect, which is the current theoretical challenge.

4.1.2 T_c in liquid helium

As a highly correlated liquid, the properties of liquid helium are not expected to be described well by the dilute weakly-interacting Bose gas model. However, the deviation of the lambda transition temperature (2.2K), from the BEC temperature of an ideal gas with the same density (3.1K), can be mostly explained using a simple effective mass description. In other words, the experimental $T_{\rm c}$ is reproduced quite closely by replacing the helium atomic mass m by a quasiparticle effective mass m^* in the ideal gas expression $T_{\rm c}^{(0)} = \left(2\pi/m[\zeta(\frac{3}{2})]^{2/3}\right) n^{2/3}$.

The effective mass in the liquid exceeds the bare mass, because of the inertia of the medium which has to make way for any atom to move. This results in a lowering of $T_{\rm c}$. The effect is essentially equivalent to that described in the last section for a momentum-dependent interaction.

Feynman [187] calculated a partition function for liquid helium that takes into account the effective mass enhancement described above. Using this partition function, Chapline calculated [210] a $T_{\rm c}$ quite close to the actual λ-point of helium. This is equivalent to using the effective mass in the ideal gas expression for $T_{\rm c}^{(0)}$.

According to current understanding, therefore, one expects the following variation of $T_{\rm c}$ with $n^{1/3}a$: an initial linear-like increase, followed by a maximum and then a decrease due to the effective-mass mechanism described above, until eventually $\Delta T_{\rm c}$ becomes negative (Fig. 4.1). This kind of dependence has been seen in the Monte-Carlo calculation of Grüter, Ceperley, Laloë [199, 200], and in Reppy's experiments [85] described below.

4.1.3 Relevant experiments

One could think of studying the $T_{\rm c}$ problem experimentally, e.g., in the context of trapped atomic Bose gases. Unfortunately, trap effects dominate in the dependence of $T_{\rm c}$ on the interaction for these systems. Turning on a repulsion for a trapped Bose gas causes an expansion of the gas, reducing the density and hence decreasing the transition temperature $T_{\rm c} \propto n^{2/3}$ [211, 212]. In contrast to the intrinsic effect under discussion in this thesis, trapping effects on $T_{\rm c}$ are found at mean-field level already.

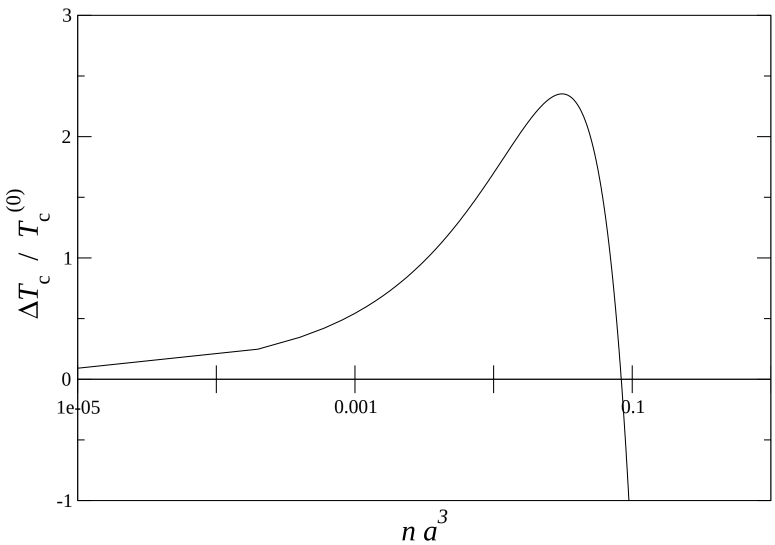


Figure 4.1: Expected qualitative behavior of the relative T_c shift for ^{4}He, as a function of na^3, shown schematically. The qualitative form is based on the experimental results in Ref. [85] and the numerics of Ref. [199]. The numbers are indicative of order-of-magnitude only.

The ^{4}He-Vycor system explored by Reppy *et al.* has been proposed as an independent experimental realization of the weakly non-ideal bose gas [82–85]. Here the small-$an^{1/3}$ regime is reached by enormously decreasing the average ^{4}He density. Despite the lack of consensus on the WIBG interpretation of the ^{4}He-Vycor system, it is worth noting the variation of T_c in this system with the effective $an^{1/3}$.

Reppy's investigation of the transition temperature shows a linear-like increase of T_c at low $an^{1/3}$, followed by a maximum and a decrease to values lower than $T_c^{(0)}$. The general behavior is satisfying, but for the initial increase, the data is not yet good enough to choose between $\sim a$, $\sim a \ln a$, or $\sim a\sqrt{\ln a}$-like behavior.

4.2 Comments on Various Approaches

4.2.1 Baym and collaborators

The work of Baym, Laloë and collaborators [190, 194–200] establishes definitely that a repulsive interaction *increases* T_c, and that the dependence of ΔT_c on the interaction is of linear order, as opposed to quadratic or $\sim \sqrt{a}$.

One should note, however, that the Baym group's work relies heavily on power-counting and dimensional arguments. This kind of argument typically does not capture logarithmic corrections. The possibility of leading non-analytic dependence of ΔT_c on the interaction, therefore, needs to be examined further.

It has been suggested [213] that the large-N calculation [196] is a reliable indicator that the leading shift is purely linear, because of intuition gained from other systems where large-N calculations capture all logarithms. It should be pointed out, however, that the $N \neq 2$ models belong to a different universality class, and have different excitation spectra as compared to the usual ($N = 2$) Bose gas. More care is therefore required in interpreting results from the large-N approach.

The Baym-Laloë collaboration seems to have been the first to relate the single-particle spectrum at T_c to the shift ΔT_c. Another noteworthy aspect is the use of Ursell operators [190, 198], and a corresponding diagram technique, to describe particle correlations. (As far as we can tell, for this problem, the Ursell operator technique does not give any extra information compared to the usual Green function techniques.)

4.2.2 Direction of shift, infrared spectrum

Once the shift in T_c is related to the modification of the single-particle spectrum, the direction of the shift can be argued using well-known facts about the spectrum of the Bose gas.

We will use the following expression for the shift, derived in section 5.3 and in Ref. [2]:

$$\frac{\Delta T_c}{T_c^{(0)}} \approx -\frac{2}{3n}\int \frac{d^3k}{(2\pi)^3}\,[f_c(\xi_{\mathbf{k}}) - f_c(\epsilon_{\mathbf{k}})] \ . \tag{4.2}$$

Here f_c is the Bose function at the critical point; $f_c(x) = (e^{x/kT_c} - 1)^{-1}$; and $\xi_{\mathbf{k}}$ is the quasiparticle spectrum at T_c, i.e., $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} + \Sigma(\mathbf{k}, \xi_{\mathbf{k}}; T = T_c)$.

The expression used by Baym *et al.* [190, 195], $\Delta T_c \propto \int dkU(k)/[k^2 + U(k)]$, is the high-temperature (zero Matsubara frequency) version of (4.2). Noting that the major contribution to (4.2) comes from the IR, one can write approximately

$$\Delta T_c \propto \int_0^{\text{cutoff}} dk\ k^2[\epsilon_{\mathbf{k}}^{-1} - \xi_{\mathbf{k}}^{-1}] \ . \tag{4.3}$$

This equation might introduce or leave out logarithmic corrections; however the following argument is unchanged for (4.2).

If the quasiparticle spectrum $\xi_{\mathbf{k}}$ is "harder" than the bare spectrum $\epsilon_{\mathbf{k}} = k^2/2m$, i.e., if $\xi_{\mathbf{k}}$ is sub-quadratic in k or quadratic with an effective mass $m^* < m$, then the $\epsilon_{\mathbf{k}}$ integral dominates and the shift is positive. On the other hand if the spectrum at the transition point is "softened" by the interaction, then Eqn. (4.3) [or (4.2)] leads to a decrease in the transition temperature.

Since the weakly interacting Bose gas is known to have a linear (Bogoliubov) IR spectrum below the critical temperature, and a quadratic spectrum above T_c, it is extremely reasonable to assume, even without any RG or perturbation-theoretic calculations, that the spectrum at the critical point should be something intermediate between linear and quadratic. As pointed out above, this means a *positive* shift in the transition temperature.

The long-wavelength spectrum at the transition point should be $\xi_{\mathbf{k}} \sim k^{2-\eta}$, where η is the anomalous dimension. The $2 - \eta$ exponent follows from the definition of η, which stipulates that the correlation function (Green's function) at criticality has IR behavior $G(\mathbf{k}) \sim k^{\eta-2}$ [96, 214, 215].

The present system falls in the same universality class as the XY model, or the $N = 2$ quantum rotor model. For this universality class, large-N and ϵ-expansion calculations give respectively $\eta \approx 0.14$ and $\eta \approx 0.02$, and a recent Monte Carlo calculation [216] has produced the value $\eta \approx 0.038$.

Second-order treatments of the self-energy (e.g., the Baym group's [190, 195] and our work in chapter 5 and [2]) over-estimate the modification of the spectrum, giving a $k^{3/2}$ IR behavior ($\eta = 0.5$). On the other hand, some other treatments (such as Stoof's [16] and Schakel's [102, 103]) neglect the modification altogether ($\eta = 0$).

4.2.3 Renormalization group treatment (Stoof)

The RG flow equations derived by Bijlsma & Stoof [16] allow one to numerically locate the transition point. Under RG flow, the physical effective chemical potential flows to positive and negative values, if one starts out respectively in the Bose-condensed and the symmetry-unbroken phase. The critical point is therefore identified by finding parameters for which the chemical potential flows to zero. Stoof finds [16, 204] that the dependence of ΔT_{c} on a/λ has more structure than purely linear; $\Delta T_{\mathrm{c}} \propto (a/\lambda)\ln(a/\lambda)$.

The differences between Stoof's and the Baym group's results merit further study. Stoof has suggested [204] that the source of the difference is not the infrared but the *ultraviolet*. The correction to the spectrum, $U(k) \propto \Sigma(\mathbf{k}, E_{\mathbf{k}}) - \Sigma(0, 0)$, has an ultraviolet $\ln k$ behavior in the Baym group's treatment [190, 195]. In the RG framework, the corresponding quantity is the running coupling constant $\mu(\Lambda)$, which shows no UV logarithms in the calculation of Ref. [16]. This leads to a logarithmic difference in the two predictions for T_{c}. Stoof also suggests [204] that an improved RG calculation, including the momentum dependence of the effective two-body interaction at the transition point, would shed light on the difference between the two approaches.

4.2.4 Schakel: IR regularization

In treating his effective-action theory near the transition point, Schakel [102, 103, 217] is also faced with infrared divergences. One of the remarkable features of this work is an innovative analytic continuation procedure used to regulate IR problems.

Recently, it has been reported [213, 218] that the original contribution to $T_{\rm c}$ found by Schakel at one-loop order is cancelled exactly at second loop. A proper discussion is yet to appear in the literature.

Baym *et al.* have remarked [190] that Schakel's IR prescription (ζ-function regularization) must be incorrect because it gives an incorrect result for the compressibility of the ideal gas. This argument, unfortunately, is derived from an equation that Schakel derived explicitly for the nontrivial broken-symmetry ground state [103, 213] of an interacting gas ($\mu > 0$). The same equations are not applicable to the noninteracting case ($\mu \leq 0$) for which the structure of the vacuum is different (more trivial). It thus appears that the regularization procedure itself is quite possibly sound.

In addition, another consideration brought forth by Schakel's work is the possible importance of nonzero Matsubara frequencies, $\omega_n \neq 0$, which he does not neglect in his effective-action treatment. While we have formulated a quasiparticle description summing all ω_n (section 5.2), using the full expression without making a high-temperature ($\omega_n = 0$) approximation remains a formidable task.

One concern with Schakel's derivation [103] is that, just below the transition point (broken-symmetry phase), a quadratic spectrum is used. A more accurate treatment would use the phonon-like spectrum $E_{\mathbf{k}} \approx \sqrt{\epsilon_{\mathbf{k}}^2 + 2Un_0(T)\epsilon_{\mathbf{k}}}$, or perhaps the critical-point spectrum $E_{\mathbf{k}} = \xi_{\mathbf{k}}$. The effect such a spectrum would have on the thermodynamic potential or on $T_{\rm c}$ is not clear. (Redoing Schakel's $T_{\rm c}$ calculation with a phonon-like spectrum seems quite difficult.)

4.2.5 Canonical ensemble: role of $N_0(T_c)$.

In Wilkens *et al.*'s paper [193], calculations have been performed using a canonical ensemble of N particles, and the shift ΔT_c obtained is linear and *negative* in the scattering length. It has been pointed out [194] that the error lies in evaluating quantities (derivatives) at $N_0 = 0$, instead of evaluating them at the expectation value of $N_0(T_c)$. Mueller *et al.* [194] have also used finite-size scaling ideas in combination with a first-order calculation in the canonical ensemble, and obtain a positive linear shift.

The idea that the zero-momentum state occupancy N_0 becomes "microscopic" at and above the transition needs to be clarified. One finds $\lim_{N\to\infty}(N_0/N) = 0$ for $T \geq T_c$, where N is the total number of particles. However, this does not necessarily imply $N_0 \sim \mathcal{O}(1)$. In fact, for both the ideal and the interacting Bose gas, $N_0(T_c) \sim N^{2/3}$, which is large compared to 1 though small compared to N.

In view of the continuing appearance of the $N_0(T_c) \sim \mathcal{O}(1)$ misconception, we give here a simple derivation of $N_0(T_c) = \gamma N^{2/3}$, together with an expression for the coefficient γ, for the ideal-gas case.

Let us consider the ideal-gas relationship $N = N_0 + \frac{V}{\lambda^3} g_{3/2}(e^{-\alpha})$, where $\alpha = -\beta\mu$, and g_ν refers to a Bose-Einstein integral function [115]. Since α is small near the transition, one can use the Robinson expansion [115] for $g_{3/2}$. Also, inverting $N_{\mathbf{k}} = 1/(e^{\beta\epsilon_{\mathbf{k}}+\alpha} - 1)$ for $\mathbf{k} = 0$, one gets $\alpha \approx 1/N_0$ (since $\epsilon_0 = 0$). Combining these yields, near the transition,

$$N_0^{3/2} + \left[\frac{V}{\lambda^3}\,\zeta\left(\tfrac{3}{2}\right) - N\right]\sqrt{N_0} - \frac{V}{\lambda^3}(2\sqrt{\pi}) \;\approx\; 0\ .$$

Since $n\lambda^3 = \zeta(\frac{3}{2})$ at the transition, the coefficient of $\sqrt{N_0}$ needs to vanish. Therefore

$$N_0(T_c) \;\approx\; \left[\frac{2\sqrt{\pi}}{\zeta(3/2)}\right]^{2/3} N^{2/3}\ . \tag{4.4}$$

The $\mathbf{k} = 0$ occupancy N_0 can actually be calculated for all T, using an extra Lagrange multiplier μ_0 conjugate to N_0, in addition to usual chemical potential

μ coupling to N. Using this "two chemical potentials" formalism [219], one can show that for the ideal gas $\langle N_0 \rangle = 1$ occurs at a temperature $T \approx 2.6\, T_{\rm c}^{(0)}$, well above the transition point.

For the interacting-gas case, an equation corresponding to (4.4) is difficult to derive, but finite-size scaling arguments [194, 220] show that $N_0 \sim N^{2/3}$ continues to hold, even though the non-ideal Bose gas belongs to a different universality class.

The definition of the transition point used in Ref. [193] is in terms of the probability

$$P_n(T, N) \propto \text{tr}\left[\delta_{\hat{N}_0,n} e^{-\beta \hat{H}} \delta_{\hat{N},N}\right] \tag{4.5}$$

for having a condensate occupancy $N_0 = n$. The claim is that $P_n(T, N)$ decreases monotonically with n for all temperatures above the transition, and has a peak at nonzero n for in the condensed phase, and therefore the transition should be defined as the thermodynamic limit of the zero of $D(T, N) = P_0(T, N) - P_1(T, N)$.

The correspondence between the canonical particle-counting statistics, and the grand-canonical ideas of average occupancies, is not trivial. One important point to note is that a monotonically decreasing P_n does not imply a zero expectation value for N_0, even though P_n has a maximum at $n = 0$. (For example, using $P_n \propto e^{-\alpha n}$ gives $\langle N_0 \rangle = \sum_n n P_n = 1/[e^\alpha - 1]$.) The stipulation of a monotonically decreasing P_n therefore does not imply a zero or $\mathcal{O}(1)$ value of N_0 at the transition.

4.3 Kanno's Approach: Using Modified Thermodynamic Potential

Kanno developed an unusual approach [188, 189] to study many-body systems, and also used this approach to calculate Bose gas properties [125, 189].

4.3.1 Formalism

Kanno's calculation of thermodynamic quantities involves a "quasilinear" canonical transformation [188, 189] of the free-particle operators $\hat{c}$ ($\hat{c}^\dagger$):

$$\hat{c}_i = \hat{\gamma}_i + \sum_{j;k,l} A_i(j;k,l)\hat{\gamma}_j^\dagger\hat{\gamma}_k\hat{\gamma}_l + \sum_{j,k;l,m,n} A_i(j,k;l,m,n)\hat{\gamma}_j^\dagger\hat{\gamma}_k^\dagger\hat{\gamma}_l\hat{\gamma}_m\hat{\gamma}_n + \ldots \ , \quad (4.6)$$

resulting in a Hamiltonian having a particular (diagonal-like) form. The formalism involves examining the properties of the coefficients A, and the coefficients appearing in the quasi-diagonal Hamiltonian. Applying this formalism to the weakly repulsive Bose gas [125, 189], one can derive the entropy and pressure above and below the transition, the Bogoliubov spectrum, and many other quantities.

Kanno does not provide comparisons to the usual language of perturbative many-body theory. It is appropriate to point out that many of his physical results, if not all, are equivalent to results obtained in more mainstream approaches. For example, the quantities $A(\mathbf{p})$ and $B(\mathbf{p})$, appearing in [189], are equivalent to the normal and anomalous self-energies (Σ_{11} and Σ_{12}) described in chapter 1.

4.3.2 Shift in T_c

Kanno defines [125] a modified thermodynamic potential

$$\Omega'(N_0, \mu', T) = -\frac{1}{\beta}\ln \mathrm{tr}'\, e^{-\beta(\hat{H}-\mu'\hat{N}')} \ ,$$

with an unusual Lagrange multiplier μ' conjugate to the number of non-condensate particles, $N' = N - N_0$, as opposed to the usual chemical potential which is conjugate to the total particle number N. The prime on the trace indicates that the condensate ($\mathbf{k} = 0$) mode is to be excluded while taking the trace.

From the definition of Ω', it follows that

$$N - N_0 \ = \ N' \ = \ -\frac{\partial \Omega'(N_0, \mu', T)}{\partial \mu'} \ . \tag{4.7}$$

Once Ω' is calculated in some approximation, comparing this relation to $N - N_0 = V g_{3/2}(e^{\beta\mu})/\lambda^3$ for the free Bose gas at the transition point yields the shift $\Delta T_{\rm c}$, in terms of the quantity $\mu'(T_{\rm c})$.

The quantity μ' itself is obtained from a consideration of the free energy $F(N_0, N, T) = \Omega'(N_0, \mu', T) + \mu' N'$. The actual value of N_0 chosen by the system is the one that minimizes $F(N_0, N, T)$, i.e.,

$$0 = \left.\frac{\partial F}{\partial N_0}\right|_{T=T_{\rm c}} = \left.\frac{\partial \Omega'}{\partial N_0}\right|_{T=T_{\rm c}} - \mu'(T_{\rm c}) \ . \tag{4.8}$$

This equation allows one to express $\mu'(T_{\rm c})$ in terms of the interaction strength.

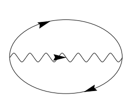
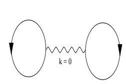
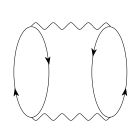
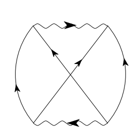
4.3.3 Re-formulation

As advertised at the beginning of this chapter and in section 4.1, we show here how the modified thermodynamic potential Ω' (which is the quantity relevant to the $T_{\rm c}$ problem) can be obtained from a diagrammatic second-order perturbative calculation.

The Feynman rules for a diagrammatic perturbative calculation of the modified thermodynamic potential $\Omega'(\mu', N_0, T)$ are very similar to those for the usual $\Omega(\mu, T)$. One has to use a momentum-dependent chemical potential

$$\mu_{\mathbf{k}} = \begin{cases} \mu' & \text{if } \mathbf{k} = 0, \\ 0 & \text{if } \mathbf{k} \neq 0, \end{cases}$$

and separate out $\mathbf{k} = 0$ from the momentum-sums at each stage. Two first order and two second-order diagrams need to be calculated:

$\Omega' \approx \Omega'^{(0)} +$  $+$  $+$  $+$  .

Other second-order diagrams are excluded by considering the chemical potential to be renormalized up to first order, as discussed in section 4.1.1.

The zeroth and first order terms can be evaluated exactly. UV divergences appearing at second order can be removed by eliminating the bare potential U in favor of the 2-body t-matrix, as done for the self-energy in chapter 5 or [2]. The result is

$$\Omega' = -\frac{V}{\beta\lambda^3} g_{5/2}(e^{\beta\mu'}) + \frac{t}{V}N^2 + 4t^2 N_0 \int_{\mathbf{p}}\int_{\mathbf{q}} \frac{f(\epsilon_{\mathbf{p}}-\mu')f(\epsilon_{\mathbf{q}}-\mu')}{\epsilon_{\mathbf{p}}-\epsilon_{\mathbf{q}}-\epsilon_{\mathbf{p}-\mathbf{q}}}$$
$$+ 2t^2 V \int_{\mathbf{p}}\int_{\mathbf{q}}\int_{\mathbf{k}} \frac{f(\epsilon_{\mathbf{p}}-\mu')f(\epsilon_{\mathbf{k}}-\mu')f(\epsilon_{\mathbf{q}}-\mu')}{\epsilon_{\mathbf{p}}+\epsilon_{\mathbf{k}}-\epsilon_{\mathbf{q}}-\epsilon_{\mathbf{p}+\mathbf{k}-\mathbf{q}}} \,. \quad (4.9)$$

The expression for Ω' obtained in Ref. [125] is essentially the same as Eqn. (4.9).

4.3.4 Comparison

The parameter $\mu'(T_c)$ corresponds to the infrared cutoff used in the the non-selfconsistent regularization of the next chapter. Eqn (4.8) gives the same result for the parameter $\mu'(T_c)$ as that obtained in 5 for the chemical potential μ using $\mu = \Sigma(0,0)$. The reason for this is that $\partial F/\partial N_0$ represents the energy of a zero-momentum quasiparticle with respect to the chemical potential, i.e., $\Sigma(0,0) - \mu$. Therefore Eqn. (4.8) is equivalent to the usual condition $\Sigma(0,0) = \mu$ for the Bose-condensation point.

The shift ΔT_c obtained using eqs (4.7), (4.8), (4.9) is the same as that obtained by our non-selfconsistent regularization scheme, described in section 5.4 and Ref. [2].

The distinctive idea in Kanno's work is the introduction of a modified thermodynamic function Ω' such that $-\partial\Omega'/\partial\mu'$ replaces the right side in the criticality condition $N = (V/\lambda^3)\zeta(\frac{3}{2})$. This bypasses the issue of self-consistency and the modification of the spectrum at T_c, and the use of the chemical potential as a cutoff follows naturally.

4.4 Summary, Further Calculations

To summarize, this chapter reviews aspects of the calculation of the T_{c} shift due to a small repulsive interaction in a Bose gas. This discussion provides the background against which we present our own calculations, involving a quasiparticle picture, in the next chapter. We conclude this chapter with some comments regarding unresolved issues and calculations that remain to be performed.

The outstanding issue appears to be the possibility of a logarithm-like correction to the leading shift in T_{c}. Presumably, a better understanding of the spectrum at the critical point (both at the infrared and the ultraviolet) should help settle this issue. Before outlining improvements to $\xi_{\mathbf{k}}$ in 4.4.1, I will first list some other work that would be interesting to the T_{c} community.

- One possibility is to use finite-size scaling ideas, in conjunction with second-order perturbation theory in the grand-canonical ensemble. This would be complementary to the calculation of Mueller & Baym [194], which uses finite-size scaling with first-order perturbation theory in the canonical ensemble.

- The controversy surrounding Schakel's work needs to be resolved. First, a clear proof of the contribution to ΔT_{c} being cancelled at higher order has not appeared in the literature. As for the calculation of the thermodynamic potential in [103], just below T_{c}, it would be nice to see this treated more properly, either with a Bogoliubov-like spectrum or with the $E_{\mathbf{k}}(T_{\mathrm{c}}) \sim k^{2-\eta}$ spectrum.

- As mentioned in section 4.1, a detailed comparison is required between the δ-expansion work [206–209] and other approaches to T_{c}.

4.4.1 Toward the critical-point spectrum

Finally, we will reflect on possible improvements to our current understanding of the quasiparticle spectrum $\xi_{\mathbf{k}} = E_{\mathbf{k}}(T_c)$. Two directions need to be pursued in this regard.

First, as suggested by Stoof, an improved RG calculation, taking into account the momentum dependence of the effective interaction, would improve the ultraviolet behavior of the spectrum in his formalism. Also, it might be worth retaining the wavefunction renormalization (critical index η), which changes the IR spectrum at T_c from $\sim k^2$ to $\sim k^{2-\eta}$. This is neglected in [16] on the grounds that η is small.

A second approach suggests itself when we note that the quasiparticle treatments [2, 190, 195] have all calculated self-energies in terms of the two-body t-matrix $t = 4\pi a/m$. This quantity describes all possible collisions between *two* particles, without taking into account the fact that the surrounding medium has an effect on these collisions. Many-body corrections arising from the surrounding gas can be treated by using the many-body $\mathcal{T}$-matrix, which is the sum of ladder diagrams with full (self-consistent) internal propagators (section 1.2.3). The two-body t-matrix has no temperature-dependence, while the temperature-dependence of the effective potential (vanishing at T_c) is captured by the many-body $\mathcal{T}$-matrix [14–16]. Stoof suggests [204] that this might explain why the quasiparticle analysis (perturbative treatment of spectrum modification) over-adjusts the infrared spectrum.

Determining the self-energies and quasiparticle spectra in terms of the many-body $\mathcal{T}$-matrix has been attempted [219], using the relations provided by Shi & Griffin [15] between the bare potential U, the t-matrix, and the many-body $\mathcal{T}$-matrix. Our attempt was unsuccessful due to the difficulty of disentangling multiple frequency and momentum inter-dependences at second order.

Chapter 5

Quasiparticle Approach to $T_{\rm c}$

After setting the background in chapter 4, we are now ready to present, in this chapter, our quasiparticle framework for analyzing the Bose-Einstein condensation temperature ($T_{\rm c}$) in a WIBG. Our approach to the $T_{\rm c}$ problem is based on a perturbative quasiparticle picture of the system in the un-condensed (high-temperature) side.

The main result of this and related investigations is that $T_{\rm c}$ increases with the scattering length a, with the leading dependence being either linear or log-linear in a. The calculation of $T_{\rm c}$ reduces to that of computing the excitation spectrum near the transition. We describe two approaches to regularizing the infrared divergence at the transition point. One leads to a $a\sqrt{|\ln a|}$-like shift in $T_{\rm c}$, and the other allows numerical calculations for the shift.

5.1 Introduction

For several decades, the quasiparticle picture has been a central paradigm for studying correlated many-body systems. Even in a highly correlated Bose liquid like liquid helium, a quasiparticle description accounts for the transition temperature extremely well; e.g., the lambda point temperature can be obtained (Sec. 4.1.2) from the ideal gas expression $T_{\rm c}^{(0)} = \left(2\pi/m[\zeta(\frac{3}{2})]^{2/3}\right) n^{2/3}$ by replacing the helium atomic mass m by a quasiparticle effective mass, m^{*}.

The question we would like to address is the following: how much information can one obtain about $T_{\rm c}$ from a quasiparticle description of the weakly interacting

Bose gas? The simplest treatment, a Hartree correction, does not provide any shift to the transition temperature. The exchange contribution at mean field level (Fock), does not have any effect on T_c for a momentum-independent interaction, but gives a negative shift when $U(\mathbf{k})$ contains explicit momentum dependence (Sec. 4.1.1, also Ref. [6]). The Fock contribution to T_c depends on the details of the interaction, and for small interactions is weaker than the leading shift we are examining. The present understanding is that there is no mean field effect on the leading shift, and that higher-order correlations are required to calculate this leading shift in T_c.

As introduced in Sec. 1.2.3, interaction effects in the non-ideal Bose gas can be described in terms of the two-body t-matrix, i.e., the vacuum scattering amplitude. This quantity describes all possible collisions between *two* particles, without taking into account the fact that the medium has an effect on these collisions. Many-body corrections arising from the surrounding gas can be treated by using the many-body $\mathcal{T}$-matrix, which is the sum of ladder diagrams.

In this chapter, we approach the calculation of T_c for a Bose gas using perturbation theory in the two-body t-matrix. This is to be regarded as a first step toward a calculation using the many-body $\mathcal{T}$-matrix. The idea is that, if one can regard the system as a weakly interacting gas of quasiparticles, then perturbation theory in the quasiparticle interactions should give a reasonable starting point, and the effects of the surrounding medium can be treated as a correction to these results.

The work in this chapter is not meant to be a definitive determination of T_c for the weakly interacting bose gas, but instead pursues the more limited aim of calculating T_c within a quasiparticle framework. As is typical for descriptions of critical-point properties, our approach faces infrared divergences. We will elaborate on two ways to regularize these divergences, in Secs. 5.4 and 5.5.

5.2 The Quasiparticle Approach

The starting point of our analysis is the expansion of the single particle Green function to leading nontrivial order in quasiparticle interactions, namely:

$$\begin{aligned} G(\mathbf{k},z) &= \frac{1}{z-(\epsilon_{\mathbf{k}}-\mu+\Sigma_{\mathrm{R}}(\mathbf{k},E_{\mathbf{k}}))-(\Sigma(\mathbf{k},z)-\Sigma_{\mathrm{R}}(\mathbf{k},E_{\mathbf{k}}))} \\ &\approx \frac{1}{z-E_{\mathbf{k}}} + \frac{[\Sigma(\mathbf{k},z)-\Sigma_{\mathrm{R}}(\mathbf{k},E_{\mathbf{k}})]}{[z-E_{\mathbf{k}}]^2} + \ldots \\ &= \frac{1}{z-E_{\mathbf{k}}}\left[1+\int\frac{d\omega}{\pi}\frac{\Sigma_{\mathrm{I}}(\mathbf{k},\omega)}{(\omega-z)(\omega-E_{\mathbf{k}})}\right]+\ldots . \end{aligned} \tag{5.1}$$

Here $\Sigma_{\mathrm{R,I}}(\mathbf{k},\omega)=\mathfrak{Re},\mathfrak{Im}\Sigma(\mathbf{k},\omega+i0^+)$ are the real and imaginary parts of the self energy, and $E_{\mathbf{k}}=\epsilon_{\mathbf{k}}-\mu+\Sigma_{\mathrm{R}}(\mathbf{k},E_{\mathbf{k}})$ is the quasiparticle energy. This expression has a number of attractive properties which would not appear in simple perturbation theory in the inter-particle interactions.

In the quasi-particle approximation (5.1), the single-particle spectral function $\mathcal{A}(\mathbf{k},\omega)\equiv-2\mathfrak{Im}G(\mathbf{k},\omega+i0^+)$ is given by:

$$\mathcal{A}(\mathbf{k},\omega) \approx 2\pi\delta(\omega-E_{\mathbf{k}})\left[1+\int\frac{d\omega'}{\pi}\frac{\Sigma_{\mathrm{I}}(\mathbf{k},\omega')}{(\omega'-E_{\mathbf{k}})^2}\right]-2\frac{\Sigma_{\mathrm{I}}(\mathbf{k},\omega)}{(\omega-E_{\mathbf{k}})^2} . \tag{5.2}$$

Note that the form (5.2) automatically satisfies the spectral sum-rule, $\int d\omega\mathcal{A}(\mathbf{k},\omega)=2\pi$.

We proceed by writing the particle number N as a momentum sum over the convolution of the spectral function and the Bose thermal distribution function, $f(\omega)=1/(e^{\beta\omega}-1)$.

$$\begin{aligned} N &= \sum_{\mathbf{k}}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\mathcal{A}(\mathbf{k},\omega)f(\omega) = N_0 + V\int\frac{d^3k}{(2\pi)^3}f(E_{\mathbf{k}}) \\ &+ V\int\frac{d^3k}{(2\pi)^3}\int\frac{d\omega}{\pi}\frac{[f(E_{\mathbf{k}})-f(\omega)]}{(\omega-E_{\mathbf{k}})^2}\Sigma_{\mathrm{I}}(\mathbf{k},\omega) . \end{aligned} \tag{5.3}$$

To proceed further we must resort to an explicit expression for the self-energy. We make the assumption that our quasi-particles interact weakly, so that we

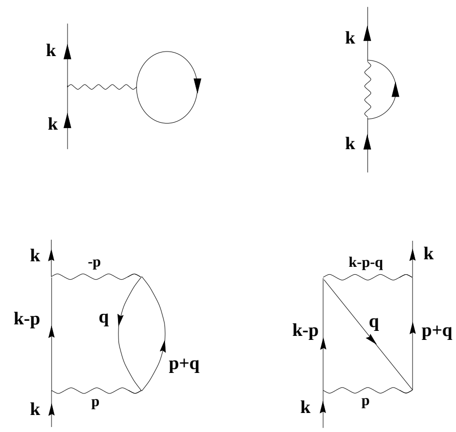


Figure 5.1: The two first-order and two second-order self-energy diagrams included in our calculation. Of the six topologically distinct second-order diagrams, only the two shown here are necessary for momentum-independent interaction, because the other four can be obtained by 1st-order self-energy insertions in the first order diagrams.

can work perturbatively. We limit ourselves to terms up to second order in the interaction.

5.2.1 Perturbation theory

We first perform perturbation theory for the self-energy in the bare potential $U(\mathbf{k})$, retaining explicit momentum-dependence. The diagrams that need to be calculated are shown in figure 5.1. After performing the Matsubara sums, we get:

$$\begin{aligned}\Sigma(\mathbf{k},z) \;\approx\; & \int_{\mathbf{p}} \left[U(\mathbf{k}-\mathbf{p})+U(0)\right] f(\tilde{\epsilon}_{\mathbf{p}}) \\ + \; & \int_{\mathbf{p}}\int_{\mathbf{q}} \frac{\left[f(\tilde{\epsilon}_{\mathbf{q}})+f(\tilde{\epsilon}_{\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p+q}})+f(\tilde{\epsilon}_{\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p-k}})-f(\tilde{\epsilon}_{\mathbf{p+q}})f(\tilde{\epsilon}_{\mathbf{p-k}})\right]}{z-(\tilde{\epsilon}_{\mathbf{p+q}}+\tilde{\epsilon}_{\mathbf{p-k}}-\tilde{\epsilon}_{\mathbf{q}})} \times \\ & U(\mathbf{p})\left[U(-\mathbf{p})+U(\mathbf{k}-\mathbf{p}-\mathbf{q})\right] \; . \qquad (5.4)\end{aligned}$$

Here $\tilde{\epsilon}_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu$, and the notation $\int_{\mathbf{p}} \equiv \int \frac{d^3p}{(2\pi)^3}$ has been introduced for momentum integrals.

For a momentum-independent $U(\mathbf{k}) = U$, the first-order Hartree and Fock terms give equal contributions, as do the two second-order diagrams. However, using $U(\mathbf{k}) = U$ directly in (5.4) causes ultraviolet (UV) problems in the second-order self-energy, due to the term with a single Bose function. The prescription for eliminating such divergences is to switch to a description in terms of the vacuum scattering amplitude t, which is a more physical quantity than the bare potential.

Up to second order, the bare potential $U(\mathbf{k})$ is related to the t-matrix by

$$\begin{aligned} U(\mathbf{k}) &= \mathfrak{Re}\langle \mathbf{p}-\mathbf{k},\mathbf{q}+\mathbf{k}\left|\hat{t}\right|\mathbf{p},\mathbf{q}\rangle \\ &\quad - \int_{\mathbf{k}'} \frac{\mathfrak{Re}\left[\langle \mathbf{p}-\mathbf{k},\mathbf{q}+\mathbf{k}|\,\hat{t}\,|\mathbf{p}-\mathbf{k}',\mathbf{q}+\mathbf{k}'\rangle\langle \mathbf{p}-\mathbf{k}',\mathbf{q}+\mathbf{k}'|\,\hat{t}\,|\mathbf{p},\mathbf{q}\rangle\right]}{\epsilon_{\mathbf{p}}+\epsilon_{\mathbf{q}}-\epsilon_{\mathbf{p}-\mathbf{k}}-\epsilon_{\mathbf{q}+\mathbf{k}}} \end{aligned} \tag{5.5}$$

(Lippman-Schwinger equation, Sec. 1.2.3). When Eqn. (5.5) is used to replace the bare potential by the t-matrix in Eqn. (5.4), we get an $\mathcal{O}(t^2)$ term from the first-order self-energy that exactly cancels the UV-divergent term appearing at second order.

At this stage, there is no longer any difficulty in using a low-momentum limit in which the t-matrix reduces to a momentum-independent constant $t = 4\pi\hbar^2 a/m$. The real (on-shell) and imaginary parts of the self-energy now reduce to:

$$\Sigma_{\mathrm{R}}(\mathbf{k},\tilde{\epsilon}_{\mathbf{k}}) \;\approx\; 2tn \;+\; 2t^2 \int_{\mathbf{p}}\int_{\mathbf{q}} \frac{f(\tilde{\epsilon}_{\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p}+\mathbf{q}})+f(\tilde{\epsilon}_{\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p}-\mathbf{k}})-f(\tilde{\epsilon}_{\mathbf{p}+\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p}-\mathbf{k}})}{\tilde{\epsilon}_{\mathbf{q}}+\tilde{\epsilon}_{\mathbf{k}}-\tilde{\epsilon}_{\mathbf{p}+\mathbf{q}}-\tilde{\epsilon}_{\mathbf{p}-\mathbf{k}}}\;, \tag{5.6}$$

and

$$\begin{aligned} \Sigma_{\mathrm{I}}(\mathbf{k},\omega) \;\approx\; &-2\pi t^2 \int_{\mathbf{p}}\int_{\mathbf{q}} \delta\left(\omega-[\tilde{\epsilon}_{\mathbf{p}+\mathbf{q}}+\tilde{\epsilon}_{\mathbf{p}-\mathbf{k}}-\tilde{\epsilon}_{\mathbf{q}}]\right) \\ &\times\left[f(\tilde{\epsilon}_{\mathbf{q}})+f(\tilde{\epsilon}_{\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p}+\mathbf{q}})+f(\tilde{\epsilon}_{\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p}-\mathbf{k}})-f(\tilde{\epsilon}_{\mathbf{p}+\mathbf{q}})f(\tilde{\epsilon}_{\mathbf{p}-\mathbf{k}})\right]\;. \end{aligned} \tag{5.7}$$

5.2.2 Self-consistent perturbation theory

We will use (5.6) and (5.7) to define a self-consistent quasi-particle approximation, formally second order in the vacuum scattering amplitude. This is accomplished by replacing the free-particle dispersion functions ($\epsilon_{\mathbf{q}}$'s) on the right hand sides of (5.6),(5.7) by the quasiparticle dispersions, $E_{\mathbf{q}}$'s. The imaginary part now becomes

$$\Sigma_{\rm I}(\mathbf{k},\omega) \approx -2\pi t^2 \int_{\mathbf{p}}\int_{\mathbf{q}} \delta\left(\omega - [E_{\mathbf{p+q}} + E_{\mathbf{p-k}} - E_{\mathbf{q}}]\right)$$
$$\times \left[f(E_{\mathbf{q}}) + f(E_{\mathbf{q}})f(E_{\mathbf{p+q}}) + f(E_{\mathbf{q}})f(E_{\mathbf{p-k}}) - f(E_{\mathbf{p+q}})f(E_{\mathbf{p-k}})\right] \ . \quad (5.8)$$

When this expression for the imaginary part is used in Eqn. (5.3), the $\Sigma_{\rm I}$ contribution cancels out exactly, resulting in

$$n = \int \frac{d^3k}{(2\pi)^3} f\left(E_{\mathbf{k}}\right) \ ; \quad (5.9)$$

i.e., the number of quasiparticles is equal to the number of particles. Since we will be using these results above and at the transition, we have neglected the zero-momentum occupancy N_0 as compared to N.

We note that the imaginary part of the self-energy, i.e., effects of quasiparticle broadening, has dropped out of the expression for particle number and hence does not play any role in our determination of the transition temperature.

Finally, making the real part of the self energy (eq 5.6) self-consistent, we get the following expression for the quasi-particle energy:

$$E_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu + \Sigma_{\rm R}(\mathbf{k}, E_{\mathbf{k}}) = \epsilon_{\mathbf{k}} - (\mu - 2tn)$$
$$+ 2t^2 \int_{\mathbf{p}}\int_{\mathbf{q}} \frac{f(E_{\mathbf{q}})f(E_{\mathbf{p+q}}) + f(E_{\mathbf{q}})f(E_{\mathbf{p-k}}) - f(E_{\mathbf{p+q}})f(E_{\mathbf{p-k}})}{E_{\mathbf{q}} + E_{\mathbf{k}} - E_{\mathbf{p+q}} - E_{\mathbf{p-k}}} \ . \quad (5.10)$$

5.3 Critical Temperature and Spectrum

To derive an expression for the change in the critical temperature due to quasiparticle interactions, we first consider the density of an ideal Bose gas, both at its transition temperature $T_{\rm c}^{(0)} = \left(2\pi/m[\zeta(\frac{3}{2})]^{2/3}\right) n^{2/3}$, and at temperature $T_{\rm c}$ which is the transition point of the *interacting* Bose gas.

$$\int_{\mathbf{k}} \frac{1}{e^{\epsilon_{\mathbf{k}}/k_B T_{\rm c}} - 1} = n^{(0)}(T_{\rm c}) \approx \left(\frac{T_{\rm c}}{T_{\rm c}^{(0)}}\right)^{3/2} n^{(0)}(T_{\rm c}^{(0)})$$
$$\approx \left(1 + \frac{3}{2}\frac{\Delta T_{\rm c}}{T_{\rm c}^{(0)}}\right) n^{(0)}(T_{\rm c}^{(0)}) \ . \quad (5.11)$$

Here $T_c = T_c^{(0)} + \Delta T_c$. We are looking for the difference of transition temperatures of an interacting and a non-interacting gas at the *same density*. Therefore, $n^{(0)}(T_c^{(0)}) = n(T_c) = n$. Evaluating (5.9) at T_c and subtracting Eqn. (5.11) we obtain the leading correction to T_c due to interactions:

$$\frac{\Delta T_c}{T_c^{(0)}} \approx -\frac{2}{3n}\int \frac{d^3k}{(2\pi)^3}\,\left[f_c(\xi_{\mathbf{k}}) - f_c(\epsilon_{\mathbf{k}})\right]. \tag{5.12}$$

Here

$$f_c(x) = \frac{1}{e^{x/kT_c}-1}$$

is the Bose distribution function evaluated at the critical temperature. Also, $\xi_{\mathbf{k}}$ is the single particle excitation spectrum at T_c, i.e.,

$$\xi_{\mathbf{k}} = E_{\mathbf{k}}(T_c) = \epsilon_{\mathbf{k}} - \mu(T_c) + \Sigma_{\mathbf{R}}(\mathbf{k},\xi_{\mathbf{k}}) = \epsilon_{\mathbf{k}} + \Sigma_{\mathbf{R}}(\mathbf{k},\xi_{\mathbf{k}}) - \Sigma_{\mathbf{R}}(0,0)\,.$$

The problem of calculating the relative shift has thus been reduced to calculating the spectrum at the critical point, $\xi_{\mathbf{k}}$.

Eqn. (5.12) allows us to predict the direction of the shift in T_c. Based on this equation, in chapter 4 (Sec. 4.2.2) a simple argument has been presented for the *increase* of T_c due to the addition of a repulsive interaction.

5.3.1 Spectrum from perturbation theory

It follows from (5.10) that $\xi_{\mathbf{k}}$ satisfies the nonlinear integral equation,

$$\begin{aligned}\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} + 2t^2\int_{\mathbf{p}}\int_{\mathbf{q}} &\frac{f(\xi_{\mathbf{p}})f(\xi_{\mathbf{q}}) + f(\xi_{\mathbf{p}})f(\xi_{\mathbf{p+k-q}}) - f(\xi_{\mathbf{q}})f(\xi_{\mathbf{p+k-q}})}{\xi_{\mathbf{p}} + \xi_{\mathbf{k}} - \xi_{\mathbf{q}} - \xi_{\mathbf{p+k-q}}} \\ - 2t^2\int_{\mathbf{p}}\int_{\mathbf{q}} &\frac{f(\xi_{\mathbf{p}})f(\xi_{\mathbf{q}}) + f(\xi_{\mathbf{p}})f(\xi_{\mathbf{p-q}}) - f(\xi_{\mathbf{q}})f(\xi_{\mathbf{p-q}})}{\xi_{\mathbf{p}} - \xi_{\mathbf{q}} - \xi_{\mathbf{p-q}}}\end{aligned} \tag{5.13}$$

In the perturbative quasiparticle approach, Eqns. (5.12) and (5.13) completely determine ΔT_c.

If the quasiparticle dispersion $\xi_{\mathbf{k}}$ were quadratic at small momenta, the self-energy integral in Eqn. (5.13) would be divergent in the infrared. A power-counting of this equation shows that, for self-consistency, the dispersion at small momenta should be sub-quadratic, $\xi_{\mathbf{k}} \sim k^{3/2}$, modulo possible logarithmic factors which power-counting cannot predict. The power-counting arguments were first devised by Patashiskii and Pokrovskii [221].

Unfortunately, this approach using second-order perturbation theory over-modifies the infrared spectrum. The long-wavelength spectrum at the transition point should be $\xi_{\mathbf{k}} \sim k^{2-\eta}$, where η is the anomalous dimension. The present system falls in the same universality class as the $N = 2$ quantum rotor model. For this universality class, a recent Monte Carlo calculation [216] has produced the value $\eta \approx 0.038$. In contrast, Eqn. (5.13) gives $\eta = 0.5$.

Acknowledging that the present framework over-modifies the spectrum at T_{c}, we proceed to analyze Eqns. (5.12) and (5.13) in order to extract the shift in the critical temperature due to interactions.

5.3.2 Infrared divergence

There are several ways of dealing with the infrared divergence. One is to treat Eqn. (5.13) self-consistently and determine $\xi(\mathbf{k})$ numerically. This procedure is outlined in Sec. 5.5. This approach has also been explored by Baym and collaborators.

In the next section, we take a different approach, motivated by the work of Kanno ([125] and Sec. 4.3). We make an expansion in $\Sigma_{\mathrm{R}}(\mathbf{k}, \xi_{\mathbf{k}})$ that allows us to use the chemical potential at the critical point, $\mu(T_{\mathrm{c}}) = \mu_{\mathrm{c}}$, as an infrared cutoff. This approach allows us to express the relative shift in the transition temperature as an expansion in $\beta_{\mathrm{c}}\mu_{\mathrm{c}} = \mu_{\mathrm{c}}/k_{\mathrm{B}}T_{\mathrm{c}}$, which itself can in turn be related to the scattering length. This leads to a transition temperature shift ΔT_{c} that is $a\sqrt{\mathcal{K}(a)}$ to leading order, where $\mathcal{K}(x)$ is a function approximately like $|\ln(x)|$, to be specified in the next section. The appropriateness of this procedure is difficult to gauge, but it provides strong support to the possibility of non-analytic corrections to the transition temperature at leading order.

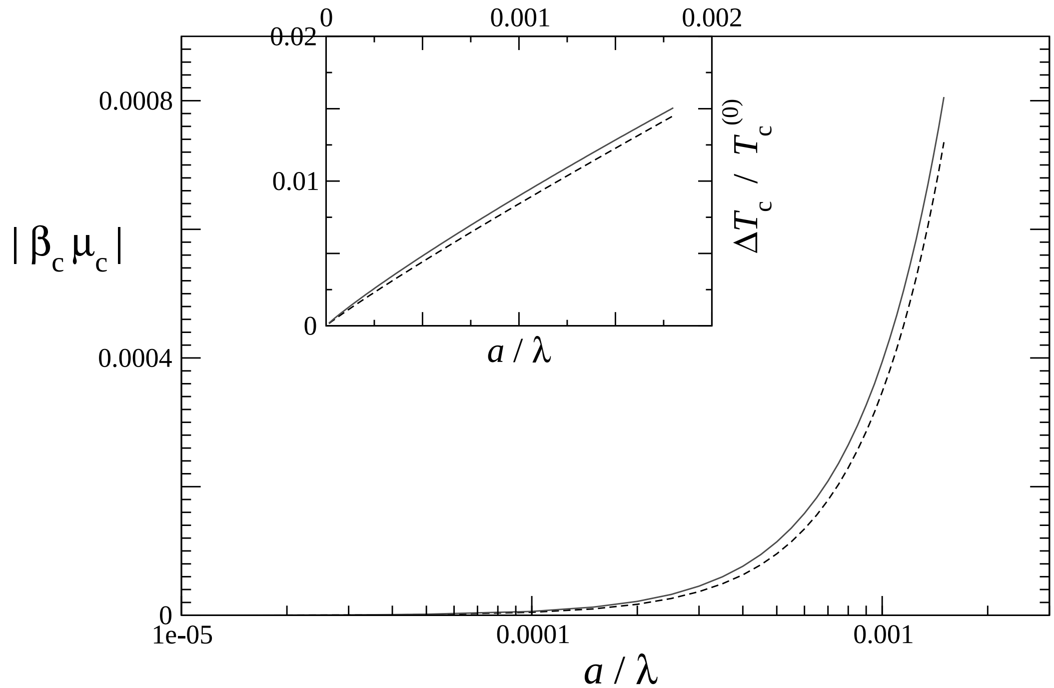


Figure 5.2: Approximation for Eqn. (5.17). Solid curve is exact numeric solution; dashed curve shows approximation $|\beta_c\mu_c| \approx 16\pi(a/\lambda_c)^2|\ln(a/\lambda_c)|$. Inset shows comparison for the relative shift of T_c; solid curve is numeric solution and dashed curve is the $\sim a\sqrt{|\ln(a)|}$ approximation.

Another infrared regularization procedure that has appeared in the literature is A.M.J. Schakel's prescription of analytic continuation [103]. The connection between Schakel's treatment and other calculations awaits further study.

5.4 Non-SelfConsistent Calculation

We expand $f_c(\xi_{\mathbf{k}}) = f_c(\epsilon_{\mathbf{k}} - \mu_c + \Sigma_R(\mathbf{k}, \xi_{\mathbf{k}}))$ around $\epsilon_{\mathbf{k}} - \mu_c$:

$$f_c(\xi_{\mathbf{k}}) \approx f_c(\epsilon_{\mathbf{k}} - \mu_c) + f_c'(\epsilon_{\mathbf{k}} - \mu_c)\Sigma_R(\mathbf{k}, \epsilon_{\mathbf{k}} - \mu_c)) \ . \tag{5.14}$$

This provides us with an infrared cutoff, $-\mu_c$, for our integrals. The approximation is non-selfconsistent in the sense that a modification of the spectrum to a non-quadratic form no longer plays any role. Using this expansion, the relative shift in T_c becomes

$$\frac{\Delta T_\mathrm{c}}{T_\mathrm{c}^{(0)}} \approx -\frac{2}{3n}\int_\mathbf{k}\left[f_\mathrm{c}(\epsilon_\mathbf{k}-\mu_\mathrm{c})-f_\mathrm{c}(\epsilon_\mathbf{k})\right] - \frac{2}{3n}\int_\mathbf{k} f_\mathrm{c}'(\epsilon_\mathbf{k}-\mu_\mathrm{c})\Sigma_\mathrm{R}(\mathbf{k},\epsilon_\mathbf{k}-\mu_\mathrm{c})\,. \quad (5.15)$$

Both integrals can be expanded in $|\beta_\mathrm{c}\mu_\mathrm{c}|$, as detailed in appendix A, each integral being $\mathcal{O}(\sqrt{|\beta_\mathrm{c}\mu_\mathrm{c}|})$ at leading order. The result is

$$\begin{aligned}\frac{\Delta T_\mathrm{c}}{T_\mathrm{c}^{(0)}} &\approx +\frac{2}{3n}\left[\frac{\sqrt{\pi}}{\lambda_\mathrm{c}^3}\sqrt{-\beta_\mathrm{c}\mu_\mathrm{c}}\right]+\mathcal{O}(\beta_\mathrm{c}\mu_\mathrm{c}) \\ &= \frac{2}{3}\left[\zeta(\tfrac{3}{2})\right]^{-1}\sqrt{\pi|\beta_\mathrm{c}\mu_\mathrm{c}|} + \mathcal{O}(\beta_\mathrm{c}\mu_\mathrm{c})\,. \quad (5.16)\end{aligned}$$

The infrared cutoff needs to be expressed in terms of the scattering length; this is achieved by using the condition for the transition, $\mu_c = \Sigma_\mathrm{R}(0,0)$. The relation is

$$\beta_\mathrm{c}\mu_\mathrm{c} \approx \frac{16\pi a^2}{\lambda_\mathrm{c}^2}\ln(-\beta_\mathrm{c}\mu_\mathrm{c})\,, \quad (5.17)$$

also derived in appendix A. This equation expresses the IR cutoff $\beta_\mathrm{c}\mu_\mathrm{c}$ in terms of the dimensionless interaction parameter a/λ, albeit in an implicit form.

The implicit function $y = \alpha_1 x^2 \ln(-y)$ (small x, y) is well approximated by $y \approx \alpha_1 x^2 \ln|x|$. With $|\beta_\mathrm{c}\mu_\mathrm{c}|$ having dependence similar to $\sim (a/\lambda_\mathrm{c})^2|\ln(a/\lambda_\mathrm{c})|$, we thus have a shift approximately of the form $\Delta T_\mathrm{c} \sim a\sqrt{|\ln a|}$.

The approximation of Eqn. (5.17) by $|\beta_\mathrm{c}\mu_\mathrm{c}| \approx 16\pi(a/\lambda_\mathrm{c})^2|\ln(a/\lambda_\mathrm{c})|$ is displayed in figure 5.2.

5.5 Self-Consistent Approach: Numerics

A self-consistent treatment of (5.13) involves a calculation of the (sub-quadratic) quasiparticle dispersion $\xi(\mathbf{k})$. Unfortunately, we are unaware of any procedure for an analytic selfconsistent solution of (5.13), other than information obtained from power-counting.

Eqn. (5.13) involves principal values, which are difficult to treat numerically in general, and even more so in the present case because the principal-value integral is the innermost among four integrations, and also because the denominator consists of functions ($\xi_{\mathbf{k}}$) that are imperfectly determined at intermediate stages of the calculation.

A "high-temperature" approximation, obtained by using $f(\xi) \approx 1/\beta\xi$, produces a simpler version without the zeroes of the principal value form:

$$\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - 2t^2 \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \left(\frac{1}{\xi_{\mathbf{p}}\xi_{\mathbf{q}}\xi_{\mathbf{p}+\mathbf{k}-\mathbf{q}}} - \frac{1}{\xi_{\mathbf{p}}\xi_{\mathbf{q}}\xi_{\mathbf{p}-\mathbf{q}}} \right) . \tag{5.18}$$

We present here numeric results based on this approximation.

Since the self-consistent spectrum has infrared behavior $\xi(\mathbf{k}) \sim k^{3/2}$, we look for solutions of the form $\xi(\mathbf{k}) = k^{3/2}\sigma(k) + k^2$ where $\sigma(x)$ is some smooth function that decays to zero for large x, and is constant or logarithm-like at small x. (Here the momenta and ξ have been rescaled to be dimensionless, by factors $\sqrt{\beta/2m}$ and β respectively). A more detailed account of the numeric work is in Appendix A.

Figure 5.3 shows a numerically determined spectrum $\xi(\mathbf{k})$. Once $\xi(\mathbf{k})$ has been computed for a particular a/λ, the relative shift in T_c can be computed using our basic Eqn. (5.12). The ΔT_c vs. a plot in the inset shows a definite linear-like increase, as opposed to, say, a $\sqrt{a}$ or $a^{3/2}$ dependence. However the existence, or lack thereof, of a logarithm-like factor is difficult to prove from numeric data.

We note that the high-temperature version (5.18), while dimensionally equivalent to the full Eqn. (5.13), might already be missing logarithm-like factors. In other words, even though the two equations should have solutions of the same type $\xi(\mathbf{k}) = k^{3/2}\sigma(k) + k^2$, the form of the interpolation function $\sigma(k)$ can be expected to be quite different. The effect on ΔT_c from a nonanalytic correction to $\sigma(k)$ is difficult to predict, but quite possibly non-analytic.

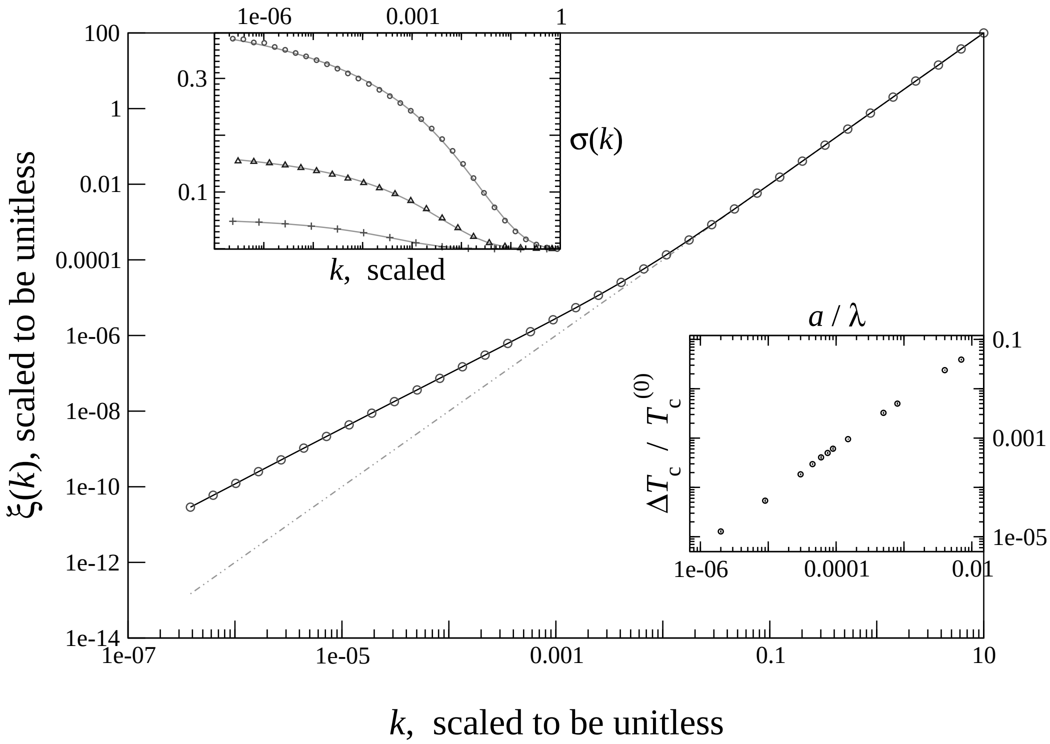


Figure 5.3: Quasiparticle spectrum, for $(a/\lambda) = 0.0005$. (The "fit" indicates quality of iteration – line shows penultimate, and dots show final, iteration). Crossover from $k^{3/2}$ to k^2 behavior is clear; dotted line is the noninteracting k^2 spectrum for comparison. Upper inset shows the crossover function $\sigma(k)$ for several a/λ values (9×10^{-5}, 8×10^{-4} and 4×10^{-3}). Lower inset plots relative shift of T_c against interaction.

5.6 Concluding Remarks

To summarize, we have outlined the calculation of T_c from a perturbative expansion in the two-body t-matrix, or equivalently the scattering length a. The transition temperature is expressed in terms of the quasiparticle spectrum at T_c, and hence the problem is transformed to one of calculating the critical-point spectrum $\xi_{\mathbf{k}} = E_{\mathbf{k}}(T_c)$.

In developing the formalism, we find that the use of the two-body t-matrix removes the ultraviolet divergences that appear in a perturbative calculation in terms of a momentum-independent bare potential U. We also find that the width of the quasiparticle peak (Σ_I) does not affect T_c. Whether this is true at all orders, or a peculiarity of second-order, and whether this is true for a more accurate treatment in terms of the many-body $\mathcal{T}$-matrix, remain open questions.

One new feature of this work is an unusual approach to dealing with the infrared divergence that appears in treating the spectrum at T_c. This regularization scheme bypasses the issue of spectrum modification, and results in a non-analytic ($\sqrt{\ln a}$-like) correction factor to the linear shift of the transition temperature.

The major remaining issue in the T_c problem seems to be the possible existence of a logarithm-like factor in the leading a-dependence. The resolution of this problem should lie in a clear treatment of the spectrum at T_c. Of the two major approaches, the perturbative approach alters the quadratic spectrum too strongly, while the RG calculation [16] neglects spectrum alteration.

The reason for perturbation theory over-estimating the spectrum modification remains to be examined in detail. It has been suggested [204] that the reason is that perturbation in a or t does not take into account the fact that the effective interaction strength *flows to zero* under renormalization group flow, at the critical point. An alternate way to view this is to note that the many-body $\mathcal{T}$-matrix (sum of ladder diagrams) vanishes [14–16] at the transition temperature. A perturbative calculation in terms of the many-body $\mathcal{T}$-matrix, as opposed to one in terms of the two-body t-matrix that we have considered here, may therefore be expected to give a more realistic modification of the quasiparticle spectrum at T_{c}. A calculation of $\xi_{\mathbf{k}}$ in terms of the many-body $\mathcal{T}$-matrix has not yet appeared in the literature.

Chapter 6

Conclusion and Outlook

To conclude, we will try to put the material discussed in the previous chapters in a broader context.

The study of the uniform interacting Bose gas model continues to bring forth new surprises and results today, half a century after the first mean field analysis.

One case in point is the behavior of the low-energy many-body $\mathcal{T}$-matrix, near the transition temperature, as shown schematically in Fig. 1.6 (section 1.3.2). This violent temperature-dependence was discovered very recently, in 1997 [14]. The repercussions of this result, in calculations of T_c and of various quantities near the transition, are just beginning to be explored.

The application of ideas of quantum optics to study the quantum state of the Bose gas is another area only starting to be examined. Concepts of quantum optics, not available to WIBG researchers until a decade or so ago, open up many new questions concerning the WIBG. A handful of studies have appeared [76–80, 142, 222, 223], but only the surface of this emerging field appears to have been scratched.

The first topic discussed in this thesis, squeezing in the WIBG, is part of this expanding area of research. In addition to clarifying some of the squeezing properties of a uniform Bose gas, our work in chapter 3 exposes a new set of queries that remain to be addressed. Some such open questions are identified in Sec. 3.4.

Our treatment of the second topic, the T_c problem, involves an analysis and re-formulation of some of the literature, and the development of a quasiparticle description in terms of the t-matrix. The calculations presented here should be regarded as a first step toward a treatment that takes into account the full many-body $\mathcal{T}$ matrix.

Associated with the T_c problem is another fundamental question concerning the Bose gas: the quasiparticle spectrum at the condensation point ($\xi_{\mathbf{k}}$). While numeric studies on the universality class (XY model, e.g., [216]) provide the infrared behavior of the spectrum, the ultraviolet part of the spectrum may also have an effect on T_c [190, 195, 204]. Thus the problem of calculating the non-universal aspects of $\xi_{\mathbf{k}}$ is closely coupled to the problem of calculating T_c. A proper calculation of the critical-point spectrum would also have to take into account the behavior of the $\mathcal{T}$ matrix.

Thus, in calculating quantities associated with the BEC transition of the non-ideal Bose gas, the task seems to be to formulate descriptions in terms of the many-body $\mathcal{T}$ matrix. As mentioned, work in this direction has barely begun, and this is expected to be a major theme in calculations involving the BEC transition in the coming years.

Appendix A

$T_{\rm c}$ Calculations: Details

Details of caclulations reported in chapter 5 are given in this appendix. Section A.1 has the omitted details of the non-selfconsistent calculation of 5.4, and section A.2 has a more elaborate account of the numeric procedures used to calculate the self-consistent spectrum, reported in Sec. 5.5.

A.1 Shift in $T_{\rm c}$, Using Infrared Cutoff

In this appendix we evaluate the two contributions to $\Delta T_{\rm c}/T_{\rm c}^{(0)}$ in Eqn. (5.15), and show how Eqn. (5.17) follows from the criticality condition $\Sigma(0,0) = \mu$.

The first integral in (5.15) is $\int_{\mathbf{k}} [f_{\rm c}(\epsilon_{\mathbf{k}} - \mu_{\rm c}) - f_{\rm c}(\epsilon_{\mathbf{k}})] = \lambda_{\rm c}^{-3} g_{3/2}(e^{\beta_{\rm c}\mu_{\rm c}}) - \lambda_{\rm c}^{-3}\zeta(\frac{3}{2})$. Here the $g_{3/2}$'s are the Bose-Einstein integral functions [115], $g_n(x) = \sum_i (x^i/i^n)$, and $\zeta(n) = g_n(1)$. We are using the notation $\int_{\mathbf{k}} \equiv (2\pi)^{-3} \int d^3k$. Using the Robinson expansion [115] for $g_{3/2}$, we get

$$\int_{\mathbf{k}} [f_{\rm c}(\epsilon_{\mathbf{k}} - \mu_{\rm c}) - f_{\rm c}(\epsilon_{\mathbf{k}})] = \frac{1}{\lambda_{\rm c}^3}[-2\sqrt{\pi(-\beta_{\rm c}\mu_{\rm c})} + \mathcal{O}(\beta_{\rm c}\mu_{\rm c})] \ .$$

For the second integral in (5.17), using the second-order self-energy from Eqn. (5.6), one obtains after some variable transformations:

$$\begin{aligned}\int_{\mathbf{k}} f'_{\rm c}(\tilde{\epsilon}_{\mathbf{k}})\, \Sigma_{\rm R}(\mathbf{k}, \epsilon_{\mathbf{k}}) \ &= \ -2t^2 \frac{\partial}{\partial \mu_{\rm c}} \int_{\mathbf{p}} \int_{\mathbf{q}} \int_{\mathbf{k}} \frac{f_{\rm c}(\tilde{\epsilon}_{\mathbf{p}}) f_{\rm c}(\tilde{\epsilon}_{\mathbf{q}}) f_{\rm c}(\tilde{\epsilon}_{\mathbf{k}})}{\epsilon_{\mathbf{p}} + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{q}} - \epsilon_{\mathbf{p}+\mathbf{k}-\mathbf{q}}} \\ &\approx \ \frac{1}{\lambda_{\rm c}^5}\, 16\pi^{3/2} a^2 \, \frac{\ln(-\beta_{\rm c}\mu_{\rm c})}{\sqrt{-\beta_{\rm c}\mu_{\rm c}}} \ .\end{aligned}$$

Here $\tilde{\epsilon}_{\mathbf{p}} = \epsilon_{\mathbf{p}} - \mu_{\rm c}$. In combination with the criticality condition (5.17), this leads to

$$\int_{\mathbf{k}} f_c'(\tilde{\epsilon}_{\mathbf{k}})\Sigma_R(\mathbf{k},\epsilon_{\mathbf{k}}) \approx \frac{1}{\lambda_c^3}\left[\sqrt{\pi(-\beta_c\mu_c)} + \mathcal{O}(\beta_c\mu_c)\right] \ .$$

The condition (5.17) for the transition temperature is obtained from $\mu_c = \Sigma_R(0,0)$ by using the second-order perturbative result for the self-energy (e.g., obtained from Eqn. (5.6) by setting $\mathbf{k} = 0$):

$$\mu_c = 4t^2 \int_{\mathbf{p}}\int_{\mathbf{q}} \frac{f(\tilde{\epsilon}_{\mathbf{p}})f(\tilde{\epsilon}_{\mathbf{q}})}{\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{q}} - \epsilon_{\mathbf{p}-\mathbf{q}} - \mu_c} \approx -\frac{8ma^2}{\pi^2\beta_c^2}\ln(-\beta_c\mu_c) \ .$$

A.2 Numeric Procedure for Spectrum $\xi_{\mathbf{k}}$

We describe here the numeric procedure for calculating the spectrum self-consistently from the "high-temperature" approximation for the critical point spectrum $\xi_{\mathbf{k}}$ (Eqn. 5.18):

$$\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - 2t^2 \int \frac{d^3p}{(2\pi)^3}\int \frac{d^3q}{(2\pi)^3} \left(\frac{1}{\xi_{\mathbf{p}}\xi_{\mathbf{q}}\xi_{\mathbf{p}+\mathbf{k}-\mathbf{q}}} - \frac{1}{\xi_{\mathbf{p}}\xi_{\mathbf{q}}\xi_{\mathbf{p}-\mathbf{q}}}\right) \ . \qquad \text{(A.1)}$$

We work in terms of dimensionless quantities, $\tilde{p} = \sqrt{(\beta/2m)}p$ and $\tilde{\xi} = \beta\xi$. In the rest of this section all momenta and ξ's actually refer to their dimensionless tilde versions. We are looking for solutions of the form

$$\xi(\mathbf{k}) = k^{3/2}\sigma(k) + k^2 \ , \qquad \text{(A.2)}$$

where $\sigma(x)$ is some smooth function that decays to zero for large x. The basic idea is to start with a function of this form in the right side of (A.1), and iterate until convergence. We make several notes on the procedure:

- A simple iteration unfortunately gives oscillatory behavior, with ξ_k's moving away from the actual solution. This problem was resolved by using, at each step, a function intermediate between the current and previous iterations.
- One could use ξ values at a discretized set of k-values as the representation of $\xi_{\mathbf{k}}$, at each step of the iteration. However, this method involves an interpolation procedure when using the discretized set as input for the next

iteration. In order to avoid making interpolations (especially the very important interpolation between $(k,\xi) = (0,0)$ and the smallest grid-point), we use functional representations for $\sigma(k)$, and hence for $\xi(k)$.

- At each iteration step, the previous $\xi(k)$ is used to generate the values of the new $\xi(k)$ at a logarithmically discretized set of k-points. Logarithmic discretization makes sense here because the low-k regime is particularly importnant.

- At each iteration, the previous and new forms of the interpolation function $\sigma(k)$ are compared. This gives a much more sensitive indication of iteration than the whole spectrum $\xi(k)$, whose qualitative behavior does not change much.

- Surprisingly, the iteration is found to be quite sensitive to the form of the interpolation function $\sigma(k)$; not just the scale of the $k^{3/2}$-k^2 crossover. It seemed difficult to model σ as an algebraic function without a prohibitively large number of parameters. One such attempt is shown in Fig. A.1.

- The interpolation functions we found suitable were of the form $\sigma(k) = (c_2 - c_0 k^{\nu}) \exp\left[-(k/s_1)^{s_2}\right]$. Examples of $\sigma(k)$ found with this form are shown in the upper inset of Fig. 5.3. Some more examples are shown in Fig. A.2.

- While the quality of "fits" between penultimate and final iterations are good, it should be noted that we did not investigate ultraviolet behavior. If there is any logarithmic deviation of $\xi_{\mathbf{k}}$ from k^2 at high k, our method might not capture such behavior, since the form (A.2) enforces a free-particle (k^2) UV dependence.

Once the $\xi_{\mathbf{k}}$ have been computed for a particular a/λ, the relative shift in T_{c} can be computed using Eqn. (5.12), which in terms of the dimensionless variables

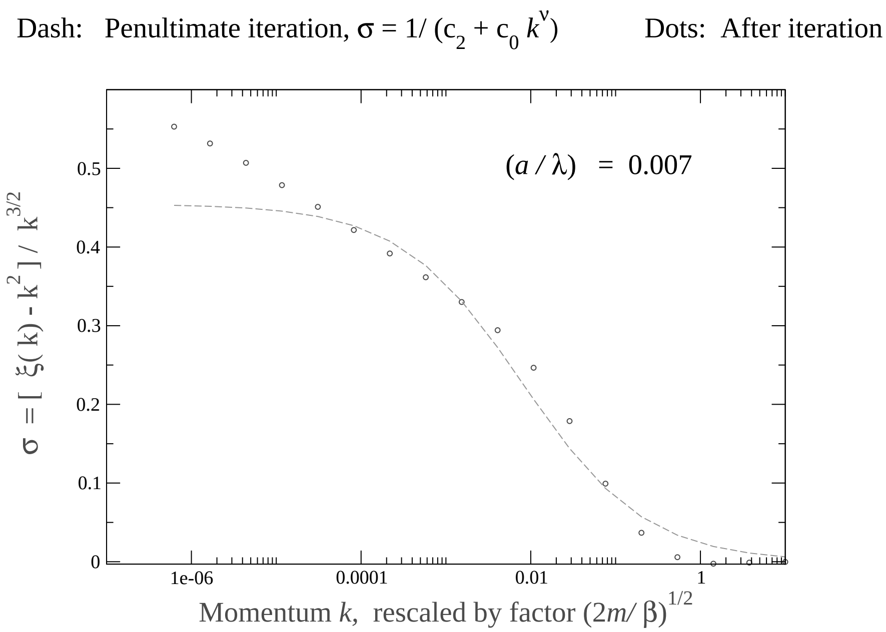


Figure A.1: Attempt to use $\sigma(k) = 1/(c_2 + c_0 k^\nu)$. The best fit for this form of $\sigma(k)$ is shown here. The dashed line is a $\sigma(k)$ before iteration, having the algebraic form with $\nu = 0.6$, $c_0 = 40$, $c_2 = 2.2$. The dots show the $\sigma(k)$ obtained by iteration from the algebraic form. Obviously, a different form is required for the iteration.

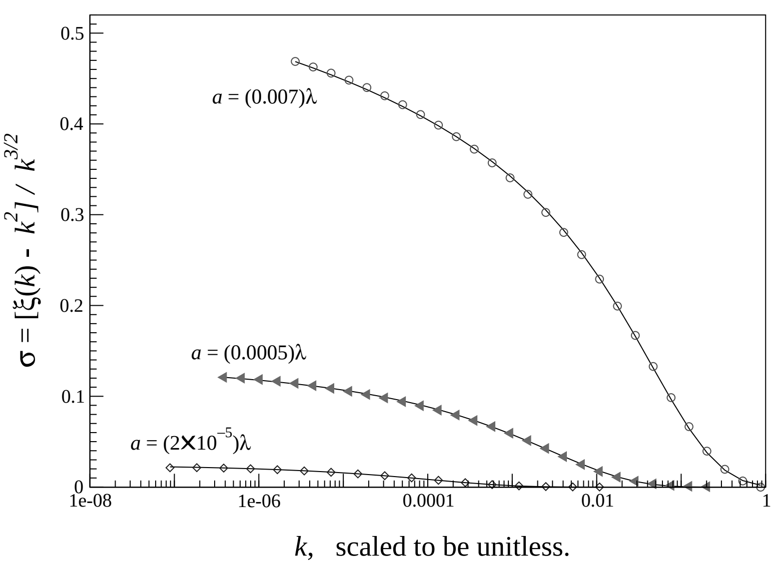


Figure A.2: Solutions for $\sigma(k)$, using the form $\sigma(k) = (c_2 - c_0 k^\nu) \exp\left[-(k/s_1)^{s_2}\right]$. Solid lines show the $\sigma(k)$ used for the last iteration, and points are obtained in this final iteration. The quality of the fit indicates the quality of the solutions.

is

$$\frac{\Delta T_{\mathrm{c}}}{T_{\mathrm{c}}^{(0)}} = \frac{8}{3\sqrt{\pi}\zeta(\frac{3}{2})} \int dk\ k^2 \left[\frac{1}{e^{\xi_{\mathbf{k}}} - 1} - \frac{1}{e^{k^2} - 1} \right] \ .$$

The $\Delta T_{\mathrm{c}}/T_{\mathrm{c}}^{(0)}$ calculated by performing this integral numerically are plotted against a/λ in the lower inset of Fig. 5.3.

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